The Decoupled Threshold VAR: Mean and Variance Regimes in State-Dependent Fiscal Multipliers*

Valeria Gargiulo†

This version: November 9, 2026

Click here for the latest version

Abstract

This paper proposes a novel approach to the estimation of threshold VARs in which the regimes govern the conditional variance as well as the conditional mean. We introduce the *decoupled threshold VAR*, in which the two sets of regimes are driven by different observable state variables with separate thresholds, both estimated from the data; the standard *joint-switching* model, in which the two states and thresholds coincide, is a special case, consistently selected and estimated under our general framework. Existing threshold estimators locate the regimes from the conditional mean alone. We show that this is inconsistent when there is a regime change in the variance – a common feature of macroeconomic data – and inefficient when both moments change, and we develop a likelihood-based estimator that uses both. We establish consistency, derive the non-standard limiting distributions of the thresholds, and show that their super-consistent rate leaves inference on the regime parameters unaffected. The framework delivers a new reading of U.S. government spending multipliers. Wars and major fiscal events form a separate high-variance regime. Once it is isolated, significant state dependence emerges even at the thresholds imposed in earlier work, and post-WWII multipliers are larger in expansions than in recessions. The result reconciles apparently conflicting findings in the literature.

*I am grateful to my advisors Katerina Petrova and Barbara Rossi for their constant guidance and support, and to Davide Debortoli for the invaluable discussions which considerably improved the paper. I also thank Christian Brownlees, Alessandro Ciancetta, Paula Donaldson, Kirill Evdokimov, Luca Gambetti, Ankita Ghosh, Oriol González-Casasús, Adam Lee, Daniel Lewis, Marta Morazzoni, Tatevik Sekhposyan, Elena Pellegrini, Michele Piffer, Elias Tzavalis, Robert Wojciechowski, and seminar participants at BI Norwegian Business School, CREi and UPF for their useful comments. All errors are my own.

†Universitat Pompeu Fabra and Barcelona School of Economics. Email: valeria.gargiulo@upf.edu

1 Introduction

Nonlinearities in macroeconomic data are estimated by letting parameters differ across periods or regimes. A central lesson of this literature over the last twenty years is that the second moments change more than the first. The Great Moderation was primarily a fall in the volatility of shocks; the drifting-parameter models of Cogley and Sargent (2005) and Primiceri (2005) attribute most of the time variation in U.S. monetary dynamics to stochastic volatility rather than to the coefficients; and Sims and Zha (2006) find that, among regime-switching models of monetary policy, the best fit is obtained when only the variances switch. Nor is volatility only a nuisance parameter: it is an economic object with real effects of its own – uncertainty shocks (Bloom, 2009) and, closest to our application, fiscal volatility shocks (Fernández-Villaverde et al., 2015) – worth measuring and dating.

This paper proposes a novel threshold VAR (TVAR) in which the variance has regimes of its own, and develops its estimation theory. Like the regimes in the dynamics, the variance regimes are dated by an observable state variable crossing an estimated threshold. The motivation is structural analysis with regimes: how the transmission of an identified shock differs across states of the economy. Two different features of the economy can define a regime. One is the state of the economy – slack, the zero lower bound, financial stress – which changes how a shock propagates. The other is the shock environment – a war, an oil crisis, a pandemic – which changes how large the shocks are. The two need not coincide: the Great Moderation changed the second with little change in the first, and the pandemic quarters are treated in current practice as a change in the size of the shocks rather than in the dynamics (Lenza and Primiceri, 2022). A model with a common regime structure for the mean and the variance must represent both features with one partition of the sample. This has two possible consequences. If only the propagation is allowed to differ – the standard TVAR, with a single regime structure in the mean – the innovation covariance pools two shock environments into one. The impact response of every identified shock is computed from that covariance, so it is misestimated in every regime, before any dynamics enter; and the information the variance carries on the location of the regime boundary is lost.[1] If both are allowed to differ but along a single boundary – a regime structure common to the mean and the variance – the estimated boundary is determined by whichever of the two changes is more pronounced in the data, and a comparison of transmission across states becomes, in part, a comparison across shock environments.

Two classes of models allow the volatility of the shocks to change over time. On the one hand, time-varying-parameter models with stochastic volatility and Markov-switching

[1] Variance switches carry identifying information of a second kind, about the structural shocks themselves, when the impact matrix is held fixed across regimes (Rigobon, 2003; Lewis, 2021). Our impact vectors are regime-specific, and the variance regimes locate the regime structure and separate the shock environment from the state of the economy.

models let the second moments change, but treat the regime change as latent and, in their classic formulations, as exogenous in its timing: the model infers when the regime changed, and the economist interprets the result afterwards. In contrast, threshold and smooth-transition models tie regime changes to an observable state variable, so the regimes have an economic interpretation by construction. Threshold models make the change a step at an estimated cutoff, so that the model is linear within each regime and the cutoff is the only nonstandard object to estimate; smooth-transition models (Teräsvirta, 1994; Auerbach and Gorodnichenko, 2012) replace the step with a calibrated transition function, so that the coefficients vary continuously with the state. TVARs with regime-specific covariance matrices are standard practice in state-dependent macroeconomics (Tsay, 1998; Auerbach and Gorodnichenko, 2012; Caggiano et al., 2015; Canova and Forero, 2024). In all of them, however, the variance regime coincides with the mean regime – the same state variable, the same cutoff, switching together – and the theory of the threshold estimator locates the cutoff from the conditional mean alone. This paper bridges the two classes: it keeps the observable states and estimated thresholds of the second, and gives the variance regimes of its own, as the first allows. It separates the two regime structures and extends the estimation theory to the resulting model.

The fiscal multiplier literature is the natural place to illustrate why this matters. Whether government spending is more effective in bad times is a question addressed with regime-dependent models, and with few exceptions the regimes are imposed: the researcher chooses the state variable and the cutoff that separates expansions from recessions, as in Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018). The identifying variation in this literature, moreover, often comes with wars and military build-ups, which are both episodes of unusual shock volatility and of unusual macroeconomic states – near-full employment in the mobilizations, deep slack in the Depression years that precede WWII. A model with a single notion of regime cannot tell the two apart, and in this literature what it picks up is the war, not the recession. Fiscal policy is therefore the scenario in which volatility regimes are both particularly relevant, and consequential to ignore.

The paper makes two contributions: methodological and empirical. The methodological contribution is the *decoupled* threshold VAR, in which mean and variance regimes are driven by different observable state variables with separate thresholds, together with its estimation theory. Mean regimes govern how shocks propagate – the *propagation regimes*; variance regimes govern the environment in which the identifying shocks arrive – the *shock regimes*; the model dates the two separately. We propose a likelihood-based threshold estimator that converges at the rate of the sample size – faster than the usual square-root rate – with nonstandard limiting distributions of compound-Poisson-argmax form, and that leaves standard inference on the regime parameters – and, under any identification scheme, on

the structural objects derived from them – unaffected. The model nests the standard *joint* model, in which the same state variable and the same cutoff govern both moments, as well as the linear VAR and the specifications with regimes in the mean only or in the variance only. The proposed unrestricted estimator recovers the joint model when it holds, and a likelihood rule selects it consistently, so the researcher need not decide ex ante whether the mean and the variance switch together. The empirical contribution reconciles the disagreement on state-dependent fiscal multipliers: in the data and framework of Ramey and Zubairy (2018), henceforth RZ2018, wars and major fiscal events form a separate high-variance regime, and once it is isolated, state dependence emerges even at the thresholds imposed in earlier work, with post-WWII multipliers larger in good times than in bad.

The methodological contribution addresses a limitation in existing threshold estimation: the threshold is located using the conditional mean alone, even when the variance also changes. The threshold is selected first as the value of the state that optimizes the criterion function over a grid of candidates; then, conditional on it, the regime parameters are obtained from one linear model per regime. Under regularity conditions, the threshold estimator is super-consistent at the fast rate T, so its estimation error does not enter the asymptotic distribution of the regime parameters, which is what justifies the two steps. The asymptotic theory behind this – as opposed to the models it is applied to – is univariate, and it identifies the threshold through the conditional mean alone (Tong and Lim, 1980; Chan, 1993; Hansen, 2000; Samia and Chan, 2011): the sum of squared residuals of Chan (1993) and Hansen (2000) is a pure measure of mean fit, on which a regime change in the variance leaves no systematic imprint, and the likelihood criterion of Samia and Chan (2011) could in principle use second moments but its theory does not incorporate regime-dependent variances in the threshold step. In a threshold model estimated this way, the location of the regimes is chosen as if the variance never changed, even when the covariance is allowed to differ in the second estimation step. The threshold step therefore ignores the variance, at a cost that depends on where the regimes are: inconsistency when the change is in the variance alone, and a loss of precision when both moments change.

We close the gap for the decoupled and the joint model with a likelihood-based threshold estimator in the spirit of Samia and Chan (2011). Given the thresholds and the covariances, the maximum-likelihood estimator of the regime coefficients is the GLS estimator, and a single covariance update is asymptotically equivalent to the exact profile likelihood, which makes the decoupled model no harder to estimate than a linear VAR on each combination of mean and variance regime. The two thresholds are asymptotically independent, the variance threshold admits a near-pivotal plug-in interval, and the results provide the frequentist foundations for the Bayesian threshold estimation of Gargiulo et al. (2026). A Monte Carlo study delineates when second moments matter by comparing the likelihood

estimator with the least-squares criterion in three cases: (i) when only the mean switches, the two are equivalent; (ii) when both moments switch, the likelihood halves the threshold error at identical regime-parameter precision, as super-consistency implies; (iii) when only the variance switches, least squares is inconsistent while the likelihood estimator retains the T-rate. A single threshold imposed on a decoupled process leaves the unmodeled dimension unestimated at any sample size. Using the second moments never reduces precision, and when only the variance switches it is the only way to locate the regime boundary.

The applied contribution is motivated by a disagreement. Auerbach and Gorodnichenko (2012) find government spending multipliers larger in recessions than in expansions; RZ2018 find no evidence of larger multipliers in times of slack and, under their baseline identification, no significant differences across regimes; Alloza (2022) finds multipliers larger in expansions, and attributes the disagreement to the definition of the recession regime. In all three the regimes are imposed, which raises the question of whether the disagreement reflects regime misspecification. We address it by estimating the regime thresholds. We do so in the data and framework of RZ2018, for their *slack exercise*, in which the state variable is lagged unemployment, and the imposed threshold is 6.5%, on their three samples: the full historical sample and the sample excluding WWII – the two historical samples – and the post-WWII sample. RZ2018 estimate regime-dependent local projections instrumented with the military news shock of Ramey (2011). Our estimator is likelihood-based and needs a model of the conditional covariance, which a local projection does not supply, so we estimate the same system as a TVAR with the news series ordered first, which, by the equivalence result of Plagborg-Møller and Wolf (2021), identifies the same impulse responses as their LP-IV.[2]

The empirical analysis delivers three results. First, the imposed threshold is not the problem: the least-squares search confirms RZ2018's cutoff (6.95% against 6.5%). The likelihood estimator departs from it only because it incorporates the second moments, and the identified regime is not a state of the economy but a shock environment: a small set of wartime quarters at near-full employment, in which the volatility of fiscal innovations is two to three times higher, and whose indicator is nearly collinear with the largest realizations of the instrument. What looked like a state is a shock. Second, the decoupled estimator separates the two. It assigns a separate threshold to each dimension, so that the impact of the shock is a property of the variance regime and its propagation a property of the mean regime, and state dependence is read across the two mean regimes within the same variance regime.[3] With the absolute value of the lagged instrument as the variance state it isolates a *fiscal-event regime* of 33 quarters – the World Wars, Korea, Vietnam, the Reagan build-up,

[2]RZ2018's second exercise, at the zero lower bound (ZLB), is reported in Appendix C.

[3]Isolating the fiscal-event regime also strengthens RZ2018's weak first stage in the low-unemployment state, from an F-statistic of 2.2 to 11.6, because the wartime quarters carry a contemporaneous news–spending relation of the opposite sign (Appendix E).

the Gulf War, 2002, 2008–2013 – at a threshold that is remarkably stable across samples and for both mean states. Outside these quarters, state dependence is in the data, and its direction depends on the sample: on the historical samples, multipliers are larger in the high-unemployment regime – at the 6.5% threshold imposed by RZ2018, once the fiscal-event regime is isolated, and significantly so when WWII is excluded – and after WWII they are larger in the low-unemployment regime, a ranking of point estimates that is not significant in this exercise and becomes so in the robustness exercise below. Third, the shock regime is not an artifact of the instrument. Because the variance state is built from the news series, one may object that the fiscal-event regime merely tracks the heteroskedasticity of the proxy equation; a robustness exercise in the framework of Auerbach and Gorodnichenko (2012) removes the instrument from the system, identifies recursively, and lets the second moments of the macroeconomic aggregates alone select the variance threshold. It recovers the same 15 fiscal-event quarters on the common 1949–2008 sample, and on post-WWII data the larger multipliers in good times – there, the high-growth regime – become a statistically significant difference.

The disagreement thus decomposes into three ingredients – the shock environment, the sample period, and the definition of the regimes – of which only the last is addressed by the existing debate. The relevant notion of a regime is two-dimensional: no choice of recession definition can absorb the shock-environment dimension, which, held fixed at RZ2018's own regime definition, is by itself sufficient to restore state dependence on the historical samples. The researcher chooses the state variables, and the framework provides criteria for that choice: the location of the cutoff, whether the regimes are in the dynamics or in the volatility, and the stability of the partition across samples. Government spending itself, the natural alternative, meets none of them, because it classifies quarters by their response to shocks rather than by their arrival (Appendix I).

Related literature. The paper contributes first to the econometrics of threshold models (Tong and Lim, 1980; Chan, 1993; Hansen, 2000; Tsay, 1998; Hansen, 2011). The closest paper is Samia and Chan (2011), who establish consistency and the compound-Poisson-argmax limit of the maximum-likelihood threshold estimator in a univariate generalized threshold regression; their argument is the template for ours. Their theory does not cover regime-dependent variances in the threshold step, so that the threshold is still identified from the conditional mean. We extend their approach to multivariate systems in which the covariance switches with the mean, so that the second moment contributes to the location of the threshold, and to the decoupled model, in which mean and variance regimes have separate states and thresholds, each estimated at rate T. The double-threshold ARCH model of Li and Li (1996) is a univariate precursor.

Relative to the latent-regime approaches – Markov switching (Hamilton, 1989; Sims and

Zha, 2006), drifting parameters with stochastic volatility (Cogley and Sargent, 2005; Primiceri, 2005; Chan and Eisenstat, 2018; Justiniano and Primiceri, 2008), and the endogenous Markov chains of Kim et al. (2008) and Chang et al. (2017) – threshold models tie the regimes to lagged observable states, and we extend this principle to the variance dimension. The time-varying-parameter panel local projections of Gargiulo et al. (2025) let the coefficients vary freely over time, with no law of motion imposed, but the variation is indexed by calendar time rather than by an observable state, so it dates when a response changed without linking the change to an observable state. Relative to the state-dependent local-projection literature (Jordà, 2005; Ramey and Zubairy, 2018; Gonçalves et al., 2024), we provide the estimation theory for the regime structure on which those projections condition.

On the empirical side, the paper contributes to the literature on the state dependence of fiscal multipliers (Blanchard and Perotti, 2002; Ramey, 2011): Auerbach and Gorodnichenko (2012, 2013) find larger multipliers in downturns; Owyang et al. (2013) and RZ2018 find little or no slack dependence on long historical samples with imposed thresholds; Fazzari et al. (2015) estimate the threshold, on capacity utilization; Bernardini and Peersman (2018) condition on private debt; Ghassibe and Zanetti (2022) show that the source of the fluctuations underlying the state matters; and Alloza (2022) attributes the disagreement to the definition of the recession regime. Rather than proposing another state variable or cutoff, we estimate the regime structure. Closest to us, Barnichon et al. (2022) locate the nonlinearity in the *sign* of the shock; our shock-environment dimension is its second-moment counterpart, and the two mechanisms can coexist.

The rest of the paper is organized as follows. Section 2 presents the models, the estimators and the asymptotic theory; Section 3 the Monte Carlo evidence; Section 4 the application to RZ2018, with the robustness exercise in the framework of Auerbach and Gorodnichenko (2012) in Section 4.7; Section 5 concludes. Proofs and additional results are in the Appendix.

2 Threshold VARs with Decoupled Mean and Variance Regimes

Threshold autoregressions, introduced by Tong (1977) and developed by Tong and Lim (1980) and Chan (1993), model nonlinearity through a small number of regimes selected by an observable state variable that is measurable with respect to $\mathcal{F}_{t-1}$, the information set generated by the data up to $t-1$. The resulting framework is parsimonious and interpretable; it was extended to vector systems by Tsay (1998) without, so far, an estimation theory for the threshold itself. Estimation proceeds in two steps: first, the threshold is estimated by optimizing a criterion over a grid of possible values, then the regime parameters are obtained conditionally on it. The choice of criterion in the first step matters: the sum of squared residuals (SSR) of Chan (1993) and Hansen (2000) identifies the threshold from

breaks in the conditional mean alone; Samia and Chan (2011) replace the SSR criterion with the likelihood, which can in principle carry information from the entire conditional distribution. Regime-dependent variances – which carry threshold information of their own and which neither framework accommodates – are the starting point of this paper.

The economic motivation is the one of the introduction: the state of the economy and the shock environment are distinct sources of regime dependence, and there is no reason for them to be governed by the same observable, the same cutoff, or the same number of regimes. A threshold VAR in which mean regimes and variance regimes are *decoupled* – driven by their own state variables and thresholds, with regime-specific coefficient and covariance matrices – formalizes the two mechanisms separately and lets the data locate both.

This section develops the likelihood-based estimation theory for that model. The model in which both switch together at a common threshold is the standard threshold VAR with regime-specific covariance matrices; its likelihood estimator is the one of Gargiulo et al. (2026). It is the special case of Section 2.5, where we show that the restriction can be selected rather than imposed. Further, the proposed model also nests the mean-only, variance-only and linear specifications. The theory is organized in three layers, each with its own conditions: (i) geometric ergodicity and moments of the data-generating process; (ii) identification and consistency of the estimator; (iii) regularity local to the thresholds, delivering the convergence rates and limiting distributions. Sections 2.1–2.2 define the model and the estimator; Section 2.3 states the conditions, grouped by layer, with remarks collected after each group; Section 2.4 states the results.

2.1 The Model

Let $y_t \in \mathbb{R}^n$ follow a decoupled threshold VAR (D-TVAR) of order p with k_μ mean regimes and k_σ variance regimes,

$$y_t = \sum_{i=1}^{k_\mu} \left(\Phi_{i,0} + \sum_{j=1}^{p} \Phi_{i,j}\, y_{t-j} \right) \mathbb{I}\{q^\mu_{t-d_\mu} \in \mathcal{R}^\mu_i\} \;+\; \sum_{\ell=1}^{k_\sigma} \Sigma_\ell^{1/2}\, \mathbb{I}\{q^\sigma_{t-d_\sigma} \in \mathcal{R}^\sigma_\ell\}\, \varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, I_n), \tag{1}$$

where q^μ_t and q^σ_t are scalar, $\mathcal{F}_{t-1}$-measurable state variables observed at delays $d_\mu, d_\sigma \in \{1, \dots, p\}$, $\mathbb{I}\{\cdot\}$ denotes the indicator function, and the regimes are the intervals $\mathcal{R}^\mu_i = [\gamma^\mu_{i-1}, \gamma^\mu_i)$ and $\mathcal{R}^\sigma_\ell = [\gamma^\sigma_{\ell-1}, \gamma^\sigma_\ell)$ generated by the ordered threshold vectors $\gamma^\mu \in \mathbb{R}^{k_\mu - 1}$ and $\gamma^\sigma \in \mathbb{R}^{k_\sigma - 1}$, with the conventions $\gamma^\mu_0 = -\infty$ and $\gamma^\mu_{k} = +\infty$. At each t the observation belongs to the joint cell (i, ℓ). Let $x_t := (1, y'_{t-1}, \dots, y'_{t-p})' \in \mathbb{R}^K$ with $K := np + 1$, and let $B_i := (\Phi_{i,0}, \Phi_{i,1}, \dots, \Phi_{i,p}) \in \mathbb{R}^{n \times K}$, so that the conditional mean in regime i is $B_i x_t$. The parameter vector is

$$\theta := \left(\gamma^{\mu\prime}, \gamma^{\sigma\prime}, \beta'_1, \dots, \beta'_{k_\mu}, \mathrm{vech}(\Sigma_1)', \dots, \mathrm{vech}(\Sigma_{k_\sigma})' \right)' \in \Theta \subset \mathbb{R}^{d_\theta},$$

where $\beta_i := \mathrm{vec}(B_i) \in \mathbb{R}^{nK}$ and $d_\theta = (k_\mu - 1) + (k_\sigma - 1) + k_\mu nK + k_\sigma n(n+1)/2$. When the state variables are elements of y_t, $q_t^\mu = e'_{s_\mu} y_t$ and $q_t^\sigma = e'_{s_\sigma} y_t$, where e_s denotes the s-th standard basis vector of $\mathbb{R}^n$, with a one in position s and zeros elsewhere, (1) is a self-exciting threshold VAR; the regime assignment is governed by the scalar states, not by the full lagged vector. Let θ_0 denote the true parameter value. Gaussianity in (1) is the assumption under which the estimator is derived, and it enters the theory in two ways. First, it defines the estimation criterion: the estimator of Section 2.2 maximizes the Gaussian likelihood. Second, the limit laws of Section 2.4 are derived under Gaussian innovations, although they generalize: as in Chan (1993) and Samia and Chan (2011), the compound-Poisson-argmax form of the threshold limits does not depend on the innovation distribution, which enters only the distribution of the jumps; the framework can be implemented with other innovation distributions, and we leave the formal extension for future work. The stability result on which the theory rests (Theorem A1), by contrast, requires only that ε_t be i.i.d. with mean zero, identity covariance, a positive continuous density and finite fourth moments (**A1**). Let $\pi_{i\ell} := P_\pi\big(q^\mu_{t-d_\mu} \in \mathcal{R}^\mu_i,\, q^\sigma_{t-d_\sigma} \in \mathcal{R}^\sigma_\ell\big)$ be the stationary probability of joint cell (i,ℓ) under the invariant law π of Theorem A1, and $t_{i\ell} := \big|\{t : q^\mu_{t-d_\mu} \in \mathcal{R}^\mu_i,\, q^\sigma_{t-d_\sigma} \in \mathcal{R}^\sigma_\ell\}\big|$ the corresponding sample count.

Both the variables and the delays may differ across the two regime structures. The leading cases are two different elements of y_t, the same element at two different lags ($q_t^\mu = q_t^\sigma = e'_s y_t$ with $d_\mu \neq d_\sigma$), and an economic state for the mean regimes paired with a measure of shock size for the variance regimes, as in the applications of Section 4 and Appendix J. A shared state variable at a common delay is also admissible: with distinct threshold values the two regime structures form a staircase of feasible cells, and the theory below applies (Remark 9); with a common threshold value the two structures coincide and (1) reduces to the joint mean–variance model of Section 2.5, which we show is nested in estimation.

2.2 The Estimator

Under the Gaussian criterion in (1), an observation in joint cell (i,ℓ) has conditional mean $B_i x_t = (x_t' \otimes I_n)\,\mathrm{vec}(B_i)$ – by the identity $\mathrm{vec}(B_i x_t) = (x_t' \otimes I_n)\,\mathrm{vec}(B_i)$, which is what makes the likelihood linear in β_i – and conditional covariance Σ_ℓ. We estimate θ by maximizing the resulting likelihood, concentrating the regime parameters out in closed form for each candidate pair of thresholds. The average log-likelihood is

$$L_T(\theta) = \frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{k_\mu}\sum_{\ell=1}^{k_\sigma}\left(-\frac{1}{2}\ln\det\Sigma_\ell - \frac{1}{2}\varepsilon_{it}'\Sigma_\ell^{-1}\varepsilon_{it}\right)\mathbb{I}(q^\mu_{t-d_\mu} \in \mathcal{R}^\mu_i)\mathbb{I}(q^\sigma_{t-d_\sigma} \in \mathcal{R}^\sigma_\ell), \qquad (2)$$

where $\varepsilon_{it} = y_t - (x_t' \otimes I_n)\,\mathrm{vec}(B_i)$ and $x_t = [1, y_{t-1}', \ldots, y_{t-p}']'$. Concentrating out $\{B_i\}$ and $\{\Sigma_\ell\}$ yields the profile likelihood. Conditional on $(\gamma^\mu, \gamma^\sigma)$, the coefficients B_i are estimated at the mean-regime level using variance-regime-specific precision weights, while the covariance

matrices Σ_ℓ are estimated at the variance-regime level by pooling residuals across all mean regimes within regime ℓ. The first-order condition for $\beta_i := \mathrm{vec}(B_i)$ is

$$\sum_{\ell=1}^{k_\sigma} \sum_{t \in S_{i\ell}} (x_t \otimes \Sigma_\ell^{-1}) \left[y_t - (x_t' \otimes I_n)\beta_i \right] = 0, \tag{3}$$

where the Kronecker form follows from $\partial \varepsilon_{it} / \partial \beta_i' = -(x_t' \otimes I_n)$ and $(x_t \otimes I_n)\Sigma_\ell^{-1} = x_t \otimes \Sigma_\ell^{-1}$, and

$$S_{i\ell} := \left\{ t : \; q^\mu_{t-d_\mu} \in \mathcal{R}_i^\mu, \; q^\sigma_{t-d_\sigma} \in \mathcal{R}_\ell^\sigma \right\}.$$

Equation (3) is both the first-order condition of the Gaussian likelihood in β_i and the GLS normal equation: conditional on the covariances, the maximum-likelihood estimator of the regime coefficients is the GLS estimator, with the precision weights Σ_ℓ^{-1} of each observation's variance regime. Solving (3) gives the feasible GLS estimator

$$\widehat{\beta}_i^{GLS}(\gamma^\mu, \gamma^\sigma) = \left[\sum_{\ell=1}^{k_\sigma} \sum_{t \in S_{i\ell}} (x_t x_t' \otimes \widehat{\Sigma}_\ell^{-1}) \right]^{-1} \left[\sum_{\ell=1}^{k_\sigma} \sum_{t \in S_{i\ell}} (x_t \otimes \widehat{\Sigma}_\ell^{-1}) y_t \right], \tag{4}$$

where $\widehat{\Sigma}_\ell$ is a first-step consistent estimate of Σ_ℓ (Remark 1). Define $\widehat{\beta}_i^{GLS} := \mathrm{vec}(\widehat{B}_i^{GLS})$, $\widehat{\beta}_i^{(0)} := \mathrm{vec}(\widehat{B}_i^{(0)})$. We use $\widehat{\beta}_i$ inside the GLS normal equations and $\widehat{B}_i$ for the corresponding matrix-valued coefficient blocks.

Similarly, the first-order condition for Σ_ℓ yields the variance-regime-specific residual covariance estimator, obtained by pooling residuals across all mean regimes within the same variance regime ℓ

$$\widehat{\Sigma}_\ell(\gamma^\mu, \gamma^\sigma) = \frac{1}{t_\ell} \sum_{i=1}^{k_\mu} \sum_{t \in S_{i\ell}} \widehat{\varepsilon}_{it} \widehat{\varepsilon}_{it}', \qquad t_\ell := \sum_{i=1}^{k_\mu} t_{i\ell}, \tag{5}$$

where $\widehat{\varepsilon}_{it} = y_t - (x_t' \otimes I_n)\widehat{\beta}_i^{GLS}$ are the GLS residuals.

Substituting $\{\widehat{B}_i^{GLS}\}_{i=1}^{k_\mu}$ and $\{\widehat{\Sigma}_\ell(\gamma^\mu, \gamma^\sigma)\}_{\ell=1}^{k_\sigma}$ into the concentrated Gaussian log-likelihood[4] yields

$$L_T^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) = -\frac{1}{2} \sum_{\ell=1}^{k_\sigma} \frac{t_\ell}{T} \ln \det \widehat{\Sigma}_\ell(\gamma^\mu, \gamma^\sigma) + \mathrm{const}, \tag{6}$$

and the threshold estimator is

$$(\hat{\gamma}^\mu, \hat{\gamma}^\sigma) = \arg \max_{(\gamma^\mu, \gamma^\sigma) \in \Gamma^\mu \times \Gamma^\sigma} L_T^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma). \tag{7}$$

where $(\hat{\gamma}^\mu, \hat{\gamma}^\sigma)$ denotes the *smallest maximizer*; the convention is discussed in Section 2.4.

Remark 1 (Feasible profile-GLS implementation). The weights in (4) are computed in one step. For each candidate pair $(\gamma^\mu, \gamma^\sigma)$, preliminary coefficients are obtained by OLS within each mean regime; their residuals, pooled within each variance regime, give $\widehat{\Sigma}_\ell^{(0)}$; the GLS

[4] L_T^{p} as defined in (6) uses the feasible one-step GLS plug-ins. Consistency is established directly for this feasible criterion (Lemma T5-F and Step 4 of Appendix A.3); its first-order equivalence with the exact profile $\sup_\vartheta L_T(\gamma^\mu, \gamma^\sigma, \vartheta)$ at the $1/T$ localization scale is Appendix A.5, Steps 3–4.

estimator (4) is computed with the weights $(\widehat{\Sigma}_\ell^{(0)})^{-1}$; and one update of the covariances, (5) at the GLS residuals, enters the profile criterion (6) (the steps are listed in Appendix A.6). The estimator retains the two-step structure of threshold models: the thresholds maximize the feasible profile criterion, and the regime parameters are (4)–(5) evaluated at them. One covariance update suffices: by Theorem 1 and Theorem 2(i) the preliminary covariances are consistent, and the one-step estimator is first-order equivalent to the infeasible GLS estimator (Appendix A.5), the standard justification for one-step FGLS under heteroskedasticity (Hamilton, 1994, Chapter 7).[5] On the diagonal of the threshold space (Section 2.5), where mean and variance regimes coincide, the profile is closed-form and the one-step and exact maximizers coincide.

Throughout, ℓ_t denotes a per-observation contribution to the log-likelihood, $L_T = T^{-1}\sum_t \ell_t$ the sample criterion in all its arguments, and $\ell = \mathbb{E}_\pi[\ell_t]$ its population counterpart, π being the stationary law of Theorem A1; a superscript p marks the profiled versions L_T^{p}, ℓ^{p}, obtained by concentrating out the regime parameters; $\mathcal{L}_T$ denotes a normalized local process and $l := T\,L_T$ the unnormalized likelihood, which appears only in Lemma 7. All $O_p(\cdot)$ statements refer to this normalization.

2.3 Assumptions

We state the conditions in the two groups in which they are used: those guaranteeing geometric ergodicity and moments of the data-generating process, and those delivering identification, the convergence rates and the limiting distributions of the estimator. Remarks collected after each group discuss their role and their relation to the literature.

2.3.1 Geometric Ergodicity

To study the stability of (1), we represent it as a Markov chain on $\mathbb{R}^{np}$. We stack the state

$$X_t := \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}.$$

and assume that the threshold variables $q^{\mu}_{t-d_\mu}$ and $q^{\sigma}_{t-d_\sigma}$ are measurable functions of the stacked state. This covers the empirical specifications of interest, such as the lagged level and the lagged change of a component of y_t, provided the relevant lags are contained in X_t. Hence no enlargement of the state space is needed. For each mean regime $i = 1, \ldots, k_\mu$ let A_i denote the usual $np \times np$ companion matrix built from the regime-i coefficient blocks,

[5] The estimator used throughout is this one-step version: GLS at the pooled residual covariance, then a single update of each Σ_ℓ. Iterating to the Gaussian maximum likelihood conditional on the regimes raises the log-likelihood by 8 to 9 points at the estimated thresholds on the two historical samples of Section 4 and by less than 2 elsewhere, and moves the multipliers reported there by at most 0.1 (0.2 in one cell); the profile criterion for the thresholds is evaluated at the one-step estimates.

and let $b_i \in \mathbb{R}^{np}$ collect the regime intercept placed in the first block. Let H_ℓ denote the $np \times n$ noise-loading matrix in regime ℓ (H_ℓ has only the first $n \times n$ block nonzero and equal to $\Sigma_\ell^{1/2}$).

Write the one-step evolution in terms of the (random) active regime $i^*(X_t)$[6] and $\ell^*(X_t)$:

$$X_{t+1} = A_{i^*(X_t)} X_t + b_{i^*(X_t)} + H_{\ell^*(X_t)} \varepsilon_{t+1} \tag{8}$$

where $i^*(X_t)$ selects the mean-regime via $q^{\mu}_{t-d_\mu}$ and $\ell^*(X_t)$ selects the variance-regime via $q^{\sigma}_{t-d_\sigma}$; when indexing by time we write $i(t) := i^*(X_{t-1})$ and $\ell(t) := \ell^*(X_{t-1})$, so that observation t lies in true cell $(i(t), \ell(t))$ and $y_t = B^0_{i(t)} x_t + \Sigma^{0,1/2}_{\ell(t)} \varepsilon_t$ under the true parameter. This is a time-homogeneous Markov chain with kernel

$$P(x, B) \;=\; \mathbb{P}\big(A_{i^*(x)} x + b_{i^*(x)} + H_{\ell^*(x)} \varepsilon \in B\big), \qquad x \in \mathbb{R}^{np},\; B \in \mathcal{B}_{np}.$$

where $\mathcal{B}_{np}$ is the Borel σ-field on $\mathbb{R}^{np}$ and P is the probability measure on the probability space on which X_0 and $\{\varepsilon_t\}_{t \geq 1}$ are defined. We need to find the conditions under which model (1) is geometrically ergodic.

Geometric ergodicity. The Markov chain $\{X_t\}$ is *geometrically ergodic* if it has a unique invariant probability measure π and there exist constants $M(x) < \infty$ and $\rho \in (0, 1)$ such that for all x and integers $t \geq 0$

$$\|P^t(x, \cdot) - \pi(\cdot)\|_{\mathrm{TV}} \leq M(x)\, \rho^t.$$

Equivalently, there exist a function $g \geq 1$ and constants $R < \infty$, $\rho \in (0, 1)$ with $\|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \leq R g(x) \rho^t$; in particular geometric ergodicity implies geometric β-mixing and exponential forgetting of initial conditions.

To achieve that goal, we make the following assumptions on model (1):

A1 (Innovations.) $\{\varepsilon_t\}$ is i.i.d., independent of X_0, with $E[\varepsilon_t] = 0$, $E[\varepsilon_t \varepsilon_t'] = I_n$, an everywhere positive and continuous Lebesgue density on $\mathbb{R}^n$, and $m_4 := E\|\varepsilon_t\|^4 < \infty$.

A2 (Markov structure.) The coefficient matrices $\Phi_{i,j}$ and intercepts $\Phi_{i,0}$ are finite constants; the regime-selection maps $i^*(\cdot)$ and $\ell^*(\cdot)$ are measurable, so that X_t is a time-homogeneous Markov chain.

A3 (Non-degeneracy.) Each Σ_ℓ is non-singular and $\sup_\ell \|\Sigma_\ell\| \leq C_\Sigma < \infty$.

A4 (Joint stability of the mean regimes.) There exist $P \succ 0$ and $\lambda_i \in (0, 1)$ such that

$$A_i' P A_i \preceq \lambda_i^2 P, \qquad \forall i = 1, \ldots, k_\mu.$$

Under **A1**–**A4** the stacked state $\{X_t\}$ is geometrically ergodic, with a unique invariant law π that has finite fourth moments (Theorem A1 in Appendix A.1, where the proof and the

[6] When indexing by time we write $i(t) := i^*(X_{t-1})$ for the regime active at t; note $i(t)$ is therefore $\mathcal{F}_{t-1}$-measurable.

discussion of the condition are collected). In words, block A asks that the process be stable and forget its past despite the switching, so that averages converge; block B asks that the thresholds be identifiable from the data: adjacent regimes must differ – in the coefficients at the mean cutoff, in the covariance at the variance cutoff – the states must cross their cutoffs with positive probability, and every cell must be visited. Theorem A1 is the premise of every result below: it supplies the stationarity, ergodicity and moment conditions that Chan (1993), Hansen (2000) and Samia and Chan (2011) assume. **A4** is the one condition of the block that is not standard. It is stronger than individual stability $\rho(A_i) < 1$ for all i: it requires all mean regimes to contract the same energy metric $x'Px$, which is what rules out instability generated by the switching itself, and for given parameter values it is verifiable by LMI optimization (Liberzon, 2003, Chapter 2). It is sufficient, not necessary – a threshold process can be globally stationary with a locally explosive regime, as the univariate characterization of Chan et al. (1985) shows – but no multivariate characterization is available, and the drift argument of Theorem A1 needs the common Lyapunov matrix. Section 4.6 reports the check at the estimated parameters of the application, together with the bootstrap distribution of the largest within-regime root: the condition holds on one of the three samples; on the other two it fails in the smallest regime, in one case within sampling uncertainty and in one – the Great Depression regime – not. No Lyapunov condition involves the variance-regime matrices Σ_ℓ, which enter (8) only through the bounded noise loading $H_{\ell^*(X_t)}$ (Remark A.2).

2.3.2 Identification, Rates and Limit Laws

We now state the assumptions that extend Chan (1993) and Samia and Chan (2011) to the multivariate threshold VAR setting with regimes in the variance.

Throughout, $\Lambda_t := 1 + \|x_t\|^2 + \|y_t\|^2$ is the moment envelope; $x_t = (q_{t-d}, x_t^{(-)})$ splits the regressor vector into a threshold variable and the remaining components, where a state is an element of x_t; $\mathbb{I}^{\mu}_{i,t} := \mathbb{I}\{|q^{\mu}_{t-d_\mu} - \gamma_i^{\mu,0}| \le \Delta\}$ and $\mathbb{I}^{\sigma}_{\ell,t} := \mathbb{I}\{|q^{\sigma}_{t-d_\sigma} - \gamma_\ell^{\sigma,0}| \le \Delta\}$ are the band indicators of half-width Δ; f_μ and f_σ are the stationary marginal densities of the two state variables; and $p_\ell(v) := P\big(q^{\sigma}_{t-d_\sigma} \in \mathcal{R}^{\sigma,0}_\ell \mid q^{\mu}_{t-d_\mu} = v\big)$, $\ell = 1, \dots, k_\sigma$, are the conditional variance-regime weights, with $p_\ell := p_\ell(\gamma_i^{\mu,0})$ when the mean threshold is clear from the context.

B1 (Compact parameter space.) $\Theta = \Gamma^\mu \times \Gamma^\sigma \times \mathcal{B} \times \mathcal{S}$ is compact, with $\Gamma^\mu = \{\gamma^\mu : \underline{\gamma} \le \gamma^\mu_1 < \cdots < \gamma^\mu_{k_\mu - 1} \le \overline{\gamma}\}$ and Γ^σ defined analogously, $\mathcal{B} = \prod_i \mathcal{B}_i$ compact and consistent with **A4**, and $\mathcal{S} = \prod_\ell \{\Sigma : \underline{\sigma} I \prec \Sigma \prec \overline{\sigma} I\}$ for $0 < \underline{\sigma} < \overline{\sigma} < \infty$.

B2 (Separation at the thresholds.) For each $i = 1, \dots, k_\mu - 1$: f_μ is continuous and positive at $\gamma_i^{\mu,0}$, and $P\big((B^0_{i+1} - B^0_i)x_t \neq 0 \mid q^{\mu}_{t-d_\mu} = \gamma_i^{\mu,0}\big) > 0$. For each $\ell = 1, \dots, k_\sigma - 1$: f_σ is continuous and positive at $\gamma_\ell^{\sigma,0}$, and $\Sigma^0_\ell \neq \Sigma^0_{\ell+1}$. Every joint cell (i, ℓ) with $\mathcal{R}^\mu_i \cap \mathcal{R}^\sigma_\ell \neq \emptyset$

has $\pi_{i\ell} > 0$.

B3 (Regimes are different.) $B_i^0 \neq B_{i'}^0$ for $i \neq i'$ and $\Sigma_\ell^0 \neq \Sigma_{\ell'}^0$ for $\ell \neq \ell'$.

B4 (Interior parameters.) The true parameter θ_0 is interior to Θ.

B5 (Local ρ-mixing.) There exists $\Delta > 0$ such that, for each i and ℓ, the localized observation vector $V_t^D := \big(\mathbb{I}_{i,t}^\mu,\ x_t'\mathbb{I}_{i,t}^\mu,\ y_t'\mathbb{I}_{i,t}^\mu,\ \mathbb{I}_{\ell,t}^\sigma,\ x_t'\mathbb{I}_{\ell,t}^\sigma,\ y_t'\mathbb{I}_{\ell,t}^\sigma\big)'$ is ρ-mixing with summable mixing coefficients.

B6 (Conditional moments near the thresholds.) There exist $\tau > 0$ and $M < \infty$ such that $E[\Lambda_t^2 \mid q_{t-d_\mu}^\mu = v] \leq M$ for a.e. v within τ of each $\gamma_i^{\mu,0}$, and $E[\Lambda_t^2 \mid q_{t-d_\sigma}^\sigma = v] \leq M$ for a.e. v within τ of each $\gamma_\ell^{\sigma,0}$.

B7 (Weak continuity at the mean thresholds.) For each i, the conditional law of the pair $\big(x_t^{(-)},\ \mathbb{I}\{q_{t-d_\sigma}^\sigma \in \mathcal{R}_\ell^{\sigma,0}\}_{\ell=1,\ldots,k_\sigma}\big)$ given $q_{t-d_\mu}^\mu = v$ is weakly continuous at $v = \gamma_i^{\mu,0}$; in particular the weights $p_\ell(v)$ are continuous there.

B8 (Bounded joint density near the threshold pairs.) For each pair (i,ℓ), $\big(q_{t-d_\mu}^\mu, q_{t-d_\sigma}^\sigma\big)$ admits a joint density that is bounded in a neighborhood of $(\gamma_i^{\mu,0}, \gamma_\ell^{\sigma,0})$.

B9 (Uniform candidate feasibility.) Every candidate regime has positive stationary probability, uniformly over the threshold space: $\inf_{\gamma^\mu \in \Gamma^\mu} \min_i P_\pi(q_{t-d_\mu}^\mu \in \mathcal{R}_i^\mu(\gamma^\mu)) > 0$, and likewise $\inf_{\gamma^\sigma \in \Gamma^\sigma} \min_\ell P_\pi(q_{t-d_\sigma}^\sigma \in \mathcal{R}_\ell^\sigma(\gamma^\sigma)) > 0$.

Remark 2 (Separation). The mean-side clause of **B2** requires that adjacent mean regimes differ with positive probability *at* the threshold, not merely near it: it is the multivariate form of Condition 4 of Chan (1993) and Assumption 1 of Samia and Chan (2011). It is what delivers the strictly negative drift of the local process behind the T-rate (Theorem 2). What matters is that the two regimes be distinguishable at the threshold with positive probability, not that the conditional mean of y_t jumps there.[7] The variance-side clause needs no conditioning, since covariance differences separate the laws everywhere. The cell clause is the population counterpart of every feasible cell being populated, so that the GLS normal equations and the pooled covariances are well defined; it is required only of structurally feasible cells (Remark 9). Taken together, **B2**–**B3** ensure that distinct regimes are not observationally equivalent either at the threshold boundaries or away from them.

Remark 3 (Parameter bounds). The feasible estimator's population plug-in values must lie in $\mathcal{B} \times \mathcal{S}$ for the profile to dominate the plug-in criterion (Lemma T5-F(ii), Appendix A.3);

[7] Band-level separation, which is all consistency needs, is automatic from **B3** and the everywhere-positive stationary density (Lemma 3(iii)): when $B_i^0 \neq B_{i+1}^0$ the set $\{(B_{i+1}^0 - B_i^0)x_t = 0\}$ is Lebesgue-null. Separation that degraded at the threshold would collapse the uniform negative drift behind the T-rate.

since these values form a bounded family, continuous on the compact threshold space, a finite enlargement of the bounds in **B1** always suffices.

Remark 4 (Candidate feasibility). **B9** concerns *marginal* regime probabilities under *candidate* thresholds; empty joint cells at candidate pairs are harmless, since the GLS normal equations and the pooled covariances aggregate over cells within a mean (respectively, a variance) regime. With two regimes per dimension it holds whenever the marginal state densities are positive on $\mathbb{R}$ (Lemma 3(iii) for elements of y_t); with more regimes it requires a minimum regime width in $\Gamma^{\mu}, \Gamma^{\sigma}$, the population counterpart of the minimum-observations guard in the implementation and of the trimming condition of the structural-break literature (Bai and Perron, 1998).

Remark 5 (Local mixing). **B5** is the multivariate analogue of Assumption 7 of Samia and Chan (2011): it controls the dependence of the observations local to the thresholds, where the limiting distributions are decided, and it is stated for the generating localized vector so that every localized summand in the proofs is a fixed measurable function of V_t^D and inherits its mixing coefficients.[8] It is dispensable when both state variables are elements of y_t at fixed lags: Lemma 4 (Appendix A.2) derives the two consequences of **B5** used in the proofs from geometric ergodicity and the bounded conditional density of Lemma 3, at the cost of strengthening **B6** to the conditional-moment condition **B6*** stated there, and Theorems 2 and 3 then hold with **B5** replaced by **B6***. The classes of state variables for which **B5**–**B7** hold are discussed in Appendix A.6.

Remark 6 (Conditional moments).**B6** is the analogue of Assumption 6 of Samia and Chan (2011), imposed at both thresholds. The likelihood jumps incurred by reassigning an observation across regimes (defined before Theorem 2) are dominated by $C\,\Lambda_t$, so **B6** gives them finite conditional second moments near the thresholds, which the tightness and uniform-integrability steps of the proof of Theorem 2 require.[9]

Remark 7 (Weak continuity, mean side only). **B7** is the analogue of Assumption 8 of Samia and Chan (2011). It makes the one-sided limiting jump laws at the mean thresholds well defined: the law of the reassignment loss given $q^{\mu}_{t-d_\mu} = \gamma_i^{\mu,0}$ from the right (respectively, from the left) is the limit of its conditional law given $\gamma_i^{\mu,0} \le q^{\mu}_{t-d_\mu} < \gamma_i^{\mu,0} + \delta$ (respectively, $\gamma_i^{\mu,0} - \delta \le q^{\mu}_{t-d_\mu} < \gamma_i^{\mu,0}$) as $\delta \downarrow 0$, and **B7** guarantees that these limits exist.[10] No such

[8] A stationary process $\{W_t\}$ is ρ-mixing with coefficients $\rho(k) \to 0$ if $|\mathrm{corr}(f, g)| \le \rho(k)$ for all square-integrable f measurable with respect to $\sigma(W_t, t \le j)$ and g measurable with respect to $\sigma(W_t, t \ge j + k)$ (Billingsley, 1999, §19); see Doukhan (1994) and Bradley (2005) for the relations among mixing conditions. Geometric ergodicity implies geometric β-mixing, which does not imply ρ-mixing for non-reversible chains, so **B5** is a genuine strengthening of what Theorem A1 delivers.

[9] It does not follow from the unconditional bound $\int \|x\|^4 \pi(dx) < \infty$ of Theorem A1, since an unconditional bound does not control conditional moments on the null event $\{q_{t-d} = v\}$.

[10] When the state is an element of y_t, $q^{\mu}_{t-d_\mu}$ is itself an element of x_t, so the condition is phrased on the

condition is needed at the variance thresholds: the variance-side jumps (V2) are functions of ε_t alone, which is independent of $\mathcal{F}_{t-1}$, so their law is exact whatever the conditional distribution of the state at the threshold.

Remark 8 (The role of **B8**). **B8** is the local-boundedness counterpart of the cell clause of **B2**, and its role is a counting bound: with a bounded joint density, the probability that an observation lies within ε of both a mean and a variance threshold is $O(\varepsilon^2)$, so at the $1/T$ scale of the localization, the expected number of observations reassigned in both dimensions at once is $O(1/T)$. The two local criterion processes are then driven, with probability tending to one, by disjoint sets of boundary observations, which is what makes the limiting Poisson random measure factorize and the threshold estimators asymptotically independent in Theorem 2(ii) (Remark 12). **B8** holds automatically when the two state variables are distinct elements of y_t (Appendix A.6).

Remark 9 (Shared state variables). **B8** is sufficient but not necessary: the proofs use only the coincidence bound $P(|q^{\mu}_{t-d_\mu} - \gamma^{\mu,0}_i| \le \varepsilon,\ |q^{\sigma}_{t-d_\sigma} - \gamma^{\sigma,0}_\ell| \le \varepsilon) \le C\varepsilon^2$. As stated, **B8** excludes a shared state variable, whose pair concentrates on a line; but the bound holds trivially when the state is shared and the threshold values differ, the two bands being disjoint for $\varepsilon < |\gamma^{\mu,0}_i - \gamma^{\sigma,0}_\ell|/2$. All conclusions of Theorems 1 and 2 then apply, with the cell clause of **B2** required only of the structurally feasible cells and the mean-side jump mixture of Remark 11 degenerating to the single variance regime containing the mean threshold. Only a shared state with equal threshold values falls outside Theorem 2(ii): that configuration is the joint model, whose threshold remains consistently estimable at rate T by the unrestricted estimator (Section 2.5, Proposition 1).

Remark 10 (Relation to the literature). The conditions extend Chan (1993) and Samia and Chan (2011) to a multivariate threshold VAR with regime-dependent covariances and two threshold mechanisms. Chan (1993) assumes a compact parameter space, stationarity, threshold identifiability, regime distinctness and an interior true parameter; Hansen (2000) adds the local-to-threshold conditional-moment conditions that drive the rate; the Assumptions of Samia and Chan (2011) are the likelihood counterparts, with their Assumptions 5–8 corresponding to our **B5**–**B7** at a single threshold. Stationarity, ergodicity, and the fourth moments needed for the uniform law of large numbers are not assumed but delivered by Theorem A1 under **A1**–**A4**. What has no counterpart in these papers is the second threshold mechanism: the cell clause of **B2**, **B8**, and **B9** exist because two partitions of the sample must be identified and localized at once.

conditional law of the remaining elements $x_t^{(-)}$; see Feller (1971) for weak continuity.

2.4 Results

We state the two main results in the following order: consistency first, then the rates and limiting distributions. Both rest on the geometric ergodicity and fourth moments delivered by Theorem A1 in Appendix A.1. All proofs are in the Appendix.

Theorem 1. *Under assumptions **A1**–**A4**, **B1**–**B4** and **B9**, if the data $\{y_t\}_{t=1}^T$ are generated by (1) at the true parameter θ_0, the feasible estimator of Section 2.2 satisfies*

$$\hat{\theta}_T \to_{\text{a.s.}} \theta_0 \qquad \text{as} \quad T \to \infty; \tag{9}$$

in particular $(\hat{\gamma}^\mu, \hat{\gamma}^\sigma) \to_{a.s.} (\gamma^{\mu,0}, \gamma^{\sigma,0})$.[11]

The proof of Theorem 1 can be found in Appendix A.3.

Limit objects. For the mean threshold $\gamma_i^{\mu,0}$ write $\delta_{i,t} := (B_{i+1}^0 - B_i^0)x_t$ and $\kappa_{i,t} := \delta_{i,t}'\Sigma_{\ell(t)}^{0,-1}\delta_{i,t}$, where $\ell(t) = \ell^*(X_{t-1})$ is the variance regime of observation t.[12] The log-likelihood loss from evaluating an observation of mean regime $i+1$ under the law of regime i, holding its variance regime fixed, is

$$\xi_{i,t}^{\mu,+} = -\tfrac{1}{2}\kappa_{i,t} - \delta_{i,t}'\Sigma_{\ell(t)}^{0,-1/2}\varepsilon_t, \qquad \xi_{i,t}^{\mu,+} \,\big|\, (x_t, \ell(t)) \sim \mathcal{N}\big(-\kappa_{i,t}/2,\, \kappa_{i,t}\big), \tag{V1}$$

and symmetrically for $\xi_{i,t}^{\mu,-}$ with the regime labels interchanged. The loss from evaluating an observation of variance regime $\ell+1$ under the law of regime ℓ, holding its mean regime fixed, depends on ε_t alone: with $\lambda_1^{(\ell)}, \ldots, \lambda_n^{(\ell)}$ the eigenvalues of $\Sigma_\ell^{0,-1}\Sigma_{\ell+1}^0$ and $Z_1, \ldots, Z_n$ i.i.d. standard normal,

$$\xi_\ell^{\sigma,+} \overset{d}{=} \tfrac{1}{2}\sum_{j=1}^n \big[\ln \lambda_j^{(\ell)} + (1 - \lambda_j^{(\ell)})Z_j^2\big], \qquad \xi_\ell^{\sigma,-} \overset{d}{=} \tfrac{1}{2}\sum_{j=1}^n \big[-\ln \lambda_j^{(\ell)} + (1 - 1/\lambda_j^{(\ell)})Z_j^2\big], \tag{V2}$$

with laws $\mu_\ell^{\sigma,\pm}$. The one-sided limit laws of the mean jumps are the mixtures over variance regimes

$$\mu_i^{\mu,\pm} := \sum_{\ell=1}^{k_\sigma} p_\ell \int \mathcal{N}\big(-\kappa/2,\, \kappa\big)\big|_{\kappa=\delta'\Sigma_\ell^{0,-1}\delta,\ \delta=(B_{i+1}^0 - B_i^0)x}\, dF_i^\pm(x), \tag{V3}$$

where $F_i^\pm$ is the one-sided conditional law of x_t given $q_{t-d_\mu}^\mu = \gamma_i^{\mu,0}$ and p_ℓ the variance-regime weights, both well defined by **B6**–**B7**. These laws may carry an atom at zero, when $\delta_{i,t} = 0$ with positive conditional probability at the threshold; by **B2** its mass $p_0 < 1$, and the nonzero part is absolutely continuous. Zero jumps do not move the process, which therefore coincides with a compound-Poisson process of thinned rate $f_\mu(\gamma_i^{\mu,0})(1 - p_0)$ and continuous jump law; the almost-sure uniqueness of the maximizing interval then follows as in Remark 1 of Samia and Chan (2011). All jump means are strictly negative: each equals minus a Kullback–Leibler divergence between adjacent regime laws, positive by **B2**–**B3**.

[11] In the two-regime baseline $k_\mu = k_\sigma = 2$, condition **B9** and the parameter-bound requirement of Remark 3 hold automatically, the latter after at most a finite enlargement of the bounds in **B1**.

[12] The index ℓ denotes a variance regime throughout; it carries a time argument, $\ell(t)$, when it denotes the variance regime of observation t.

For a rate $f > 0$ and mark laws (μ^+, μ^-), let $J(h)$, $h \in \mathbb{R}$, be the two-sided compound Poisson process with $J(0) = 0$, independent Poisson arrivals of rate f on each half-line, marks drawn from μ^+ for $h > 0$ and from μ^- for $h < 0$. Since both sides drift to $-\infty$, J has an almost surely unique maximizing interval $[M_-, M_+)$; we write $M_- =: \operatorname{argmax}_-(J)$ for its left endpoint, the *smallest maximizer*, which is the convention of the estimator, following Chan (1993) for the least-squares threshold estimator.[13] Define

$$M_i^\mu := \operatorname{argmax}_- \big(J_i^\mu\big), \qquad M_\ell^\sigma := \operatorname{argmax}_- \big(J_\ell^\sigma\big),$$

where J_i^μ has rate $f_\mu(\gamma_i^{\mu,0})$ and marks $\mu_i^{\mu,\pm}$, J_ℓ^σ has rate $f_\sigma(\gamma_\ell^{\sigma,0})$ and marks $\mu_\ell^{\sigma,\pm}$, and all processes $\{J_i^\mu\}_i, \{J_\ell^\sigma\}_\ell$ are mutually independent. For threshold i of the mean structure, define the normalized local criterion process $\mathcal{L}_{T,i}^\mu(h) := T\big[L_T^{\mathrm{p}}(\gamma^{\mu,0} + he_i/T,\, \gamma^{\sigma,0}) - L_T^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0})\big]$, $h \in \mathbb{R}$, with e_i the i-th unit vector, and $\mathcal{L}_{T,\ell}^\sigma$ analogously. Finally let

$$M_i^{GLS} := E_\pi\big[x_t x_t' \otimes \Sigma_{\ell(t)}^{0,-1}\, \mathbb{I}\{q_{t-d_\mu}^\mu \in \mathcal{R}_i^{\mu,0}\}\big], \qquad \pi_\ell^\sigma := P_\pi\big(q_{t-d_\sigma}^\sigma \in \mathcal{R}_\ell^{\sigma,0}\big),$$

and let D_n^+ be the Moore–Penrose inverse of the duplication matrix.

Theorem 2. *Let the estimator be that of Section 2.2 and assume **A1–A4**, **B1–B9**. Then:*

(i) *(Rates.)* $T(\hat\gamma_i^\mu - \gamma_i^{\mu,0}) = O_p(1)$ *and* $T(\hat\gamma_\ell^\sigma - \gamma_\ell^{\sigma,0}) = O_p(1)$ *for every* i, ℓ; $\sqrt{T}(\hat B_i - B_i^0) = O_p(1)$ *and* $\sqrt{T}\,\mathrm{vech}(\hat\Sigma_\ell - \Sigma_\ell^0) = O_p(1)$.

(ii) *(Thresholds.) Jointly, the local processes* $(\mathcal{L}_{T,i}^\mu)_i$ *and* $(\mathcal{L}_{T,\ell}^\sigma)_\ell$ *converge weakly to the independent compound-Poisson processes* $(J_i^\mu)_i$ *and* $(J_\ell^\sigma)_\ell$, *and*

$$\Big(T(\hat\gamma_i^\mu - \gamma_i^{\mu,0})_{i=1}^{k_\mu - 1},\; T(\hat\gamma_\ell^\sigma - \gamma_\ell^{\sigma,0})_{\ell=1}^{k_\sigma - 1}\Big) \;\Rightarrow\; \Big((M_i^\mu)_{i=1}^{k_\mu-1},\; (M_\ell^\sigma)_{\ell=1}^{k_\sigma-1}\Big),$$

with all $k_\mu + k_\sigma - 2$ *limits mutually independent.*

(iii) *(Regime parameters.) Jointly with (ii) and asymptotically independently of the threshold estimators,*

$$\sqrt{T}\,\mathrm{vec}(\hat B_i - B_i^0) \Rightarrow \mathcal{N}\big(0, (M_i^{GLS})^{-1}\big),$$
$$\sqrt{T}\,\mathrm{vech}(\hat\Sigma_\ell - \Sigma_\ell^0) \Rightarrow \mathcal{N}\big(0, \tfrac{2}{\pi_\ell^\sigma} D_n^+(\Sigma_\ell^0 \otimes \Sigma_\ell^0) D_n^{+\prime}\big),$$

with all Gaussian blocks mutually independent.

The proof of Theorem 2 can be found in Appendix A.5.

Remark 11 (The mean–variance asymmetry of the limit). The two threshold limits are not symmetric. The mean-threshold jumps are Gaussian conditionally on $(x_t, \ell(t))$, with precision $\kappa_{i,t} = \delta_{i,t}' \Sigma_{\ell(t)}^{0,-1} \delta_{i,t}$ that depends on the variance regime in force, so the limit law $\mu_i^{\mu,\pm}$ is a mixture over variance regimes with the weights p_ℓ of **B7**: the precision with which a

[13] Yu (2012) shows, in the i.i.d. threshold-regression setting, that the midpoint of the maximizing interval has smaller higher-order risk; nothing in our first-order theory depends on the choice, but the limit random variable is the left endpoint M_- under the smallest-maximizer convention.

mean threshold is located depends on which variance regime prevails near it. The variance-threshold jumps, by contrast, are free of x_t and homogeneous across mean regimes. Hence the limit law of $\hat{\gamma}_i^{\mu}$ depends on the interaction of the two regime structures, while that of $\hat{\gamma}_\ell^{\sigma}$ does not; Table 4 displays the finite-sample counterpart, with the mean threshold estimated more precisely than the variance one in every design. Both thresholds converge at rate T because the likelihood is discontinuous in each of them: moving a threshold by h/T reassigns $O_p(1)$ observations, each paying a strictly positive expected loss. Here the loss comes from a jump in the entire conditional law rather than in the conditional mean alone, so that a shift in the covariance matrices suffices (Remark 19); the rate would fall to $\sqrt{T}$ only if the regimes joined continuously at the threshold (Chan and Tsay, 1998).

Remark 12 (Why the two thresholds are asymptotically independent). Under **B8**, the probability that any observation lies within $O(1/T)$ of both thresholds is $O(1/T)$ (Remark 8), so with probability approaching one the two local processes are driven by *disjoint* sets of boundary observations, and the limiting marked point process on the four half-lines is a Poisson random measure, whose restrictions to disjoint carriers are independent. On the diagonal $q^{\mu} \equiv q^{\sigma}$ with $\gamma_i^{\mu,0} = \gamma_\ell^{\sigma,0}$ the two bands coincide, the double-crossing count is $O_p(1)$ rather than $o_p(1)$, and the factorization fails; that case is the joint model of Section 2.5. With a shared state but distinct thresholds, the bands are disjoint for all large T (Remark 9).

Remark 13 (GLS efficiency). By Theorem 2(iii) the asymptotic covariance of $\sqrt{T}\,\mathrm{vec}(\widehat{B}_i - B_i^0)$ is $(M_i^{GLS})^{-1}$, which is weakly smaller, in the positive-semidefinite order, than the sandwich $(M_i^{OLS})^{-1} V_i^{OLS} (M_i^{OLS})^{-1}$ of the estimator that ignores the variance regimes, where $M_i^{OLS} := E_\pi\big[x_t x_t' \mathbb{I}\{q_{t-d_\mu}^{\mu} \in \mathcal{R}_i^{\mu,0}\}\big] \otimes I_n$ and $V_i^{OLS} := E_\pi\big[x_t x_t' \otimes \Sigma_{\ell(t)}^0 \mathbb{I}\{q_{t-d_\mu}^{\mu} \in \mathcal{R}_i^{\mu,0}\}\big]$. The one-step feasible GLS estimator is first-order equivalent to its infeasible counterpart, so the limit in Theorem 2(iii) is also the limit of the exact profile maximum-likelihood estimator.

Remark 14 (Inference). The limit laws of the mean thresholds, M_i^{μ}, are not asymptotically pivotal: they depend on $f_\mu(\gamma_i^{\mu,0})$ and on the mixture jump laws $\mu_i^{\mu,\pm}$, hence on the conditional law of x_t at the threshold. The nonparametric bootstrap is therefore inconsistent for the thresholds under fixed-threshold-effect asymptotics (Yu, 2014), while bootstrap bands for objects computed conditionally on the threshold estimates – regime-specific coefficients, impulse responses, multipliers – remain justified by the super-consistency of Theorem 2(i). The valid routes for the thresholds themselves are subsampling (Gonzalo and Wolf, 2005), which requires only a nondegenerate limit law at a known rate, both delivered by Theorem 2, and the simulation plug-in of Li and Ling (2012), which for a mean threshold requires estimating the conditional law of x_t at the threshold together with $f_\mu(\gamma_i^{\mu,0})$; for the variance thresholds the plug-in is fully parametric (Remark 15).

Remark 15 (Near-pivotal inference for the variance thresholds). Because the variance-

jump laws $\mu_\ell^{\sigma,\pm}$ depend on $(\Sigma_\ell^0, \Sigma_{\ell+1}^0)$ only through the eigenvalues $\lambda_j^{(\ell)}$, the limit law of $T(\hat{\gamma}_\ell^\sigma - \gamma_\ell^{\sigma,0})$ is completely determined by $\big(f_\sigma(\gamma_\ell^{\sigma,0}), \Sigma_\ell^0, \Sigma_{\ell+1}^0\big)$, with no dependence on the conditional distribution of the state at the threshold. Confidence sets for $\gamma_\ell^{\sigma,0}$ follow from a parametric plug-in: estimate $\lambda_j^{(\ell)}$ from $(\hat{\Sigma}_\ell, \hat{\Sigma}_{\ell+1})$ and $f_\sigma(\gamma_\ell^{\sigma,0})$ by a univariate kernel estimate at $\hat{\gamma}_\ell^\sigma$, simulate the compound-Poisson argmax, and read off quantiles. No analogous shortcut exists for the mean thresholds, or for the joint threshold of Section 2.5, whose jump laws involve the conditional law of x_t at the threshold; there the routes of Remark 14 apply.

Remark 16 (Bayesian estimation). The results provide frequentist foundations for Bayesian estimation of both models. Gargiulo et al. (2026) estimate the joint model of Section 2.5 by combining a flat prior on the threshold over a grid with smooth conjugate priors on the regime parameters, maximizing the profile posterior over the grid and conditioning slope inference on the threshold estimate. Two properties justify the procedure. First, since the profiled parameters move by $O_p(T^{-1})$ as the threshold sweeps an $O(T^{-1})$-neighborhood (Lemma 7, Appendix A.4), a smooth prior contributes local increments of order $O_p(T^{-1})$ against the $O_p(1)$ likelihood jumps, so the profile-posterior mode is asymptotically equivalent to the maximum likelihood estimator and Theorem 3 applies to it. Second, the asymptotic independence and rate separation of Theorem 3(iv) make conditioning the posterior of the regime parameters on the threshold estimate asymptotically equivalent to conditioning on the threshold's posterior. The decoupled model admits the same treatment with two flat priors on $(\gamma^\mu, \gamma^\sigma)$: since the local likelihood factorizes asymptotically across the two threshold blocks (Theorem 2; Remark 12), the joint posterior of the normalized thresholds converges to the product of the two marginal limit posteriors regardless of the prior's dependence structure.[14]

Remark 17 (Relation to the literature). Theorem 2 is the fixed-threshold-effect (compound Poisson) limit theory of Chan (1993) and Samia and Chan (2011), extended to a multivariate Gaussian likelihood with regime-specific covariance matrices; because the likelihood is maximized, the jump laws have negative mean and the walks drift to $-\infty$, the mirror image of the minimized SSR of Chan (1993). Hansen (2000) obtains a different limit, the argmax of a two-sided Brownian motion with drift, under a threshold effect that shrinks with the sample size, which yields pivotal likelihood-ratio inference at the cost of the shrinking-effect device (Remark 14). Chan and Tsay (1998) show that a regression function continuous at the threshold gives the $\sqrt{T}$ rate with a Gaussian limit; the discontinuity guaranteed by **B2** is thus essential to both the rate and the form of Theorem 2.

Remark 18 (Structure of the proofs). The proofs follow the three layers of the condi-

[14] With flat priors, posterior-mode estimation of the thresholds is maximum likelihood in substance; a decision-theoretic treatment would use the posterior mean, which in non-regular models of this type is asymptotically efficient while the maximum likelihood estimator is not (Hirano and Porter, 2003; Chan and Kutoyants, 2012).

tions. Theorem 1 establishes a uniform law of large numbers over the compact parameter space, uniqueness of the population maximizer by the identification argument, and consistency of the feasible maximizer through the uniform validity of the one-step feasible profile (Lemma T5-F), via the argmax theorem of Newey and McFadden (1994). Theorem 2(i) extends the single-threshold localization argument of Chan (1993) and Samia and Chan (2011) to the box of candidate threshold pairs, with the feasible-GLS layer handled by the profile-stability Lemma 7; parts (ii)–(iii) establish the convergence of the local criterion processes to the compound-Poisson limits, their asymptotic independence under **B8**, and the argmax continuous-mapping step, treating one mean and one variance threshold explicitly and extending to general k_μ, k_σ at the end of Appendix A.5. Theorem 3 is the single-threshold specialization of the same arguments, with the quartic drift of **A4** over all k companion matrices and the joint jump law in place of the separate mean and variance jumps.

2.5 Special Cases and Nesting

Model (1) contains, as restrictions on $(k_\mu, k_\sigma, q^\mu, q^\sigma, \gamma^\mu, \gamma^\sigma)$, the threshold VARs used in the literature:

$\mathcal{M}_{\text{joint}}$ $q^\mu \equiv q^\sigma$, $k_\mu = k_\sigma$ and $\gamma^\mu = \gamma^\sigma$: the joint (common-threshold) model, in which one state and one cutoff govern both moments – the threshold VAR with regime-specific covariance matrices of Tsay (1998), as used in the applied literature with the threshold calibrated (Auerbach and Gorodnichenko, 2012; Caggiano et al., 2015), located by least squares (Tsay, 1998) or sampled from its posterior (Canova and Forero, 2024), and with the threshold estimated by maximizing the profile posterior over the grid in Gargiulo et al. (2026), the Bayesian form of the estimator studied here (Remark 16);

$\mathcal{M}_\mu$ $k_\sigma = 1$: regime-specific coefficients with a single innovation covariance, the multivariate counterpart of the threshold autoregression of Chan (1993) and of the threshold regression of Hansen (2000), whose least-squares criteria use the conditional mean only;

$\mathcal{M}_\sigma$ $k_\mu = 1$: one set of dynamics with a switching covariance matrix, the threshold counterpart of the variance-switching specifications of Sims and Zha (2006);

$\mathcal{M}_0$ $k_\mu = k_\sigma = 1$: the linear VAR.

Two properties should be distinguished. As a *model*, (1) nests each member of this family as a restricted case, and the Monte Carlo designs of Section 3.1 are its members (DGP1 linear, DGP2 mean-only, DGP3 variance-only, DGP4 joint, DGP5–DGP6 decoupled). As an *estimation* problem, the members differ in what the unrestricted estimator can establish about them. On $\mathcal{M}_{\text{joint}}$ both thresholds are identified: the unrestricted estimator recovers the common threshold, and a likelihood-gap rule selects the restriction consistently (Proposition 1 below). On $\mathcal{M}_\mu$, $\mathcal{M}_\sigma$ and $\mathcal{M}_0$ the thresholds of the absent regime structure are not

identified – **B3** and the cell clause of **B2** fail for that structure, so Theorems 1–2 do not apply to it – and the restriction is not selected but tested: the sup-likelihood-ratio statistic over the threshold grid, with bootstrap critical values for a parameter that is unidentified under the null (Hansen, 1996), as described in Section 3.2, where DGP1–DGP3 document the behavior of the estimators on these members. The remainder of this section treats $\mathcal{M}_{\text{joint}}$, the one member with a threshold theory of its own.

The joint model. Let $q^\mu \equiv q^\sigma =: q$ with a common delay d, $k_\mu = k_\sigma =: k$, and $\gamma^\mu = \gamma^\sigma =: \gamma$. Every observation's mean regime then coincides with its variance regime, the regimes $\mathcal{R}_i = [\gamma_{i-1}, \gamma_i)$ are shared, and model (1) reduces to the *joint mean–variance* threshold VAR

$$y_t = \sum_{i=1}^{k} \left(\Phi_{i,0} + \sum_{j=1}^{p} \Phi_{i,j} y_{t-j} + \Sigma_i^{1/2} \varepsilon_t \right) \mathbb{I}(q_{t-d} \in \mathcal{R}_i), \tag{10}$$

with parameter $\theta = \{\gamma, \operatorname{vec}(B_1)', \ldots, \operatorname{vec}(B_k)', \operatorname{vech}(\Sigma_1)', \ldots, \operatorname{vech}(\Sigma_k)'\}$ and average log-likelihood

$$\ell_T(\theta) = \frac{1}{T} \sum_{t=1}^{T} \sum_{i=1}^{k} \Big(-\tfrac{1}{2} \ln \det \Sigma_i - \tfrac{1}{2} \varepsilon_{it}' \Sigma_i^{-1} \varepsilon_{it} \Big) \mathbb{I}(q_{t-d} \in \mathcal{R}_i), \qquad \varepsilon_{it} = y_t - (x_t' \otimes I_n) \operatorname{vec}(B_i). \tag{11}$$

The profile is closed-form: with a common threshold, each mean regime contains exactly one variance regime, so the GLS weights in (4) are constant within each mean regime and the feasible GLS estimator reduces to per-regime OLS, while per-regime sample covariances are the exact conditional maximum-likelihood estimators of the Σ_i; the feasible and exact profiles of Section 2.2 therefore coincide, and the decoupled estimator nests the joint one exactly, at the level of both the criterion and the regime-parameter estimators, and the estimator is

$$\hat{\gamma} = \arg\max_{\gamma \in \Gamma} \ell_T\big(\gamma, \hat{B}(\gamma), \hat{\Sigma}(\gamma)\big), \qquad \hat{\theta}_T = \big(\hat{\gamma}, \hat{B}(\hat{\gamma}), \hat{\Sigma}(\hat{\gamma})\big), \tag{12}$$

computed by grid search over Γ and the smallest maximizer in case of ties. We call it the *joint estimator*. The model is standard; the estimator and its theory are not: its threshold is identified from breaks in the conditional means *and* covariances jointly. Gargiulo et al. (2026) use it in Bayesian form, maximizing the profile posterior under a flat prior on the threshold, which Remark 16 shows to be first-order equivalent to the maximum likelihood estimator; the applied literature otherwise calibrates the threshold, locates it by least squares, or samples it from a posterior for which no estimation theory exists.

Recovering and selecting the joint model. The restriction need not be imposed: it can be selected.

Proposition 1 (Nesting). *Suppose the data are generated by (10) with $B_i^0 \neq B_{i+1}^0$ and $\Sigma_i^0 \neq \Sigma_{i+1}^0$ for all i, and let $(\hat{\gamma}^\mu, \hat{\gamma}^\sigma)$ be the unrestricted estimator of Section 2.2 with the shared state q. Under **A1–A4**, **B1–B7** and **B9**: (i) $(\hat{\gamma}^\mu, \hat{\gamma}^\sigma) \to (\gamma^0, \gamma^0)$ a.s., with $T(\hat{\gamma}^\mu - \gamma^0) = O_p(1)$ and*

$T(\hat{\gamma}^{\sigma} - \gamma^0) = O_p(1)$*; (ii) the regime-parameter estimates are* $\sqrt{T}$*-asymptotically equivalent to those of the joint estimator; (iii) with* $\hat{L}_T^{\mathrm{F}}$ *and* $\hat{L}_T^{\mathrm{J}}$ *the maximized unrestricted and restricted criteria,* $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}}) = O_p(1)$*, whereas under distinct true thresholds on the shared state the same statistic diverges at rate* T*; hence the rule "select the restriction if* $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}}) < c_T$*" is consistent for any* $c_T \to \infty$ *with* $c_T = o(T)$*; the simulations of Section 3.3 recommend* $c_T = 2 \ln T$*.*

The proof is in Appendix A.3.3; it reuses the identification argument of Lemma 6 on the staircase of feasible cells, the localization of Theorem 2(i) with the three misclassification costs (mean only, variance only, both), and the $O_p(1)$-versus-order-T gap for the selection step, the threshold analogue of the break-number selection logic of Bai and Perron (1998). Theorem 2(ii) does not apply on the diagonal: the two localizations share their boundary observations and **B8** fails, so the limit law of the unrestricted estimator there is a coupled two-dimensional argmax whose restriction to the diagonal is the law in Theorem 3 below. We do not pursue it, because the selection step directs the practitioner to the restricted estimator whenever the restriction holds. The recommended protocol is therefore: estimate unrestricted; apply the selection rule; if the restriction is selected, impose it and use Theorem 3 for inference. A size-controlled test can be built on the same statistic: $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}})$ has a nondegenerate limit at a known rate under the restriction, which is all that subsampling critical values require (Gonzalo and Wolf, 2005); the limit is not pivotal, so no tabulated critical value exists.

Theory of the joint estimator. On the diagonal, **A4** applies to all k companion matrices; the cell clause of **B2** reduces to the positivity of each regime; **B5**–**B7** localize a single band around each γ_i^0; and **B8** is vacuous. Because the mean and variance jumps of a misclassified observation now occur together, the loss from evaluating an observation of regime $i+1$ under the law of regime i is the joint quantity

$$\xi_{i,t}^{+} := m_i(y_t) - m_{i+1}(y_t) = \tfrac{1}{2} \ln \frac{\det \Sigma_{i+1}^0}{\det \Sigma_i^0} - \tfrac{1}{2} \big(\delta_{i,t} + \Sigma_{i+1}^{0,1/2} \varepsilon_t\big)' \Sigma_i^{0,-1} \big(\delta_{i,t} + \Sigma_{i+1}^{0,1/2} \varepsilon_t\big) + \tfrac{1}{2} \varepsilon_t' \varepsilon_t, \quad \text{(J1)}$$

where m_i is the per-observation regime-i Gaussian log-likelihood at the true parameters and the expression for $\xi_{i,t}^{-}$ interchanges the regime labels; so that, conditionally on x_t, $\xi_{i,t}^{+}$ is an affine-plus-quadratic functional of ε_t with

$$E\big[\xi_{i,t}^{+} \mid x_t\big] = -\,\mathrm{KL}\Big(\mathcal{N}(B_{i+1}^0 x_t, \Sigma_{i+1}^0) \,\Big\|\, \mathcal{N}(B_i^0 x_t, \Sigma_i^0)\Big) \le 0, \quad \text{(J2)}$$

where for multivariate Gaussians

$$\mathrm{KL} = \tfrac{1}{2} \Big[\mathrm{tr}\big(\Sigma_i^{0,-1} \Sigma_{i+1}^0\big) - n - \ln \det\big(\Sigma_i^{0,-1} \Sigma_{i+1}^0\big) + \delta_{i,t}' \Sigma_i^{0,-1} \delta_{i,t}\Big]. \quad \text{(J3)}$$

Under **B2**–**B3** the divergence is strictly positive with positive probability at the threshold, so all jump means are strictly negative. With one-sided limit laws $\mu_i^{\pm}$ (well defined by **B6**–**B7**) and compound-Poisson processes J_i of rate $f(\gamma_i^0)$, f the stationary density of q_{t-d}, write

$M_i := \operatorname{argmax}_-(J_i)$, $S_i := E_\pi[x_t x_t' \mathbb{1}\{q_{t-d} \in \mathcal{R}_i^0\}]$ and $\pi_i := P_\pi(q_{t-d} \in \mathcal{R}_i^0)$. No counterpart of **B9** is required, the profile being exact.

Theorem 3 (The joint model). *Consider model (10) and the joint estimator (12), with all assumptions restricted to the diagonal as described above. (i) Under **A1**–**A4**, $\{X_t\}$ is geometrically ergodic with finite fourth moments. (ii) Under **A1**–**A4** and **B1**–**B4**, $\hat{\theta}_T \to \theta_0$ a.s. (iii) Under **A1**–**A4** and **B1**–**B7**, $T(\hat{\gamma}_i - \gamma_i^0) = O_p(1)$ for each i, $\sqrt{T}(\hat{B}_i - B_i^0) = O_p(1)$ and $\sqrt{T}\,\mathrm{vech}(\hat{\Sigma}_i - \Sigma_i^0) = O_p(1)$. (iv) Under the assumptions of (iii), jointly, the local processes converge weakly to the independent processes $(J_i)_{i=1}^{k-1}$, $\big(T(\hat{\gamma}_i - \gamma_i^0)\big)_{i=1}^{k-1} \Rightarrow (M_i)_{i=1}^{k-1}$, and, asymptotically independently of them, $\sqrt{T}\,\mathrm{vec}(\hat{B}_i - B_i^0) \Rightarrow \mathcal{N}(0, S_i^{-1} \otimes \Sigma_i^0)$ and $\sqrt{T}\,\mathrm{vech}(\hat{\Sigma}_i - \Sigma_i^0) \Rightarrow \mathcal{N}\big(0, \frac{2}{\pi_i} D_n^+ (\Sigma_i^0 \otimes \Sigma_i^0) D_n^{+\prime}\big)$, with all Gaussian blocks mutually independent.*

The proof of Theorem 3 can be found in Appendices A.1, A.3 and A.4.

Remark 19 (Threshold identification through the second moment). The drift of each jump law is minus a conditional Kullback–Leibler divergence (J2). Setting $\delta_{i,t} = 0$ in (J3) – identical conditional means – the divergence does not vanish; it collapses to the pure-covariance form

$$\mathrm{KL} = \tfrac{1}{2} \sum_{j=1}^{n} \big(\lambda_j - \ln \lambda_j - 1\big), \qquad \lambda_j := \mathrm{eig}\big(\Sigma_i^{0,-1} \Sigma_{i+1}^0\big),$$

with each summand nonnegative and zero if and only if $\lambda_j = 1$; the sum is zero if and only if $\Sigma_i^0 = \Sigma_{i+1}^0$. Whenever the regime covariance matrices differ, the local process therefore has strictly negative drift and the threshold localizes at rate T with no contribution from the conditional mean. This is the population counterpart of the Monte Carlo design DGP3 in Section 3.2, where the SSR estimator of Chan (1993), whose local process is driven by $\delta_{i,t}$ alone, is inconsistent while the likelihood estimator retains the T-rate.

Remark 20 (Inference for the joint threshold). The limit law M_i is not pivotal, and no parametric shortcut of the kind available for variance thresholds exists, because the joint jump law depends on the conditional law of x_t at the threshold through $\delta_{i,t}$; the routes of Remark 14 apply.

3 Monte Carlo Evidence

This section provides Monte Carlo evidence on the theoretical results of Section 2. The simulations illustrate the finite-sample implications of the theory: when second-moment information changes a threshold estimate, what is lost by imposing a common regime structure on a decoupled process, and how the decoupled estimator behaves when the data are generated by the joint model. They then verify the asymptotic results directly by comparing finite-sample distributions with the limit laws of Theorems 2 and 3. Section 3.1 describes the designs, all of which are instances of (1); Sections 3.2–3.4 report the comparisons; Section 3.5 verifies the limiting distributions. Appendix B collects the full statistics behind every table.

3.1 Monte Carlo Design

All designs are bivariate ($n = 2$) first-order ($p = 1$) threshold VARs with two regimes per threshold, generated for $t = 1, \ldots, T$ as

$$y_t = \Phi_{1,0}\,(1 - D_t^{\mu}) + \Phi_{2,0}\,D_t^{\mu} + 0.5\,y_{t-1} + \left[\Sigma_1^{1/2}(1 - D_t^{\sigma}) + \Sigma_2^{1/2} D_t^{\sigma}\right]\varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, I_2), \quad (13)$$

where $D_t^{\mu} := \mathbb{I}\{q_{t-1}^{\mu} \geq \gamma^{\mu,0}\}$ and $D_t^{\sigma} := \mathbb{I}\{q_{t-1}^{\sigma} \geq \gamma^{\sigma,0}\}$ are the mean- and variance-regime indicators of (1) with $d_\mu = d_\sigma = 1$: the intercept switches with D_t^{μ}, the innovation covariance with D_t^{σ}, and the autoregressive matrix $\Phi_{1,1} = \Phi_{2,1} = 0.5\,I_2$ is common to both regimes.[15] Here $B_i = (\Phi_{i,0}, \Phi_{i,1})$ is 2×3 with $x_t = (1, y_{t-1}')'$. Each path starts at $y_0 = 0$ and discards 50 burn-in observations. Sample sizes are $T \in \{200, 1000\}$ with $S = 5{,}000$ replications per design unless stated otherwise, each replication with its own random stream.[16] Performance is summarized by the root mean squared error (RMSE) of the thresholds about their true values and by the RMSE of $\hat{B}_i$ and $\hat{\Sigma}_\ell$ in Frobenius norm, averaged over regimes; classification accuracies and rate ratios for every design are in Appendix B.

Table 1: Monte Carlo designs

	$\Phi_{1,0}$	$\Phi_{2,0}$	Σ_1	Σ_2	q_t^{μ}	q_t^{σ}	$\gamma^{\mu,0}$	$\gamma^{\sigma,0}$	Regime structure
DGP1	$(0,0)'$	$(0,0)'$	I_2	I_2	$y_{1,t}$	$y_{1,t}$	0	0	none (linear VAR)
DGP2	$(1,1)'$	$(-1,-1)'$	I_2	I_2	$y_{1,t}$	$y_{1,t}$	0	0	mean only
DGP3	$(0,0)'$	$(0,0)'$	I_2	$4\,I_2$	$y_{1,t}$	$y_{1,t}$	0	0	variance only
DGP4	$(1,1)'$	$(-1,-1)'$	I_2	$4\,I_2$	$y_{1,t}$	$y_{1,t}$	0	0	joint (common threshold)
DGP5	$(1,1)'$	$(-1,-1)'$	I_2	$4\,I_2$	$y_{1,t}$	$y_{2,t}$	0	0	decoupled, system-internal states
DGP6	$(1,1)'$	$(-1,-1)'$	I_2	$4\,I_2$	u_t	v_t	0	0	decoupled, exogenous states
DGP7	$(1,1)'$	$(-1,-1)'$	I_2	Σ_2^*	u_t	u_t	0	0	joint; limit laws of Theorem 3
DGP8	$(1,1)'$	$(-1,-1)'$	I_2	Σ_2^*	u_t	v_t	0	0.3	decoupled; limit laws of Theorem 2

Notes: All designs are (13). $u_t, v_t \overset{iid}{\sim} \mathcal{N}(0,1)$, mutually independent and independent of $\{\varepsilon_t\}$; $\Sigma_2^* = \left(\begin{smallmatrix} 4 & 0.8 \\ 0.8 & 2 \end{smallmatrix}\right)$, non-proportional to Σ_1 so that the variance-side limit theory is exercised in full-matrix form. DGP1–DGP4 are the joint model (10) on a single self-exciting state. DGP5 has system-internal states, weakly correlated through the dynamics; in DGP6 neither state carries information about the other. DGP7–DGP8 use exogenous states so that the Poisson rates and regime weights of Theorems 2 and 3 are known in closed form; there $S = 5{,}000$ at $T = 200$ and $S = 2{,}000$ at $T = 1000$, and the thresholds are computed by exact maximization over observed state values (the smallest maximizer, as in the theorems) rather than on the grid.

Table 1 lists the eight designs. DGP1 has no regime and checks the estimators when the threshold is unidentified; DGP2 is the mean-shift setting for which the SSR estimator

[15] Holding $\Phi_{i,1}$ common across regimes satisfies the stability condition A4 with a trivial common Lyapunov matrix and isolates the threshold-identification problem from cross-regime stability. The implementation assigns the measure-zero event $q = \gamma$ to regime 1; with continuous states, this is immaterial.

[16] Threshold candidates are 100 equally spaced points on the observed range of each state, followed by a local refinement (51 points for one threshold, 21×21 for a pair). The refinement matters at $T = 1000$, where the resolvable spacing of order T^{-1} falls below the coarse-grid resolution and the rate could not otherwise be detected. A candidate is feasible if every mean regime contains at least $nK + 1 = 7$ (with $K := np + 1$ regressors per equation) and every variance regime at least $n(n+1)/2 + 1 = 4$ observations.

is designed; DGP3 isolates a break in the second moments; DGP4 combines both; DGP5–DGP6 are the decoupled model, with the states of DGP5 inside the system and those of DGP6 outside it. Together, DGP1–DGP6 are the members of the family of Section 2.5: linear, mean-only, variance-only, joint and decoupled. DGP7–DGP8 verify the limit laws of Theorems 3 and 2: Σ_2 is not proportional to Σ_1, so that the variance-side theory is exercised in full-matrix form; the states are exogenous, so that the Poisson rates and regime weights of the limit laws are known in closed form; and the thresholds are computed by exact maximization over the observed state values, the smallest maximizer as in the theorems, rather than on the grid.[17]

3.2 Second-Moment Information and the Joint Estimator

Table 2 compares the SSR estimator of Chan (1993), which locates the threshold from the conditional mean only, with the likelihood-based joint estimator of Section 2.5 on DGP1–DGP4. For each design and estimator, it reports three objects: the RMSE of the threshold estimate about γ^0 at $T = 200$ and $T = 1000$, together with their ratio, the *T-rate ratio* RMSE(T=200)/RMSE(T=1000), which equals 5 under convergence at rate T and $\sqrt{5} \approx 2.2$ under rate $\sqrt{T}$; the RMSE of the regime coefficients $\hat{B}_i$, in Frobenius norm and averaged over the two regimes; and the RMSE of the regime covariances $\hat{\Sigma}_i$, likewise. The rows to compare are the two estimators within each design; across designs, DGP2 is the case in which the two should coincide and DGP3 the case in which they should differ.

Table 2: RMSE of the joint estimators on DGP1–DGP4

		Threshold $\hat{\gamma}$			$\hat{B}_i$ (avg. over regimes)		$\hat{\Sigma}_i$ (avg. over regimes)	
DGP	Estimator	$T = 200$	$T = 1000$	T-rate ratio	$T = 200$	$T = 1000$	$T = 200$	$T = 1000$
DGP1	SSR	–	–	–	1.758	2.259	0.360	0.263
	joint	–	–	–	1.908	2.590	0.495	0.426
DGP2	SSR	0.025	0.004	5.50	0.334	0.147	0.246	0.109
	joint	0.025	0.004	5.62	0.334	0.147	0.247	0.109
DGP3	SSR	2.004	3.204	0.63	2.902	4.138	1.756	1.629
	joint	0.080	0.013	6.01	0.454	0.191	0.760	0.337
DGP4	SSR	0.081	0.010	8.42	0.469	0.197	0.721	0.315
	joint	0.039	0.006	6.00	0.451	0.197	0.712	0.315

Notes: S = 5,000 replications, $\gamma^0 = 0$. SSR: least-squares threshold estimator of Chan (1993); joint: likelihood-based estimator of Section 2.5. Threshold RMSE about γ^0; ratio = RMSE(T=200)/RMSE(T=1000), equal to 5 under the rate T. Parameter RMSE: root of the mean over replications of the squared Frobenius error, averaged over regimes; $\hat{B}_i$ and $\hat{\Sigma}_i$ are the per-regime OLS and residual-covariance estimates at the estimated threshold. DGP1 has no true threshold.

The headline finding is the DGP3 row. In a DGP where the threshold is identified

[17] A fixed grid resolves the *rate* but returns an interior point of the maximizing piece of the step-function criterion, displacing $T(\hat{\gamma} - \gamma^0)$ by up to one fine-grid spacing – roughly one unit at $T = 1000$, the order of the object under study. Exact maximization removes the displacement and implements the smallest-maximizer convention of the theorems, whose limit is M_-.

by the second moment alone, the SSR estimator does not converge: its threshold RMSE increases from 2.00 at $T = 200$ to 3.20 at $T = 1000$, and the empirical T-rate ratio of 0.63 is well below unity. The likelihood-based estimator, by contrast, achieves an RMSE of 0.013 at $T = 1000$ with a T-rate ratio of 6.01, consistent with the super-consistency claim of Theorem 3. The corresponding classification accuracy is 99.85% for the likelihood estimator versus 65.83% for SSR (Table A-3). Practical implication: when the candidate threshold variable plausibly drives volatility but not the conditional mean, the SSR-based threshold estimator is systematically misleading at any sample size; the likelihood-based estimator handles either source of threshold information.

The parameter columns add what the threshold columns cannot show. On DGP3 the inconsistency of the SSR threshold propagates to the regime parameters: its RMSEs for B and Σ do not fall with T and behave like those of DGP1, where no regime exists. On DGP2, where identification is carried by the conditional mean, the two estimators are indistinguishable in every column, so incorporating the covariance information entails no loss on the classical mean-shift problem. On DGP4, both estimators converge at the rate T, and the likelihood estimator halves the threshold RMSE (0.039 against 0.081 at $T = 200$); the parameter RMSEs are nevertheless identical across the two estimators. This is the content of Theorem 3(iii)-(iv): because the threshold is super-consistent, its precision has no first-order effect on the $\sqrt{T}$-asymptotics of $\hat{B}_i$ and $\hat{\Sigma}_i$, so a more precise threshold estimator improves the estimated partition, not the regime parameters. The halving itself is the identification gain of Remark 19 in finite samples: the likelihood criterion accumulates the mean break and the variance break at the same location, so its curvature around γ^0 is the sum of two contributions where the SSR criterion has one, and the threshold is located correspondingly more precisely.[18]

DGP1 deserves a comment, because its parameter RMSEs are large and do not fall with T even though no regime exists. With a *fixed* partition this could not happen: the two sub-samples would come from the same process, and per-regime OLS would estimate the common parameters consistently (on the same samples, a split at the sample median of the state gives RMSEs that fall at exactly the $\sqrt{T}$ rate, see Table A-2). The estimated partition is not fixed, however. The threshold estimator selects the split at which the two sub-sample fits differ most, so under no threshold effect the regime estimates are the extreme of many spurious comparisons rather than noisy versions of the truth: the estimated gap between the two regimes' intercepts is 1.7–2.3, against a true value of zero and a median-split gap that shrinks at the $\sqrt{T}$ rate, and it rises with T. Because the spurious gain is largest when one regime is small, the selected split increasingly isolates a tail of the state's

[18]The SSR rate ratio of 8.42 on DGP4 reflects a finite-sample bias, from the misallocation of within-regime second-moment information by the mean-only objective, that vanishes faster than $1/T$; the asymptotic rate is still T (MAD, bias and standard deviations for all designs are in Table A-1).

support – outside the central 80% of its distribution in 26–41% of replications at $T = 200$ and 49–63% at $T = 1000$, with the smaller regime's median share falling to 5–10% and its minimum pinned at the feasibility bound. The equal-weight RMSE of Table 2 gives that small regime half the weight, which is why it rises; weighted by observation share it falls with T, though more slowly than $\sqrt{T}$.[19] The practical lesson is the classical one: a test for the existence of a threshold should precede its estimation. The sup-likelihood-ratio statistic over the threshold grid, with bootstrap p-values for a parameter that is unidentified under the null (Hansen, 1996), applies to both models, the sup running over the two-dimensional grid in the decoupled case. We take the existence of regimes as given in what follows, as the questions of the applications presuppose.

3.3 Nesting the Joint Model

Proposition 1 states that, when applied with a shared state and threshold, the decoupled estimator recovers the joint model: both thresholds converge to γ^0 at rate T, the regime parameters are asymptotically those of the joint estimator, and the statistic $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}})$ is $O_p(1)$ under the restriction while diverging at rate T when the two thresholds differ, so that comparing it with any $c_T \to \infty$, $c_T = o(T)$ selects consistently between the two models. Table 3 tests each clause: the decoupled estimator is run with $q_t^\mu = q_t^\sigma = e_1' y_t$ on DGP1–DGP4 (the same samples as Table 2), and the selection rule is evaluated on DGP1–DGP6 with $c_T \in \{\ln T, 2\ln T\}$.[20]

The rule needs a penalty sequence, and it compares the likelihood gain from freeing the second threshold with a cost c_T: too small a cost and the unrestricted model is chosen under the restriction whenever the free threshold fits noise; too large and the restriction is accepted when the thresholds genuinely differ. Proposition 1 shows that any $c_T \to \infty$ with $c_T = o(T)$ avoids both errors asymptotically, because the gain is $O_p(1)$ under the restriction and of order T otherwise – the logic behind information criteria for the number of structural breaks (Bai and Perron, 1998; Liu et al., 1997). A BIC-type penalty would charge $\frac{1}{2}\ln T$ per additional parameter. We take $\ln T$ as the baseline and $2\ln T$ as the alternative because the gain from a free threshold is a supremum over candidate locations rather than a single χ^2_1 term, so it is systematically larger than a parameter count suggests (Hansen, 1996); the modified criterion of Liu et al. (1997) is heavier than BIC for the same reason.

[19] The likelihood estimator, which also searches over spurious covariance differences, lands in the tails more often than SSR (41% against 26% at $T = 200$) and is affected slightly more; $\hat{\Sigma}_i$, which a mean-targeted split barely moves, is affected less than $\hat{B}_i$. The comparison with a fixed split cannot separate pure selection from the mechanical imprecision of fitting an intercept on a small tail sub-sample; both act in the same direction.

[20] With a shared state the off-diagonal joint cells are empty by construction – the boundary nature of the joint case that Proposition 1 and Theorem 3 address – so the estimator's cell-count feasibility rule is relaxed for this experiment only. The unrestricted criterion is evaluated on the two-dimensional grid augmented with the restricted estimate $(\hat{\gamma}, \hat{\gamma})$, so that $\hat{L}_T^{\mathrm{F}} \geq \hat{L}_T^{\mathrm{J}}$ holds by construction; on DGP5–DGP6 the restricted criterion is the larger of the joint fits on q^μ and on q^σ.

DGP4, where both moments break, is the case covered by the proposition, and every implication holds: $\hat{\gamma}^{\mu}$ and $\hat{\gamma}^{\sigma}$ converge at rate T (ratios 5.2 and 5.7), the parameter RMSEs coincide with the joint estimator's, and the restriction is selected in 99.2% of replications with $c_T = \ln T$ and in all of them with $c_T = 2\ln T$. DGP2 and DGP3 show why the proposition requires both moments to break: the threshold attached to the moment that does not switch is unidentified and does not converge (RMSE 1.7–2.0 for $\hat{\gamma}^{\sigma}$ on DGP2, 2.0–2.9 for $\hat{\gamma}^{\mu}$ on DGP3), while the identified one converges at rate T. The selection rule nevertheless remains consistent there: the gap is $O_p(1)$, with quantiles that do not move between $T = 200$ and $T = 1000$, so the frequency of selecting the restriction rises only as fast as c_T – 0.80–0.95 with $c_T = \ln T$, 0.99–1.00 with $c_T = 2\ln T$. On DGP5–DGP6 the restriction is rejected in every replication at either penalty: the gap exceeds 13.8 in every sample at $T = 200$ and grows linearly in T. We therefore recommend $c_T = 2\ln T$; larger penalties begin to cost power at $T = 200$ ($3\ln T$ selects the restriction in three of 5,000 DGP5 samples; Table A-8 reports the full penalty sensitivity).[21]

Table 3: Decoupled estimator with a shared state on the joint designs: RMSE and selection rule

		Thresholds (decoupled)		$\hat{B}_i$		$\hat{\Sigma}_\ell$		Restriction selected	
DGP	T	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	dec.	joint	dec.	joint	$c_T = \ln T$	$c_T = 2\ln T$
DGP1	200	–	–	3.558	1.908	0.823	0.495	0.610	0.872
	1000	–	–	5.132	2.590	0.777	0.426	0.761	0.948
DGP2	200	0.024	1.712	0.339	0.334	0.795	0.247	0.813	0.990
	1000	0.005	1.959	0.148	0.147	0.637	0.109	0.946	0.998
DGP3	200	1.954	0.092	3.788	0.454	0.786	0.760	0.800	0.997
	1000	2.894	0.013	5.127	0.191	0.340	0.337	0.920	1.000
DGP4	200	0.050	0.078	0.464	0.451	0.720	0.712	0.992	1.000
	1000	0.010	0.014	0.198	0.197	0.315	0.315	0.999	1.000
DGP5	200	*distinct true thresholds (Table 4)*						0.000	0.000
	1000							0.000	0.000
DGP6	200							0.000	0.000
	1000							0.000	0.000

Notes: Decoupled estimator with $q_t^{\mu} = q_t^{\sigma} = e_1' y_t$ on DGP1–DGP4, next to the joint estimator on the same samples; $S = 5{,}000$. Threshold RMSE about $\gamma^0 = 0$ (undefined for DGP1). "Restriction selected" is the share of replications with $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}}) < c_T$, the statistic of Proposition 1(iii), with $\ln T = 5.30$ and 6.91 at $T = 200$ and 1000. Monte Carlo standard errors of the frequencies are below 0.006.

[21] The penalty must exceed the BIC-type $\frac{1}{2}\ln T$ per parameter because the unrestricted fit maximizes over a free, possibly unidentified threshold, so the gap behaves like a sup-type statistic rather than a χ^2_1; this is also why DGP1, where both thresholds are unidentified, has the heaviest tail (0.87 and 0.95 at $2\ln T$). When nothing breaks, either model is correct and the choice is immaterial.

3.4 Validating the Decoupled Estimator

Table 4 evaluates the decoupled estimator on DGP5–DGP6 against three single-threshold alternatives: the joint estimator with q^μ as its state (correctly oriented along the mean dimension, misspecified along the variance dimension), the joint estimator with q^σ (the mirror image), and the SSR estimator on q^μ – the estimator a practitioner would use – with per-regime OLS for B_i and a single pooled $\hat{\Sigma}$. Full statistics are in Appendix B, Tables A-4–A-5.

Three patterns emerge. First, the decoupled estimator delivers the smallest RMSE in every column, including on each threshold dimension against the joint estimator that observes the correct state for that dimension (on DGP5 at $T = 200$, 0.040 against 0.092 for $\hat{\gamma}^\mu$). The mechanism is that the joint estimator's threshold search is conducted against an inflated within-cell residual covariance: pooling observations from both true variance regimes within each q^μ-cell amplifies the noise against which the mean-shift signal is judged. The decoupled estimator obtains cleaner per-regime covariance estimates from the q^σ split, and a correspondingly sharper mean-threshold profile. Its rate ratios lie between 5.3 and 6.0 on all eight (DGP, dimension) pairs (Table A-4), consistent with Theorem 2(i), and all four parameter RMSEs decline at rate $\sqrt{T}$, as Theorem 2(iii) predicts. Second, the misspecified joint estimators and the SSR estimator are subject to misspecification floors on the dimension they do not model: the covariance of the joint estimator on q^μ – and the pooled $\hat{\Sigma}$ of SSR – converges to $(\Sigma_1 + \Sigma_2)/2 = 2.5\, I_2$, at Frobenius distance $1.5\sqrt{2} \approx 2.12$ from either truth, against observed RMSEs of 2.13 at $T = 1000$; symmetrically, $\hat{B}$ of the joint estimator on q^σ settles at distance $\sqrt{2} \approx 1.41$ from truth, against 1.42. Third, on the mean dimension, the SSR estimator is as good as the correctly oriented joint estimator (and slightly better at $T = 200$ on DGP5). However, it estimates no variance threshold at all: when the variance regime is driven by a different state, mean-only estimation is not only inefficient but uninformative about one of the two regime structures. The rates behind these comparisons rest on the limit laws of Theorems 2 and 3; Section 3.5 verifies them directly and reports the empirical coverage of the plug-in confidence intervals for the thresholds (Table A-7 in Appendix B).

3.5 Limiting Distributions

DGP7 and DGP8 check the asymptotic theory directly: the compound-Poisson argmax laws for the thresholds (Theorems 3(iv) and 2(ii)), the Gaussian limits and independence structure of the regime parameters (Theorems 3(iv) and 3(ii)–(iii)), and the plug-in inference for the variance threshold of Remark 15. The limit laws are simulated by drawing $R = 200{,}000$ realizations of the two-sided compound Poisson processes – arrival rates $\phi(\gamma^0)$ on each side, ϕ the standard normal density, jump laws (J1) for DGP7 and (V3), (V2) for the mean and

Table 4: RMSE of decoupled, joint and SSR estimators on DGP5–DGP6

		$T = 200$				$T = 1000$			
DGP	Estimator	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	$\hat{B}_i$	$\hat{\Sigma}_\ell$	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	$\hat{B}_i$	$\hat{\Sigma}_\ell$
DGP5	joint on q^{μ}	0.092	–	0.458	2.205	0.009	–	0.192	2.127
	joint on q^{σ}	–	0.138	1.616	1.167	–	0.022	1.545	0.883
	decoupled	0.040	0.103	0.379	0.828	0.008	0.018	0.162	0.363
	SSR on q^{μ}	0.051	–	0.447	2.165	0.008	–	0.192	2.127
DGP6	joint on q^{μ}	0.033	–	0.282	2.229	0.006	–	0.122	2.143
	joint on q^{σ}	–	0.094	1.459	2.139	–	0.013	1.422	2.029
	decoupled	0.027	0.052	0.228	0.832	0.005	0.009	0.097	0.363
	SSR on q^{μ}	0.032	–	0.282	2.176	0.006	–	0.122	2.132

Notes: $S = 5{,}000$; $\gamma^{\mu,0} = \gamma^{\sigma,0} = 0$. "Joint on q^{μ}" ("on q^{σ}") imposes a single threshold on the mean (variance) state for both moments; "SSR on q^{μ}" estimates a mean threshold by least squares with per-regime OLS for B_i and one pooled $\hat{\Sigma}$. $\hat{B}_i$ RMSE as in Table 2. $\hat{\Sigma}_\ell$ RMSE on a common definition for all rows: each observation is assigned the estimated covariance of the regime its estimator places it in; the squared error is the average over observations of $\|\hat{\Sigma}_{(t)} - \Sigma_{\ell(t)}\|_F^2$ with $\ell(t)$ the true variance regime, and the RMSE is the root of its mean over replications.

variance sides of DGP8 – and recording the smallest maximizer M_-.[22]

Thresholds. For each of the three thresholds – $\hat{\gamma}$ in DGP7, $\hat{\gamma}^{\mu}$ and $\hat{\gamma}^{\sigma}$ in DGP8 – the finite-sample distribution of $T(\hat{\gamma} - \gamma^0)$ is indistinguishable from its limit at Monte Carlo precision. The Kolmogorov–Smirnov distances are 0.011, 0.009 and 0.031 at $T = 200$, against a 5% critical value of 0.019, and 0.024, 0.025 and 0.012 at $T = 1000$ ($S = 2{,}000$, critical value 0.030). Figure 1 overlays the distribution functions; Table A-6 reports quantiles.

Regime parameters and independence. Table A-7 checks the Gaussian limits and the independence structure. The empirical standard deviations of $\sqrt{T}\,\mathrm{vec}(\hat{B}_1 - B_1)$ match the asymptotic values – $S_1^{-1} \otimes \Sigma_1$ for DGP7, the GLS information inverse for DGP8 – with ratios of 1.04 at $T = 200$ and 1.00 at $T = 1000$, uniformly across components. The largest correlation between a normalized threshold and any normalized slope falls to 0.03–0.05 at $T = 1000$; and the correlation between the two normalized thresholds of DGP8 is -0.007 at $T = 200$ and 0.013 at $T = 1000$. This is the distinctive implication of the four-half-line Poisson random measure in Theorem 2(ii).

Plug-in inference. Remark 15 proposes confidence intervals for $\gamma^{\sigma,0}$ by simulating the variance-side limit law. Two versions are evaluated: an oracle version, which simulates the law from the true Σ_1 and Σ_2, and the feasible version a practitioner would use, which re-simulates it in each replication from that replication's $\hat{\Sigma}_1$ and $\hat{\Sigma}_2$. We report empirical coverage rates for both procedures. At nominal 90% they are 0.859 (oracle) and 0.860 (feasible) at $T = 200$, and 0.894 and 0.899 at $T = 1000$; at nominal 95%, 0.916 and 0.915, then

[22] With system-internal threshold variables the same comparison requires estimating $f_\pi(\gamma^0)$ and the conditional regressor law from a long simulated path – the band-resampling construction of Li and Ling (2012); nothing in the theory changes. The exogenous design isolates the question of interest: whether the finite-sample distributions converge to the limits of Theorems 2 and 3.

0.942 and 0.940 (Table A-7). The two procedures are indistinguishable to within Monte Carlo error. Replacing the nuisance objects of the limit law by their estimates therefore entails no loss of coverage in finite samples: the super-consistency of the thresholds and the $\sqrt{T}$-consistency of the covariances make the plug-in procedure of Remark 15 operational as stated.

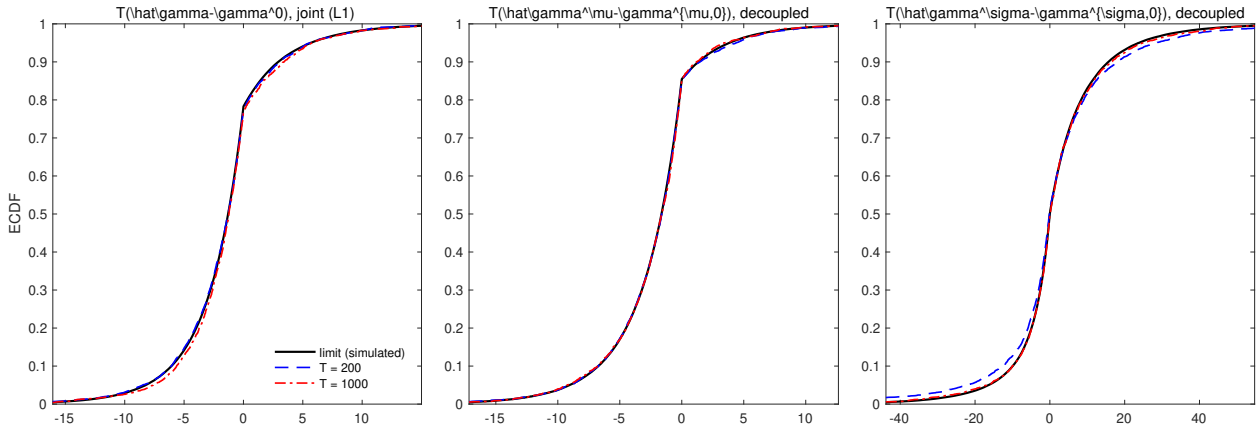


Figure 1: Empirical distribution functions of the normalized threshold estimators at $T = 200$ and $T = 1000$, overlaid on the simulated limiting distributions of Theorems 2 and 3 (solid). Left: joint model. Center and right: decoupled model, mean and variance thresholds. Note the wider scale and pronounced asymmetry of the variance-threshold law.

4 State-Dependent Fiscal Multipliers with Estimated Regimes

The empirical analysis re-examines a central question of the fiscal policy literature – whether government spending multipliers are larger in times of slack or at the ZLB – in the setting of RZ2018, which addresses it with more than a century of U.S. data. They find no state dependence: under their baseline military-news identification and with thresholds imposed at 6.5% unemployment and a 0.25% T-bill rate, multipliers are 0.60 in the high-unemployment regime and 0.59 in the low-unemployment regime, with a p-value of 0.95 for the difference; differences appear only under Blanchard–Perotti identification, and are driven by very low multipliers in the low-unemployment state.[23] The question we ask is whether this conclusion holds when the regimes are estimated rather than imposed, and when the volatility of the identifying shocks is allowed to define a regime of its own. The analysis departs from their setting in one respect at a time. Section 4.2 keeps a single threshold on unemployment, common to the dynamics and the covariance, and estimates it instead of imposing it; Section 4.3 gives the variance its own state and threshold. The multipliers are then reported at the esti-

[23] RZ2018, Table 1. At the zero lower bound they find differences on the sample excluding WWII under military-news identification, which we discuss in Appendix C.

mated regimes (Section 4.4) and, to separate the two margins on which our estimates differ from RZ2018's, at their threshold with only the shock environment modeled (Section 4.5).

4.1 Setup

The key variables are real output y_t and real government spending g_t, each normalized by a measure of potential GDP. RZ2018 employ a proxy identification scheme, where the instrument for government spending is the Ramey (2011) military news shock, normalized following the original specification.[24] RZ2018 estimate local projections instrumented with the news series; the estimator of Section 2 is built on the likelihood of a VAR, which is what makes the second moments available to the threshold step.[25] Following Plagborg-Møller and Wolf (2021), we therefore include the proxy z_t as the first variable of the system $[z_t, y_t, g_t]'$: local projections and VARs estimate the same impulse responses under the same identification, and ordering z_t first with a Cholesky factorization of the regime-specific covariance recovers the proxy-SVAR impulse responses within each regime. We set $p = 4$ lags, matching the four lags of controls in RZ2018's projections.[26]

Two exercises are considered in the original paper:

- **Slack**: the state variable is the unemployment rate. RZ2018 impose a threshold of 6.5% to capture slack in the economy.
- **ZLB**: the state variable is the interpolated three-month Treasury-bill rate. RZ2018 impose a threshold of 0.25%.

In both cases, regime assignment is based on the one-period lag of the state variable, so that it lies in the information set at the time the shock hits.

The slack exercise is estimated on three samples: (i) the full historical sample 1890Q1–2015Q4 ($T = 504$); (ii) the full sample excluding WWII rationing quarters 1941Q3–1945Q4 (labeled *Excl. WWII*); and (iii) the post-WWII sample 1947Q1–2015Q4 ($T = 276$, labeled *Post-WWII*). The ZLB exercise uses observations from 1915Q1 to 2015Q4 ($T = 404$), again with and without WWII. We let the data choose the regimes. For each sample and state variable, we first estimate the threshold in the joint mean-variance TVAR of Section 2.5. We then re-estimate the model with the decoupled estimator of Section 2.2, retaining RZ2018's

[24] RZ2018's baseline shock is the military news variable (their Section III); the multipliers and p-value quoted here are from the first panel of their Table 1. They also report Blanchard–Perotti identification and, on the full sample only, the two instruments combined (Table 1, third panel; 0.62 against 0.33, Anderson–Rubin $p = 0.23$), noting that the combination may come at a cost of exogeneity; all their subsample and robustness results use one instrument at a time. We follow the baseline.

[25] In a companion project we show that the thresholds can be equivalently estimated in a regime-dependent local-projection setup.

[26] The equivalence of Plagborg-Møller and Wolf (2021) is exact under the VAR(p) specification maintained here. Without that restriction, it would require as many lags as horizons, here 20, which the regime cells cannot support; four lags is also RZ2018's own choice.

economic state for the mean regimes while introducing the size of the identifying shock instrument as the variance-regime state. We estimate thresholds on all subsamples considered in the original paper under both state-variable specifications. The slack exercise is the focus of this section; the interest-rate exercise, in which the free estimates recover RZ2018's 0.25% cutoff on the full sample once the shock environment is isolated, is reported in Appendix C and gives the same qualitative picture.

4.2 Single-Threshold Estimation and Instrument Diagnostic

We begin with the joint estimator, in which a single threshold on the state variable governs both the dynamics and the innovation covariance – the common regime structure of the applied literature. Table 5 reports the estimates next to two benchmarks: the thresholds RZ2018 impose, and the threshold a researcher would obtain with the standard least-squares (SSR) criterion, which uses the conditional mean only.

Table 5: Joint threshold estimates: likelihood, least squares (SSR), and the imposed values

	Likelihood (common threshold)		SSR (mean only)		Imposed (RZ2018)	
Sample	$\hat{\gamma}$	Regime 1 obs. (%)	$\hat{\gamma}$	Regime 1 obs. (%)	γ	Regime 1 obs. (%)
Panel A: Slack (unemployment rate, %; Regime 1 = low unemployment)						
Full sample	3.56	57 (11.4)	6.95	343 (68.6)	6.50	319 (63.8)
Excl. WWII	3.64†	49 (10.3)	4.20	105 (22.0)	6.50	300 (62.8)
Post-WWII	5.61	139 (51.1)	5.61	139 (51.1)	6.50	189 (69.5)
Panel B: ZLB (T-bill rate, %; Regime 1 = low rate)						
Full sample	3.48	223 (55.8)	0.21‡	54 (13.5)	0.25	58 (14.5)
Excl. WWII	3.48	201 (53.2)	1.22	94 (24.9)	0.25	55 (14.6)

Notes: Thresholds estimated on a 300-point grid over the observed range of the state, with each regime required to contain more than $nK + n(n+1)/2 + 1 = 46$ ($K = np + 1$) observations. "Likelihood": the joint estimator of Section 2.5, one threshold common to both moments. "SSR": the least-squares criterion of Chan (1993), which uses the conditional mean only, on the same grid, sample and trimming. Regime 1 is the below-threshold regime. The post-WWII ZLB exercise is infeasible (too few low-rate observations). †Smallest grid point satisfying the minimum-size requirement (Regime 1 has 49 observations); the likelihood is still rising as the regime shrinks. ‡Lowest feasible grid point.

The comparison is instructive. On the full sample, the SSR estimate for slack is 6.95% – essentially RZ2018's 6.5% – and for the interest-rate exercise it is 0.21%, close to their 0.25%: the least-squares search confirms the imposed thresholds on the full sample. That agreement is not a validation of the cutoff. In RZ2018's projections, every coefficient switches with the state, the impact response included; in the VAR of Section 4.1 that is the joint model, with a regime-specific covariance. Least squares locates the threshold from the conditional mean alone and cannot see a switch in the covariance; the likelihood estimator of the joint model uses both, and it does not confirm the cutoff. It places the slack threshold at 3.56%, isolating 57 quarters of wartime near-full employment and peacetime booms, and the interest-rate threshold at 3.48%, a broad low-rate environment rather than a zero lower bound; on the

sample excluding WWII the slack estimate lies at the smallest regime the trimming allows. On the post-WWII sample, where the wartime episodes are absent, the two criteria coincide at 5.61%. When they agree, the dynamics and the covariance break at the same place and the cutoff is confirmed by both; when they disagree by more than three percentage points, as on the two historical samples, the covariance is breaking somewhere other than the dynamics, and no single threshold can serve both. The departure of the joint estimate from the imposed threshold is there attributable entirely to variance information – on the post-WWII sample the departure is a mean-fit refinement – which raises the question of what that information is.

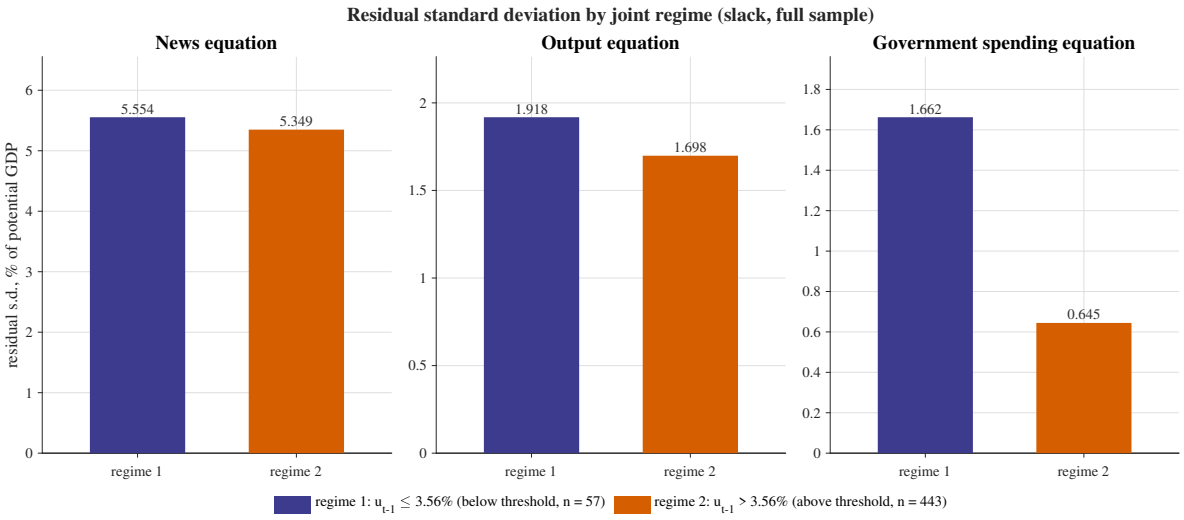


Figure 2: Residual standard deviation by joint regime: slack exercise, full sample ($\hat{\gamma} = 3.56\%$). Each panel reports the standard deviation of one equation's residual in the low-unemployment regime ($u_{t-1} \leq 3.56\%$, 57 quarters) and in the high-unemployment regime (443 quarters), from the joint TVAR estimated at the stored threshold. Units: percent of potential GDP.

Figure 2 identifies such information. It reports the residual standard deviation of each equation of the TVAR in the two regimes selected by the joint estimator on the full sample. The break is in government spending: the standard deviation of the spending-equation residual is 1.66 in the low-unemployment regime against 0.64 in the high-unemployment regime, more than twice as large. The output equation's residual volatility is modestly larger (1.92 against 1.70), and the news equation's own residual is not (5.55 against 5.35) – the largest news realizations of the sample, in 1941Q3–1942Q1, occur when unemployment is still above the threshold. The joint likelihood thus locates the threshold where the volatility of fiscal innovations changes, in the wartime spending surges, rather than where the propagation of shocks changes.The residuals are reduced-form, and the figure shows only the diagonal of the two covariance matrices; the criterion uses the whole matrix, correlations included (Table A-15 reports them by regime), and with the news series ordered first, its residual is the

identified shock itself. The instrument's own variation concentrates in the same quarters, with a lag (Figure 3: the low-unemployment regime holds 22% of the instrument's absolute mass with 11% of the observations, and 20% of its top-decile realizations). The lag is the anticipation built into the instrument: the announcements arrive while unemployment is still high, and the subsequent military mobilization lowers unemployment below the threshold in the quarters in which the announced spending is executed, which is where the spending innovations are volatile. The regime indicator and the lagged instrument therefore carry largely the same information, and the "state regime" is a proxy for a shock regime – specifically, for the wartime spending surge that the news announces. It is not a business-cycle regime: its 57 quarters pool wartime near-full employment with the peacetime booms of the 1900s and 1920s – a grouping of wars and booms rather than a partition into expansions and recessions.[27] Under the SSR split of 6.95%, as under RZ2018's own 6.5%, where 73% of the fiscal-event quarters fall in a regime with 64% of the observations, the concentration is proportionate to regime size. The distortion is therefore specific to the variance-driven split: a variance regime is being forced onto a single mean-regime partition. We therefore estimate a separate threshold for each dimension with the decoupled estimator.

For this reason, we do not report multiplier estimates from the joint specification: under this degree of concentration, regime-specific multipliers would measure the particular wartime episodes rather than the economic states.

4.3 Decoupled Estimation: Separating the Economic State from the Shock Environment

The decoupled estimator keeps the unemployment rate as the mean state and introduces a second state for the variance regimes: the absolute value of the lagged instrument, $|z_{t-1}|$. The choice of the news series as the variance state, despite Figure 2 locating the variance break in the spending equation, lies in the interest of isolating a *shock* regime, the environment in which the identifying shocks arrive. The news series is the identifying shock, whereas spending is the system's *response* to it. A variance regime defined on spending, or on its innovation, would classify quarters by the consequence rather than the cause, conflating the two dimensions the model separates. Timing makes the point concrete: announcements lead spending by quarters or years, the premise of the news identification, so a spending-based regime dates the environment after the identifying variation has entered the system. The absolute lagged news series isolates the arrival of large identified shocks, clustered into named episodes, and is known at $t-1$ like the mean state. Two features of the regime are not imposed by construction. The threshold is estimated by maximizing the likelihood over every partition that $|z_{t-1}|$ induces on the sample, and the maximum lies at the 93rd

[27] Wartime near-full employment in 1943–46, 1951–53 and 1968–69; peacetime booms in 1902–07 and 1926–29.

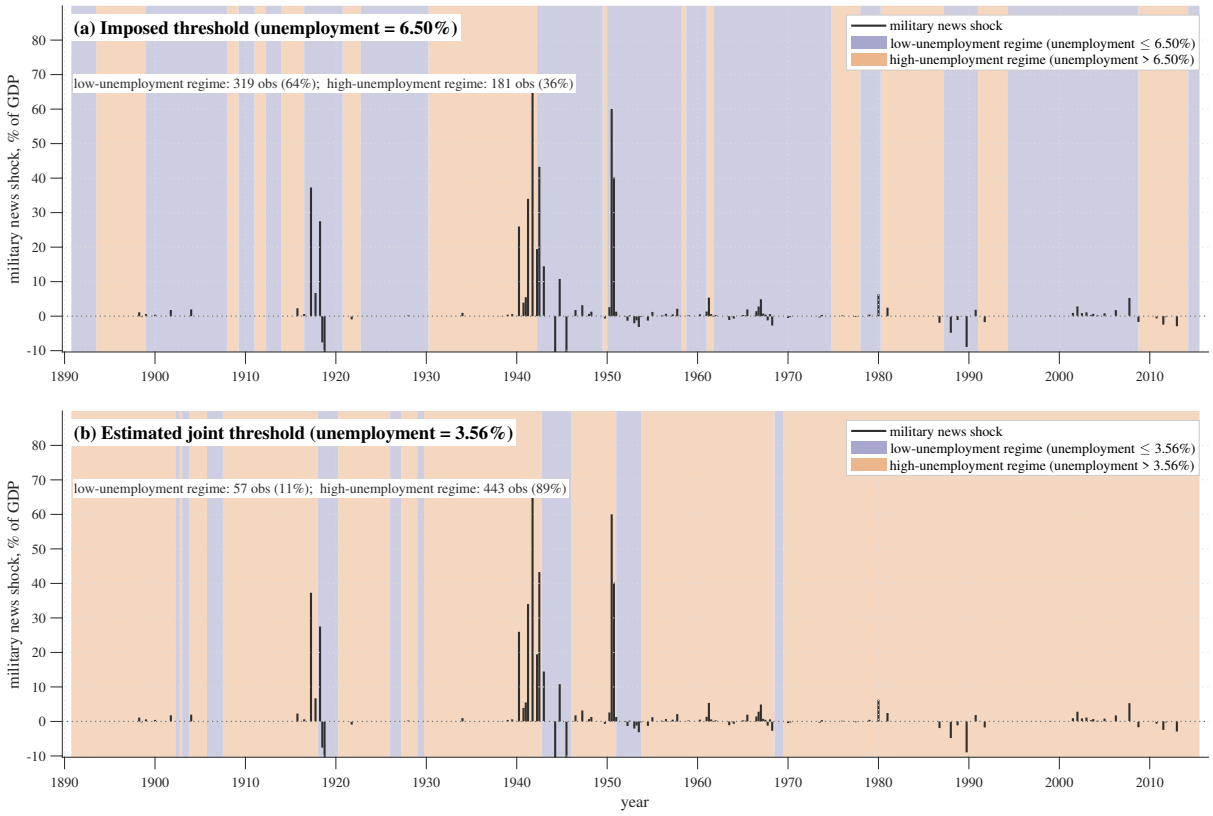


Figure 3: Unemployment regimes and the military news instrument, full sample. *Top:* the instrument colored by the imposed assignment at 6.5%; *bottom:* by the estimated joint assignment at 3.56%. The low-unemployment regime is shaded violet, the high-unemployment regime orange.

percentile of the state, leaving many sizable announcements outside the regime. The regime it selects differs from the rest of the sample in more than the size of the news: on the historical samples the standard deviation of the spending innovation is 2.6–3.7 times larger in the fiscal-event regime, that of the output innovation is not (0.9–1.2), and the correlation between the two rises from about zero to 0.4–0.5 (Table A-15); on the post-WWII sample, whose fiscal events are smaller, the spending ratio is 1.3 and the regime is mostly one of large announcements. Both alternative variance states are nonetheless examined: the spending residual in Section 4.7, where it selects a dispersion split with no event structure. Lagged spending growth (Appendix I) behaves as the argument predicts: it isolates the build-ups and demobilizations of the three major wars together with pre-1940 peacetime spending volatility, misses the announcement quarters and every post-1968 fiscal event, and in two of the three samples produces boundary partitions whose cells are too small for inference – indicating a state that classifies by consequence rather than cause.

Two features of the resulting partition matter for everything that follows. First, the variance threshold is remarkably stable, $\hat{\gamma}^{\sigma} \approx 0.024$ in every sample and for both state variables – unemployment here, the T-bill rate in Appendix C – and it isolates a small

fiscal-event regime (VR2) of large military-news quarters – on the full sample 33 quarters comprising the two World Wars (1917–19, 1940–45), the Korean mobilization (1950–51, 1953Q4), the 1961 Berlin crisis, the Vietnam build-up (1967–68), the Reagan build-up (1980–81), the late Cold War and Gulf War announcements (1988, 1990), the post-9/11 build-up (2002), and the 2008–2013 fiscal responses. Everything else forms the *normal-shock regime* (VR1). Second, once that regime is isolated, the mean threshold moves to where the propagation break lies rather than where the volatility break lies (Table 6).

Table 6: Decoupled TVAR: threshold estimates and four-cell observation counts

	Thresholds		Low unemployment		High unemployment		
Sample	$\hat{\gamma}^{\mu}$ (%)	$\hat{\gamma}^{\sigma}$	normal	fiscal-event†	normal	fiscal-event†	$\lvert z\rvert$ share (%)
Full sample	8.42	0.0243	382 (76.6)	28 (5.6)	84 (16.8)	5 (1.0)	36.8
Excl. WWII	3.44	0.0241	37 (7.8)	4 (0.8)	415 (87.0)	21 (4.4)	44.7
Post-WWII	4.23	0.0211	47 (17.3)	5 (1.8)	208 (76.5)	12 (4.4)	53.2
Imposed (RZ2018)	6.50	—		—			—

Notes: Slack exercise. $\hat{\gamma}^{\mu}$: mean threshold on the lagged unemployment rate, defining the low-unemployment ($u_{t-1} \leq \hat{\gamma}^{\mu}$, MR1) and high-unemployment (MR2) regimes. $\hat{\gamma}^{\sigma}$: variance threshold on $|z_{t-1}|$, the absolute lagged military news shock of Ramey (2011), defining the normal-shock (VR1) and fiscal-event (VR2) regimes. Cell entries: observations (% of the kept sample). "$|z|$ share": fraction of $\sum_t |z_t|$ in the fiscal-event regime. $\hat{B}_i$ by feasible GLS with variance-regime precision weights; $\hat{\Sigma}_{\ell}$ pooled within variance regime. †Too small for inference; not used below. The interest-rate exercise is in Appendix C.

The mean threshold now varies across samples, and the variation is informative. On the full sample it rises to 8.4%: the high-unemployment regime is an *extreme-slack* regime of 89 quarters, roughly three quarters of them pre-1947 (the depressions of the 1890s, 1914–15, 1921–22 and the Great Depression of 1930–41), joined by the deepest post-war recessions (1975, 1982–84, 2009–12). On the post-WWII sample it falls to 4.2%, a conventional business-cycle cutoff below RZ2018's 6.5%. On the sample excluding WWII it is 3.4%, isolating a small low-unemployment regime of 41 quarters; we read this estimate with caution, as it lies close to the trimming boundary, and the imposed-threshold exercise of Section 4.5 does not depend on it.[28]

4.4 State-Dependent Multipliers in the Normal-Shock Regime

An important element of the analysis that follows is the identification of impulse responses in the decoupled TVAR. Under internal-proxy identification, the $h = 0$ response is the first column of the Cholesky factor of $\hat{\Sigma}_{\ell}$. Since this depends only on the variance regime, the impact of a fiscal shock is the same across mean regimes within a variance regime. At the same time, its propagation – the response at $h \geq 1$ – is governed by the mean-regime coefficients $\hat{B}_i$, which differ across mean regimes but not across variance regimes. This

[28]The fiscal-event regime holds 37–53% of the instrument's absolute mass with 6–7% of the observations, as intended. The fiscal-event cells are reported in Appendix H for completeness only.

separation motivates the terminology: the variance threshold governs the impact channel, the mean threshold the propagation channel. The separation is a property of the VAR: a local projection estimates the total response at each horizon. It contains no parameter that corresponds to propagation, so the pooling that makes cells of 4 to 28 quarters informative about the impact of the shock – one impact vector across mean regimes, one propagation across variance regimes – has no counterpart in it. We report cumulative multipliers, the ratio of the cumulative output response to the cumulative spending response, because any normalization of the shock rescales both responses together and leaves the ratio unchanged. What the ratio does not remove is the impact *direction*, the relative response of output and spending, which differs across variance regimes but is common across mean regimes. Accordingly, the relevant comparison for state-dependent transmission is across mean regimes within a fixed variance regime, where any difference at $h > 0$ is propagation; cells in different variance regimes are not compared.

Inference on the multipliers is by a fixed-design wild bootstrap.[29] Residuals and fitted values come from the regime-specific regressions at the estimated thresholds; each pseudo-sample multiplies the residual vector of quarter t by a standard normal draw η_t, the same draw for the three equations, so that the cross-equation correlation and the regime-specific covariances of the residuals are preserved, and holds the regressors fixed (Gonçalves and Kilian, 2004). On each of $B = 499$ pseudo-samples the two thresholds are re-estimated on a coarser grid and the cells, coefficients, covariances and multipliers are recomputed; the bands are percentile intervals, 90% in the tables and, in the figures, 68% (dark) inside 90% (light). The re-estimation of the thresholds inside the bootstrap is not the object of the exercise – Remark 14 explains why the bootstrap is not a valid route to the thresholds themselves – but it is harmless for the multipliers: by Theorem 2(i) the threshold draws vary at rate T, below the $\sqrt{T}$ scale of the bands, so the intervals are those of the regime parameters conditional on the regime structure, which is the object of interest. In the imposed-threshold exercise of Section 4.5 the threshold pair is held fixed and only the regime parameters are re-drawn. Appendix F checks the bands against a posterior computed under a flat prior and finds them of the same width in the well-populated cells.

Table 7 reports the cumulative multipliers in the two normal-shock cells, and Figure 4 plots them together with the point estimates obtained at the imposed 6.5% threshold, to be discussed in Section 4.5. We complement the cell-level bands with a direct test of the across-regime difference: within each bootstrap replication we compute $\Delta(h) = \hat{m}_{\text{high-}u}(h) - \hat{m}_{\text{low-}u}(h)$ and report the 90% percentile interval of its bootstrap distribution (the

[29] Wild, because the innovations are heteroskedastic by construction across the variance regimes: multiplying each quarter's residual vector by its own draw preserves the conditional variance of every quarter, whatever its form, whereas resampling residuals would mix the two regimes' innovations. Fixed-design, because the regime assignment is a function of the observed states, which the pseudo-samples keep..

Δ rows). Because both cells are re-estimated from the same draw and share the normal-shock covariance $\hat{\Sigma}_1$, the two estimates are positively correlated and the interval for the difference is tighter than a comparison of the individual bands suggests; an interval excluding zero indicates a significant difference at the 10% level, paralleling the across-state tests of RZ2018 themselves.

Table 7: Cumulative multipliers in the normal-shock regime at the estimated thresholds

		Horizon h (quarters)			
Sample	Regime	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Full sample	low unemployment	0.44	0.46	0.48	0.45
		[-0.15, 0.80]	*[0.12, 0.71]*	*[0.23, 0.73]*	*[0.20, 0.73]*
	high unemployment	1.24	1.06	0.99	0.98
		[0.25, 1.95]	*[0.31, 1.86]*	*[0.21, 1.88]*	*[-0.10, 1.98]*
	Δ (high − low)	0.80	0.59	0.51	0.53
		[0.01, 1.41]	*[-0.14, 1.45]*	*[-0.33, 1.47]*	*[-0.69, 1.62]*
Excl. WWII	low unemployment	0.18	0.18	0.22	0.24
		[-0.15, 0.86]	*[-0.16, 0.77]*	*[-0.21, 0.89]*	*[-0.33, 1.30]*
	high unemployment	0.94	0.78	0.66	0.55
		[-0.87, 3.91]	*[-0.14, 2.19]*	*[-0.12, 1.91]*	*[-0.14, 1.76]*
	Δ (high − low)	0.77	0.59	0.44	0.31
		[-1.01, 3.35]	*[-0.38, 1.76]*	*[-0.50, 1.65]*	*[-0.90, 1.61]*
Post-WWII	low unemployment	3.45	2.86	3.08	4.47
		[0.64, 11.24]	*[0.62, 8.11]*	*[0.32, 10.71]*	*[-11.41, 25.88]*
	high unemployment	1.56	1.31	1.04	0.72
		[-0.43, 7.51]	*[-0.25, 5.05]*	*[-0.79, 5.09]*	*[-0.82, 3.27]*
	Δ (high − low)	−1.89	−1.55	−2.03	−3.75
		[-8.34, 3.86]	*[-6.11, 2.99]*	*[-9.07, 4.14]*	*[-25.59, 13.77]*

Notes: Slack exercise, normal-shock regime ($|z_{t-1}| \leq \hat{\gamma}^{\sigma}$), mean regimes at the estimated $\hat{\gamma}^{\mu}$ of Table 6. Point estimates with 90% wild-bootstrap intervals below ($B = 499$). The multiplier at horizon h is $\sum_{s=0}^{h} \hat{\phi}_{Y,s} / \sum_{s=0}^{h} \hat{\phi}_{G,s}$, the ratio of cumulative responses of output and government spending to the identified news shock. The $h = 0$ multiplier is identical across mean regimes within the normal-shock regime (common $\hat{\Sigma}_1$, hence common impact vector). Δ rows: difference across mean regimes, with intervals computed from the same bootstrap draws as the cell bands, so the correlation between cells is accounted for; an interval excluding zero indicates a significant difference at the 10% level. Two cells call for caution at long horizons. In the post-WWII low-unemployment cell (47 quarters) the denominator changes sign in 9% of draws at $h = 19$, which is what widens that band; in the full-sample high-unemployment cell (84 quarters) the estimated within-regime dynamics are explosive (largest root 1.58), so its multipliers are informative at short horizons only (Appendix F; see also the stability check in Section 4.6).

The results show a sample-dependent direction. On the historical samples, multipliers are larger in the high-unemployment regime: on the full sample 1.24 against 0.44 at $h = 3$ and 0.98 against 0.45 at $h = 19$, the ranking of Auerbach and Gorodnichenko (2012), driven by the pre-1947 extreme-slack variation documented above; the 90% interval for the difference excludes zero at $h = 3$ and the 68% interval through $h = 11$. On the post-WWII sample,

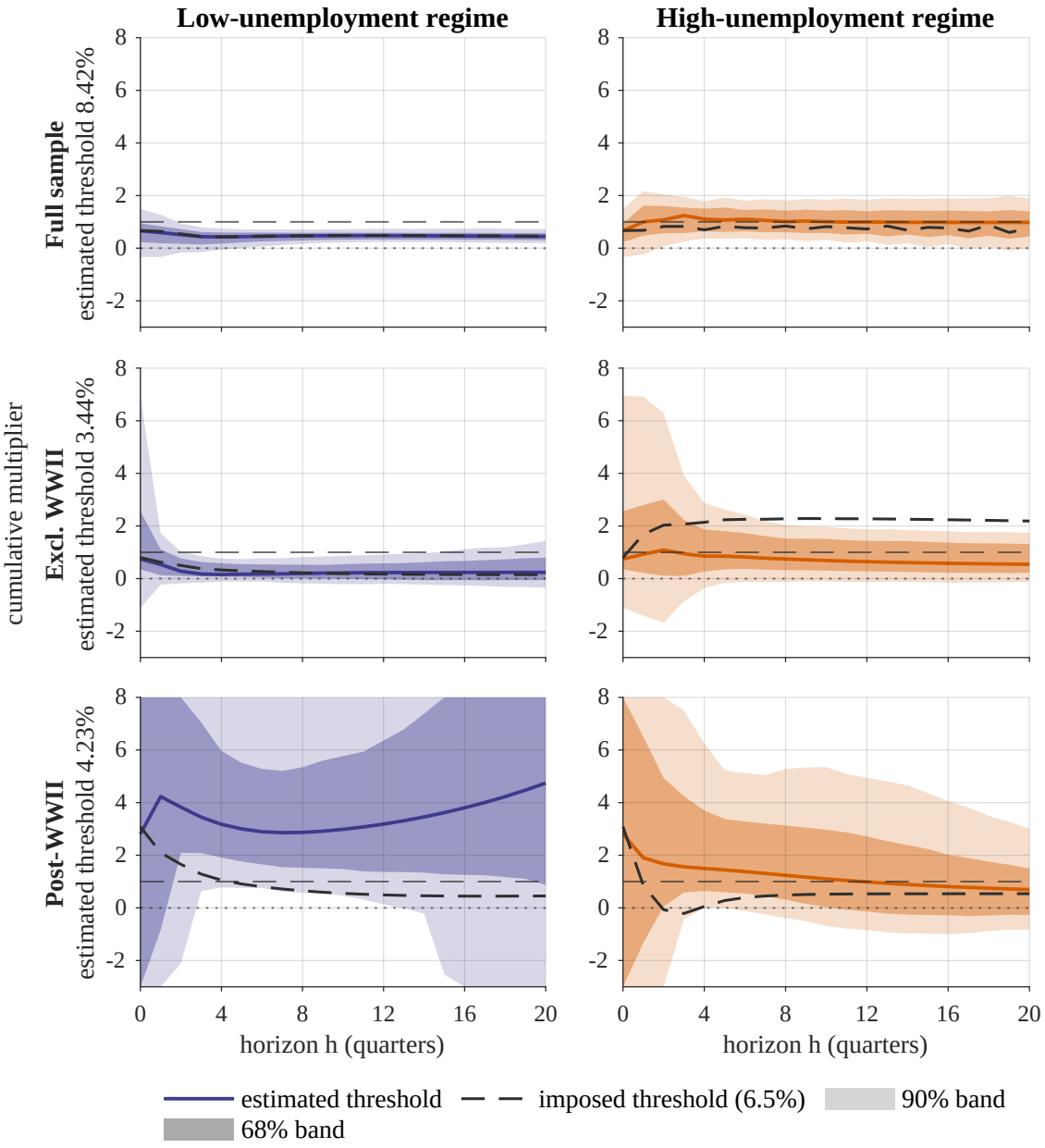


Figure 4: Cumulative multipliers in the normal-shock regime, slack exercise. Rows: full sample, excluding WWII, post-WWII; columns: low-unemployment and high-unemployment regimes at the estimated threshold, with 68% (dark) and 90% (light) wild-bootstrap bands ($B = 499$). The dashed black line is the point estimate at the imposed 6.5% threshold (Section 4.5). Bands are truncated at the display range.

the ranking reverses: 3.45 rising to 4.47 in the low-unemployment regime against 1.56 falling to 0.72 in the high-unemployment regime – larger multipliers in good times, the ranking of Alloza (2022). The post-WWII ranking is a point-estimate result in this exercise: the 90% interval for the difference includes zero at every horizon (Δ rows) and the 68% interval excludes it at $h = 3$ only, reflecting the width of the low-unemployment bands in a 47-quarter cell. Its statistically significant counterpart is the robustness exercise of Section 4.7, where the same ranking is recovered under a different identification, with the difference significant through $h = 11$. The fiscal-event cells are not examined for state dependence: with 4 to 28 quarters each, they identify the impact vector of the shock but not its propagation; their impact responses are reported in Appendix H. One further caveat concerns the small cells. The stability condition **A4** can be checked at the estimates: it holds on the sample excluding WWII and fails on the other two in their smallest cell – within sampling uncertainty in the post-WWII low-unemployment regime, and decisively only in the full-sample Great Depression regime – which is why Table 7 reads the long-horizon multipliers of those two cells with caution.[30]

The contrast with RZ2018 is substantial. At their imposed 6.5% threshold and under their baseline identification, they find 0.60 in the high-unemployment regime and 0.59 in the low-unemployment regime, with a p-value of 0.95 for the difference.[31] We find that the ranking of the two regimes is not fixed but depends on the sample – larger in the high-unemployment regime on the historical samples, larger in the low-unemployment regime after WWII – and, as the next subsection shows, that significant differences appear even at their own threshold once the fiscal-event regime is isolated.

4.5 State Dependence at RZ2018's Imposed Thresholds

Our estimates differ from RZ2018's in two respects: the location of the mean threshold, and the isolation of the fiscal-event regime. To distinguish them, we re-estimate the model with the mean threshold *imposed* at RZ2018's 6.5% and only the variance threshold estimated (at the values of Table 6). Relative to RZ2018, this keeps their threshold and changes only the treatment of the shock environment; relative to our free estimates, it changes only the

[30] **A4** restricts the data-generating process, not the estimates. Appendix G reports the check at the point estimates, the bootstrap distribution of the largest within-regime root, and the precedent: Ramey and Zubairy (2018) report explosive roots for their own threshold VAR under the slack definition of the state (their Section VI.B and footnote 42). Short-horizon multipliers are finite sums of a few coefficients and do not depend on the roots; the long-horizon multipliers in the small cells do, and are read accordingly.

[31] The VAR is not what moves the result. The joint TVAR at their 6.5% partition, with no variance regimes, gives 0.52 in the high-unemployment regime against 0.60 in the low-unemployment regime at $h = 7$ on the full sample, their null reproduced; the ranking appears once the fiscal-event regime is isolated (0.77 against 0.46, Table 8). On the sample excluding WWII, the VAR step alone already separates the regimes (1.66 against 0.42); that is also the one sample on which RZ2018's own multipliers depend on how they are computed, by summing the impulse responses or by the one-step instrumental-variables regression (their footnote 26).

location of the mean threshold. Inference is conditional on the fixed threshold pair, with only the regime parameters re-drawn in the wild bootstrap ($B = 499$).

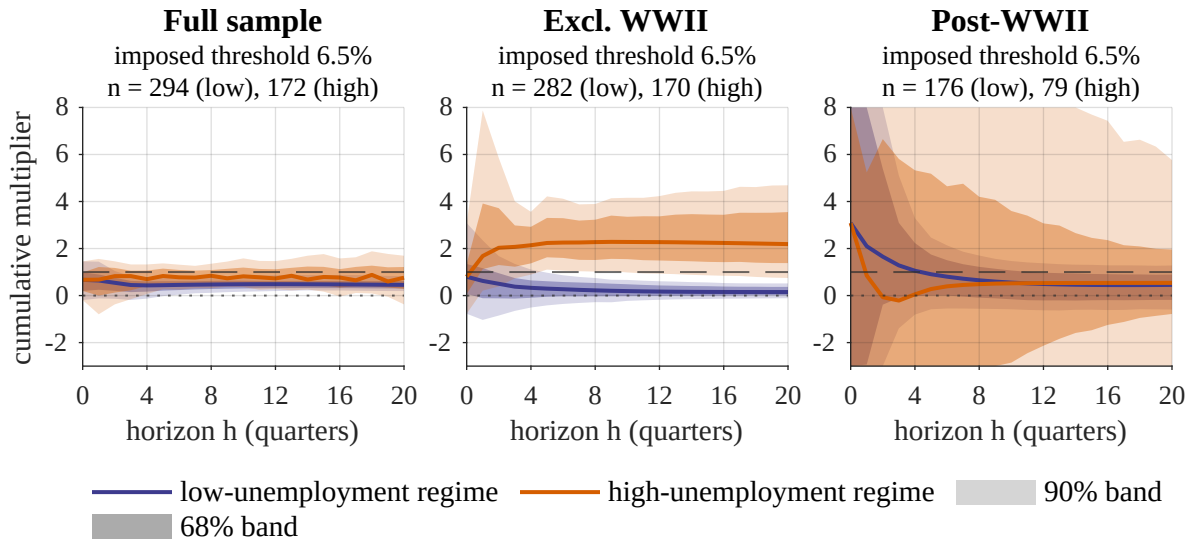


Figure 5: Cumulative multipliers in the normal-shock regime at the imposed 6.5% threshold, by sample. Each panel overlays the low-unemployment (violet) and high-unemployment (orange) regimes with 68% (dark) and 90% (light) bands conditional on the fixed threshold pair.

Figure 5 and Table 8 deliver one result per sample. On the sample excluding WWII, the two regimes separate cleanly at RZ2018's own threshold: 2.07–2.28 in the high-unemployment regime against 0.15–0.38 in the low-unemployment regime. The conventional 90% wild-bootstrap intervals for the difference exclude zero at every horizon ($\Delta = 1.7$–2.1). However, the very weak first stage in the high-unemployment normal-shock cell means that this result should be interpreted cautiously.[32] On the full sample, the ranking is the same and the gap smaller (0.83 against 0.45 at $h = 3$) and not significant; the three quarters of the U.S. entry into WWII, 1941Q3–1942Q1, are extreme leverage points inside the high-unemployment regime, and their inclusion is what separates the two historical samples (Appendix D). On the post-WWII sample, no ranking emerges at 6.5%: the high-unemployment regime at that cutoff contains 79 normal-shock quarters; its bands are wide, and the difference is indistinguishable from zero; the larger multipliers in the low-unemployment regime of Table 7 appear only at the estimated 4.2%.

[32]Two caveats for this cell. Once WWII is excluded, the high-unemployment quarters at 6.5% contain almost no news variation outside the fiscal-event regime (first-stage $F < 0.1$, Appendix E), exactly as in RZ2018's own high-unemployment state on this sample. The impact vector of the shock is estimated from the whole normal-shock regime, but the propagation coefficients of the high-unemployment regime are estimated from little identifying variation, and we read the multiplier there with the same caution that RZ2018 apply to theirs. The result is also robust to the lag length ($p = 6, 8$) at horizons up to two years and not beyond.

Table 8: Imposed-threshold exercise: normal-shock multipliers at RZ2018's 6.5%

Sample	Regime	Horizon h (quarters) $h=3$	$h=7$	$h=11$	$h=19$
Full sample	low unemployment	0.45	0.46	0.48	0.46
		[-0.19, 0.82]	*[0.10, 0.78]*	*[0.18, 0.87]*	*[0.19, 0.84]*
	high unemployment	0.83	0.77	0.78	0.60
		[-0.15, 1.32]	*[0.31, 1.26]*	*[0.25, 1.47]*	*[-0.05, 1.77]*
	Δ (high − low)	0.38	0.30	0.29	0.14
		[-0.29, 0.76]	*[-0.16, 0.81]*	*[-0.31, 0.99]*	*[-0.63, 1.31]*
Excl. WWII	low unemployment	0.38	0.24	0.18	0.15
		[-0.65, 1.35]	*[-0.31, 0.79]*	*[-0.20, 0.63]*	*[-0.12, 0.52]*
	high unemployment	2.07	2.26	2.28	2.20
		[0.73, 4.01]	*[1.03, 3.88]*	*[0.95, 4.16]*	*[0.76, 4.68]*
	Δ (high − low)	1.69	2.01	2.10	2.05
		[0.77, 3.37]	*[0.86, 3.56]*	*[0.75, 3.95]*	*[0.59, 4.53]*
Post-WWII	low unemployment	1.28	0.72	0.52	0.45
		[-1.39, 5.10]	*[-0.55, 1.87]*	*[-0.61, 1.41]*	*[-0.59, 1.27]*
	high unemployment[§]	−0.21	0.45	0.53	0.54
		[-23.51, 21.58]	*[-13.41, 17.65]*	*[-8.62, 13.18]*	*[-3.85, 6.34]*
	Δ (high − low)	−1.50	−0.27	0.01	0.09
		[-24.49, 20.93]	*[-14.38, 17.83]*	*[-9.14, 13.10]*	*[-4.34, 5.62]*

Notes: Mean threshold imposed at 6.5%; variance threshold at the estimated value of Table 6; normal-shock regime only. 90% wild-bootstrap intervals conditional on the fixed threshold pair ($B = 499$, all draws valid). Normal-shock cell counts (low / high unemployment): 294 / 172 (full), 282 / 170 (excl. WWII), 176 / 79 (post-WWII). Δ rows as in Table 7. [§]The wide intervals reflect bootstrap draws in which the cumulative spending response, the denominator of the multiplier, changes sign: 23% of draws at $h = 3$ and 13–14% at $h \geq 7$ in this 79-quarter cell, against at most 1.6% in every other cell of the table. They are a feature of the ratio in a small cell with a weak first stage (Appendix E), not evidence of unstable dynamics, and the posterior bands of Appendix F reproduce them.

Overall, the exercise shows that imposing a threshold has first-order consequences for the conclusions, and that the consequences differ by sample. On the historical samples the imposed threshold is not the problem: holding it at 6.5%, the ranking appears on both samples as soon as the fiscal-event regime is isolated, and on the sample excluding WWII the difference between regimes is significant at every horizon, suggesting that the null result in RZ2018 there reflects the shock environment, not the regime definition. On the post-WWII sample the opposite holds: at 6.5% no ranking emerges, and it is the estimated threshold that reveals the larger multipliers in the low-unemployment regime, so there the regime definition is the binding margin. The two margins bind in different parts of the sample.[33]

The imposed-threshold exercise also answers a question about the instrument itself. RZ2018 document that the military news series is a weak instrument in their high-unemployment

[33]RZ2018's own Blanchard–Perotti estimates on the full sample find significantly higher multipliers in the high-unemployment regime (0.68 against 0.30), the same direction we recover on the historical samples – visible, in their paper, only under the identification that does not inherit the instrument's concentration.

state and on the post-WWII sample. We find the same, and add that in the low-unemployment state part of the weakness is produced by the fiscal-event quarters. In those quarters the contemporaneous relation between the news and government spending has the opposite sign to the rest of the sample – the spending that a large announcement signals arrives only later – so a first stage that pools 24 war quarters with 294 ordinary ones combines two relations of opposite sign. Restricting RZ2018's low-unemployment regime to its normal-shock quarters raises the first-stage F-statistic on the news from 2.2 to 11.6 on the full sample and from 0.8 to 13.7 on the sample excluding WWII (Appendix E). The fiscal-event regime is thus where the news is largest and least representative of the rest of the sample, a further reason to estimate it as a regime of its own.

4.6 Taking Stock

Two mechanisms account for the difference between our results and RZ2018's. The first is the shock environment: the instrument's variation concentrates in a few wartime quarters whose fiscal innovations are two to three times more volatile than the rest of the sample's, and a common regime structure can represent them only as a distortion of the state regimes; once isolated, state dependence appears even at the imposed thresholds. The second is sample composition: the historical samples contain extreme-slack episodes – pre-1947 depressions and the deepest post-war recessions – that dominate the high-unemployment regime and deliver the ranking of Auerbach and Gorodnichenko (2012) (larger multipliers in slack), whereas the post-WWII sample, on its own, delivers the opposite one. The regime definition of Alloza (2022) is the third ingredient, and it is the binding margin where the first is not: after WWII.

One question remains open. The variance state is the absolute value of the lagged instrument, and the instrument is one of the variables in the VAR. The fiscal-event regime could therefore reflect the heteroskedasticity of the news equation alone – a feature of the identification approach – rather than episodes of unusual volatility in the macroeconomy. Within this system the two readings cannot be separated, because the likelihood that selects $\hat{\gamma}^{\sigma}$ observes the news innovations directly. Section 4.7 separates them by removing the instrument from the system.

4.7 Robustness: Removing the Instrument from the System

We re-estimate the regime structure in the framework of Auerbach and Gorodnichenko (2012), henceforth AG2012, chosen because it removes the instrument from the system.[34]

[34] AG2012 estimate a smooth-transition VAR, but the framework is well suited to a hard threshold. In their footnote 10 they report that when the smoothness parameter of the logistic transition is estimated jointly with the regime parameters, its point estimates exceed 5–10 across specifications – which they describe as evidence that the model "is best described as a model switching regimes sharply at certain thresholds" – and they calibrate a smooth transition only because joint nonlinear estimation is imprecise in small samples.

Identification is recursive following Blanchard and Perotti (2002) in a VAR of government spending, taxes and output over 1947Q1–2008Q4; the news series enters only as a candidate variance state and is absent from the VAR, and the mean state is a backward-looking version of AG2012's growth state, known at $t-1$.[35] The likelihood that selects the variance threshold is therefore computed exclusively from the second moments of the macroeconomic aggregates. If the fiscal-event regime of Section 4.3 were a feature of the proxy, there would be no reason for this likelihood to select the same partition. Appendix J reports the full exercise; two results matter here.

First, the AG2012 likelihood selects a variance threshold of 2.04% of GDP. This corresponds to the variance threshold estimated in Section 4.3 after the unit conversion between the two news series, and reproduces the fiscal-event partition of Section 4.3 *quarter for quarter* on the common 1949–2008 window (15 of 15 quarters). The second moments of the macroeconomic aggregates themselves break at the fiscal-event quarters: the shock regimes are a property of the macroeconomy, not of the instrument. Second, in the normal-shock regime the post-WWII ranking is the one found above. Multipliers are larger in the high-growth than in the low-growth regime – 1.11 rising to 3.12 against 0.18 to 0.31 (Table A-19) – and here the difference Δ between the two is statistically significant through $h = 11$ ($\Delta = 0.93$, 1.69, 2.14 at $h = 3$, 7, 11, with intervals excluding zero). This holds under a different identification scheme, a different state variable and a different sample construction. Across the two exercises, then, the shock-regime partition is identical and the direction of state dependence invariant to how "good times" are measured; the magnitudes differ by a factor of two to three, as the identifications and the mean partitions do.

Our relationship to Alloza (2022) is therefore one of confirmation on one margin and departure on another. We confirm that state-dependent conclusions hinge on how the regimes are defined, and we estimate that definition rather than impose it. However, the imposed-threshold exercise shows that the definitional margin is not exhaustive: holding the regime definition fixed at RZ2018's, the shock-environment split alone restores state dependence on the historical samples. Two researchers using an identical definition of the regimes can reach opposite conclusions depending on whether the shock environment is modeled. The relevant notion of a regime is two-dimensional – a propagation regime defined on the macroeconomic state, and a shock regime on the environment in which the identifying shocks arrive – and the second dimension is invisible to any single-partition analysis, however the partition is defined.

The T-rate super-consistency of the threshold (Theorems 2 and 3) resolves precisely that concern.

[35] A predetermined state is also what Alloza (2022) argues for: AG2012's centered moving average embeds output growth at $t+1$ and $t+2$, and he shows that regimes defined with information available at $t-1$ reverse their ranking. Here the backward-looking state is required by Assumption B2, and the ranking we recover under recursive identification is his.

5 Conclusion

This paper has developed the estimation theory for threshold VARs in which the conditional variance switches across regimes, as well as the conditional mean. In the decoupled model of equation (1), the mean and the variance regimes have their own state variables and thresholds; the joint model, in which the same state variable and the same threshold govern both moments, is the special case in which the two thresholds coincide, and a likelihood-gap rule selects it consistently when it holds. In both models, a likelihood-based estimator of the thresholds is super-consistent at rate T, with limiting distributions of compound-Poisson-argmax form, and the regime parameters retain standard $\sqrt{T}$ Gaussian inference: estimating the regime structure entails no first-order asymptotic cost. The simulations establish when the second moments matter. When only the mean switches, the likelihood and least-squares criteria are equivalent; when both moments switch, the likelihood improves precision; when only the variance switches, the likelihood is required for consistency, because a mean-based criterion contains no information on the regime boundary. A single threshold imposed on a process with separate mean and variance regimes omits the variance dimension, and the omission does not vanish as the sample grows.

The empirical analysis shows the role of the variance dimension in the fiscal multiplier literature. There, the identifying variation arrives with wars, and wars are episodes of unusual shock volatility as much as of unusual macroeconomic states. A least-squares threshold search confirms the threshold that Ramey and Zubairy (2018) impose. The likelihood estimator departs from it only because it uses the second moments, and the regime the second moments locate is the shock environment – a regime indicator nearly collinear with the largest realizations of the instrument – rather than a new state of the economy. The decoupled estimator separates the two. It isolates a fiscal-event regime of 33 quarters at a variance threshold that is remarkably stable in every sample and for both state variables, and a specification that excludes the instrument recovers the same fiscal-event quarters on the common sample. Outside the fiscal-event quarters, the data show state dependence. On the historical samples, which include data prior to WWII, multipliers are larger in the high-unemployment regime, both at the estimated thresholds and at the threshold imposed in the earlier literature – significantly so, at the imposed threshold, once WWII is excluded. After WWII, multipliers are larger in good times, significantly so under recursive identification. The disagreement in the literature on state-dependent multipliers therefore has three sources – the shock environment, the sample period, and the definition of the regimes – of which only the last has been debated. A regime, in this literature, is two-dimensional.

Which variables define the states is an economic question; whether a candidate state defines regimes in the data is an econometric one, and the framework answers the second. A theory of state dependence suggests a state variable and a direction of the effect, and

the framework translates that suggestion into objects that can be confronted with the data: the location of the cutoff; whether the regimes are in the dynamics, in the volatility, or in both; whether the partition survives a change of sample; and the responses once the regime structure is estimated rather than assumed. The empirical analysis illustrates each of these criteria. The unemployment state and the shock-environment state satisfy them; government spending as a variance state does not, because it classifies quarters by their response to shocks rather than by the arrival of shocks. Two limitations restrict what the data can say at long horizons, and we have indicated them where they apply: the weakness of the instrument in small cells, which we share with the literature that relies on it, and the explosive within-regime dynamics of the high-unemployment regime on the century-long sample. Three extensions follow directly from the theory and are left for future work: formal tests of the existence of each regime structure and of the common-threshold restriction, whose statistics have nonstandard limits and call for subsampling critical values; confidence intervals for the estimated thresholds of Section 4 – the plug-in interval of Remark 15 for the variance threshold, which requires an estimate of the state's density at the cutoff, and subsampling for the mean threshold, whose limit law involves the conditional distribution of the regressors; and a test, in the large cells, of the restriction that propagation is common across variance regimes. Within those limits, the conclusion is the one with which the paper opened. In macroeconomic data, volatility has regimes of its own; an estimator that does not model them does not thereby avoid them, but locates them in the wrong place.

References

Alloza, Mario. 2022. "Is fiscal policy more effective during recessions?" *International Economic Review* 63 (3): 1271–1292.

An, Hong-Zhi, and Fu-Chun Huang. 1996. "The geometrical ergodicity of nonlinear autoregressive models." *Statistica Sinica* 6 (4): 943–956.

Andrews, Donald WK. 1988. "Laws of large numbers for dependent non-identically distributed random variables." *Econometric theory* 4 (3): 458–467.

Auerbach, Alan J, and Yuriy Gorodnichenko. 2012. "Measuring the output responses to fiscal policy." *American Economic Journal: Economic Policy* 4 (2): 1–27.

Auerbach, Alan J, and Yuriy Gorodnichenko. 2013. "Fiscal multipliers in recession and expansion." In *Fiscal policy after the financial crisis*, 63–98, University of Chicago Press.

Bai, Jushan, and Pierre Perron. 1998. "Estimating and testing linear models with multiple structural changes." *Econometrica* 47–78.

Barnichon, Regis, Davide Debortoli, and Christian Matthes. 2022. "Understanding the size of the government spending multiplier: It's in the sign." *The Review of Economic Studies* 89 (1): 87–117.

Bernardini, Marco, and Gert Peersman. 2018. "Private debt overhang and the government spending multiplier: Evidence for the United States." *Journal of Applied Econometrics* 33 (4): 485–508.

Billingsley, Patrick. 1961. "The Lindeberg–Lévy theorem for martingales." *Proceedings of the American Mathematical Society* 12 (5): 788–792.

Billingsley, Patrick. 1999. *Convergence of Probability Measures*. New York: Wiley, , 2nd edition.

Blanchard, Olivier, and Roberto Perotti. 2002. "An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output." *The Quarterly Journal of Economics* 117 (4): 1329–1368.

Bloom, Nicholas. 2009. "The impact of uncertainty shocks." *econometrica* 77 (3): 623–685.

Bradley, Richard C. 2005. "Basic properties of strong mixing conditions. A survey and some open questions." *Probability Surveys* 2 107–144.

Caggiano, Giovanni, Efrem Castelnuovo, Valentina Colombo, and Gabriela Nodari. 2015. "Estimating fiscal multipliers: News from a non-linear world." *The Economic Journal* 125 (584): 746–776.

Canova, Fabio, and Fernando J Pérez Forero. 2024. "Does the transmission of monetary policy shocks change when inflation is high?"Technical report, Banco Central de Reserva del Perú.

Chan, Joshua CC, and Eric Eisenstat. 2018. "Bayesian model comparison for time-varying parameter VARs with stochastic volatility." *Journal of applied econometrics* 33 (4): 509–532.

Chan, K. S., J. D. Petruccelli, H. Tong, and S. W. Woolford. 1985. "A multiple-threshold AR(1) model." *Journal of Applied Probability* 22 (2): 267–279.

Chan, Kung S, and Howell Tong. 1985. "On the use of the deterministic Lyapunov function for the ergodicity of stochastic difference equations." *Advances in applied probability* 17 (3): 666–678.

Chan, Kung-Sik. 1993. "Consistency and Limiting Distribution of the Least Squares Estimator of a Threshold Autoregressive Model." *Annals of Statistics* 21 520–533.

Chan, Kung-Sik, and Ruey S. Tsay. 1998. "Limiting properties of the least squares estimator of a continuous threshold autoregressive model." *Biometrika* 85 (2): 413–426.

Chan, Ngai Hang, and Yury A Kutoyants. 2012. "On parameter estimation of threshold autoregressive models." *Statistical inference for stochastic processes* 15 (1): 81–104.

Chang, Yoosoon, Yongok Choi, and Joon Y Park. 2017. "A new approach to model regime switching." *Journal of Econometrics* 196 (1): 127–143.

Chen, Min, and Gemai Chen. 2000. "Geometric ergodicity of nonlinear autoregressive models with changing conditional variances." *Canadian Journal of Statistics* 28 (3): 605–614.

Cogley, Timothy, and Thomas J Sargent. 2005. "Drifts and volatilities: monetary policies and outcomes in the post WWII US." *Review of Economic dynamics* 8 (2): 262–302.

Doukhan, Paul. 1994. *Mixing: Properties and Examples*. Volume 85. of Lecture Notes in Statistics, New York: Springer-Verlag.

Fazzari, Steven M, James Morley, and Irina Panovska. 2015. "State-dependent effects of fiscal policy." *Studies in Nonlinear Dynamics & Econometrics* 19 (3): 285–315.

Feigin, Paul D, and Richard L Tweedie. 1985. "Random coefficient autoregressive processes: a Markov chain analysis of stationarity and finiteness of moments." *Journal of time series analysis* 6 (1): 1–14.

Feller, William. 1971. *An Introduction to Probability Theory and Its Applications*. Volume II. New York: Wiley, , 2nd edition.

Fernández-Villaverde, Jesús, Pablo Guerrón-Quintana, Keith Kuester, and Juan Rubio-Ramírez. 2015. "Fiscal volatility shocks and economic activity." *American Economic Review* 105 (11): 3352–3384.

Gargiulo, Valeria, Atsushi Inoue, and Barbara Rossi. 2025. "A New Approach to Fiscal Multipliers: Time Variation and High Frequency Shocks."Technical report, Centre for Economic Policy Research.

Gargiulo, Valeria, Christian Matthes, and Katerina Petrova. 2026. "Monetary policy across inflation regimes." *European Economic Review* 105322.

Ghassibe, Mishel, and Francesco Zanetti. 2022. "State dependence of fiscal multipliers: the source of fluctuations matters." *Journal of Monetary Economics* 132 1–23.

Gonçalves, Sílvia, Ana María Herrera, Lutz Kilian, and Elena Pesavento. 2024. "State-dependent local projections." *Journal of Econometrics* 244 (2): 105702.

Gonçalves, Sılvia, and Lutz Kilian. 2004. "Bootstrapping autoregressions with conditional heteroskedasticity of unknown form." *Journal of econometrics* 123 (1): 89–120.

Gonzalo, Jesús, and Michael Wolf. 2005. "Subsampling inference in threshold autoregressive models." *Journal of Econometrics* 127 (2): 201–224.

Hamilton, James D. 1989. "A new approach to the economic analysis of nonstationary time series and the business cycle." *Econometrica: Journal of the econometric society* 357–384.

Hamilton, James D. 1994. *Time series analysis*. Princeton university press.

Hansen, Bruce E. 1996. "Inference when a nuisance parameter is not identified under the null hypothesis." *Econometrica: Journal of the econometric society* 413–430.

Hansen, Bruce E. 2000. "Sample splitting and threshold estimation." *Econometrica* 68 (3): 575–603.

Hansen, Bruce E. 2011. "Threshold autoregression in economics." *Statistics and its Interface* 4 (2): 123–127.

Hirano, Keisuke, and Jack R Porter. 2003. "Asymptotic efficiency in parametric structural models with parameter-dependent support." *Econometrica* 71 (5): 1307–1338.

Ibragimov, Ildar A., and Rafail Z. Has'minskii. 1981. *Statistical Estimation: Asymptotic Theory*. New York: Springer-Verlag.

Jordà, Òscar. 2005. "Estimation and inference of impulse responses by local projections." *American economic review* 95 (1): 161–182.

Justiniano, Alejandro, and Giorgio E Primiceri. 2008. "The time-varying volatility of macroeconomic fluctuations." *American Economic Review* 98 (3): 604–641.

Kallenberg, Olav. 2002. *Foundations of Modern Probability*. New York: Springer, , 2nd edition.

Kim, Chang-Jin, Jeremy Piger, and Richard Startz. 2008. "Estimation of Markov regime-switching regression models with endogenous switching." *Journal of Econometrics* 143 (2): 263–273.

Kurtz, Thomas G. 1975. "Semigroups of conditioned shifts and approximation of Markov processes." *Annals of Probability* 3 (4): 618–642.

Kushner, Harold J. 1980. "A martingale method for the convergence of a sequence of processes to a jump-diffusion process." *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 53 207–219.

Kushner, Harold J. 1984. *Approximation and Weak Convergence Methods for Random Processes, with Applications to Stochastic Systems Theory*. Cambridge, MA: MIT Press.

Lenza, Michele, and Giorgio E Primiceri. 2022. "How to estimate a vector autoregression after March 2020." *Journal of Applied Econometrics* 37 (4): 688–699.

Lewis, Daniel J. 2021. "Identifying shocks via time-varying volatility." *The Review of Economic Studies* 88 (6): 3086–3124.

Li, CW, and Wai Keung Li. 1996. "On a double-threshold autoregressive heteroscedastic time series model." *Journal of applied econometrics* 11 (3): 253–274.

Li, Dong, and Shiqing Ling. 2012. "On the least squares estimation of multiple-regime threshold autoregressive models." *Journal of Econometrics* 167 (1): 240–253.

Liberzon, D. 2003. "Switching in Systems and Control, Lecture Notes."Technical report, University of Illinois; see sections on common Lyapunov functions.

Liu, Jian, Shiying Wu, and James V Zidek. 1997. "On segmented multivariate regression." *Statistica Sinica* 497–525.

Meyn, S. P., and R. L. Tweedie. 1993. *Markov Chains and Stochastic Stability*. Springer, , Available online; see Chapters 5, 15–16 for drift/petite set results.

Montiel Olea, José Luis, and Carolin Pflueger. 2013. "A Robust Test for Weak Instruments." *Journal of Business & Economic Statistics* 31 (3): 358–369.

Newey, Whitney K., and Daniel McFadden. 1994. "Large Sample Estimation and Hypothesis Testing." Chapter in the Handbook of Econometrics (or lecture notes), standard reference for M-estimators.

Owyang, Michael T, Valerie A Ramey, and Sarah Zubairy. 2013. "Are government spending multipliers greater during periods of slack? Evidence from twentieth-century historical data." *American Economic Review* 103 (3): 129–134.

Plagborg-Møller, Mikkel, and Christian K Wolf. 2021. "Local projections and VARs estimate the same impulse responses." *Econometrica* 89 (2): 955–980.

Primiceri, Giorgio E. 2005. "Time varying structural vector autoregressions and monetary policy." *The Review of economic studies* 72 (3): 821–852.

Ramey, Valerie A. 2011. "Identifying government spending shocks: It's all in the timing." *The quarterly journal of economics* 126 (1): 1–50.

Ramey, Valerie A, and Sarah Zubairy. 2018. "Government spending multipliers in good times and in bad: evidence from US historical data." *Journal of political economy* 126 (2): 850–901.

Rigobon, Roberto. 2003. "Identification through heteroskedasticity." *Review of Economics and Statistics* 85 (4): 777–792.

Samia, Noelle I, and Kung-Sik Chan. 2011. "Maximum likelihood estimation of a generalized threshold stochastic regression model." *Biometrika* 98 (2): 433–448.

Sims, Christopher A, and Tao Zha. 2006. "Were there regime switches in US monetary policy?" *American Economic Review* 96 (1): 54–81.

Stroock, Daniel W., and S. R. Srinivasa Varadhan. 1971. "Diffusion processes with boundary conditions." *Communications on Pure and Applied Mathematics* 24 (2): 147–225.

Teräsvirta, Timo. 1994. "Specification, estimation, and evaluation of smooth transition autoregressive models." *Journal of the american Statistical association* 89 (425): 208–218.

Tong, Howell. 1977. "Some comments on the Canadian lynx data." *Journal of the Royal Statistical Society Series A: Statistics in Society* 140 (4): 432–436.

Tong, Howell. 1990. *Non-linear Time Series: A Dynamical System Approach*. Oxford: Oxford University Press.

Tong, Howell, and Keng S Lim. 1980. "Threshold autoregression, limit cycles and cyclical data." *Journal of the Royal Statistical Society: Series B (Methodological)* 42 (3): 245–268.

Tsay, Ruey S. 1998. "Testing and modeling multivariate threshold models." *Journal of the American Statistical Association* 93 (443): 1188–1202.

Yu, Ping. 2012. "Likelihood estimation and inference in threshold regression." *Journal of Econometrics* 167 (1): 274–294.

Yu, Ping. 2014. "The bootstrap in threshold regression." *Econometric Theory* 30 (3): 676–714.

Online Appendix for "The Decoupled Threshold VAR: Mean and Variance Regimes in State-Dependent Fiscal Multipliers"

A Proofs **A-2**

A.1 Geometric Ergodicity A-2

A.2 Auxiliary Lemmas A-6

A.3 Consistency A-9

A.4 Rates and Limit Laws: the Single-threshold Case A-15

A.5 Rates and Limit Laws: the Decoupled Case A-23

A.6 Discussion of the Assumptions and the Implementation A-27

B Additional Tables from Monte Carlo Exercise **A-28**

C The Interest-Rate (ZLB) Exercise **A-33**

D Leverage Points and the Full Sample **A-33**

E Instrument Strength by Cell **A-35**

F Posterior Bands as a Check on the Bootstrap **A-37**

G The Stability Condition at the Estimates **A-39**

H Fiscal-Event Cells **A-39**

I Spending Growth as the Variance State **A-42**

J The Auerbach–Gorodnichenko Exercise **A-42**

A Proofs

Proofs are organized by argument rather than by order of appearance in the text: the single-threshold arguments underlying Theorem 3 come first, because the proofs for the decoupled model (Theorems 1 and 2) extend their steps and refer to them.

A.1 Geometric Ergodicity

A.1.1 Geometric Ergodicity and Fourth Moments

Lemma 1. *Under assumptions A1 to A3, the Markov chain* $(\mathbf{x}_t)$ *defined by* (8) *is aperiodic and* ν*-irreducible with* ν *being the Lebesgue measure. Furthermore, a* ν*-non-null compact set is a small set.*

Proof. We follow Chan and Tong (1985, Section 2) and An and Huang (1996, Lemma 5.2), adapted to the multivariate threshold VAR, and verify three properties:

(i) ν-irreducibility. For $p > 1$ the one-step kernel is singular with respect to Lebesgue measure on $\mathbb{R}^{np}$ (the noise enters only the first n coordinates of X_{t+1}), so irreducibility is established through the p-step kernel. Conditional on $X_t = x$, factor the joint law of $(y_{t+1}, \ldots, y_{t+p})$ sequentially: given $(x, y_{t+1}, \ldots, y_{t+j-1})$ the active regime at $t+j$ is determined, and y_{t+j} is a deterministic shift of $\Sigma_\ell^{1/2}\varepsilon_{t+j}$ with Σ_ℓ nonsingular (A3) and ε_{t+j} having an everywhere-positive continuous density (A1), hence has an everywhere-positive continuous conditional density on $\mathbb{R}^n$. The joint density of the block is the product of these conditional densities, hence everywhere positive on $\mathbb{R}^{np}$; since X_{t+p} is a fixed permutation of the block, $P^p(x, \cdot)$ has an everywhere-positive density $q_p(x, \cdot)$ for every x, so every set of positive Lebesgue measure is reached in p steps from every starting point: the chain is ν-irreducible with $\nu = \text{Leb}$.

(ii) Aperiodicity. We must show the transition kernel has no cyclical structure. The same argument applied to $p+1$ steps shows that $q_{p+1}(x, \cdot)$ is also everywhere positive. A cycle of period $d \geq 2$ would require the sets visited at times p and $p+1$ from a common starting point to lie in distinct elements of the cyclic decomposition, contradicting the positivity of both densities on all of $\mathbb{R}^{np}$; since $\gcd(p, p+1) = 1$, the chain is aperiodic.

(iii) Compact sets are small. The p-step density involves no integration over intermediate states: X_{t+p} is exactly the block $(y_{t+p}, \ldots, y_{t+1})$, so for a target point $z = (z_p', \ldots, z_1')'$ the intermediate states χ_j (the stacked state after j steps) are deterministic functions of $(x, z_1, \ldots, z_j)$, the regime path is the deterministic path $i_j = i^*(\chi_{j-1})$, and the p-step density is the finite product

$$q_p(x, z) \;=\; \prod_{j=1}^{p} \varphi_{\Sigma_{i_j}}\big(z_j - A_{i_j}^{(y)}\chi_{j-1} - b_{i_j}^{(y)}\big),$$

where $\varphi_\Sigma(w) := \det(\Sigma)^{-1/2} f_\varepsilon(\Sigma^{-1/2} w)$ is the density of $\Sigma^{1/2}\varepsilon_t$ (continuous and strictly positive by A1, A3; Gaussian under (1)), Σ_{i_j} is the covariance of the variance regime active at step j, and $A^{(y)}, b^{(y)}$ are the first n rows of the companion objects. Fix $C = \{x : \|x\| \leq R\}$ and a compact $K \subset \mathbb{R}^{np}$ of positive Lebesgue measure, and set $R_K := \sup_{z \in K} \|z\|$. Every intermediate state satisfies $\|\chi_{j-1}\| \leq$

$R' := \max(R, R_K)$, its coordinates being coordinates of x or of z, so every argument in the product is bounded, $\|z_j - A_{ij}^{(y)}\chi_{j-1} - b_{ij}^{(y)}\| \le R_K + \max_i(\|A_i\|R' + \|b_i\|) =: R''$, and

$$\delta_0 := \min_{1\le i\le k} \inf_{\|w\|\le R''} \varphi_{\Sigma_i}(w) > 0$$

by continuity and positivity of the finitely many φ_{Σ_i} (A1, A3). Therefore $q_p(x,z) \ge \delta_0^p =: \delta > 0$ for all $x \in C$ and $z \in K$, uniformly over the (state-determined) regime path, and $P^p(x,B) = \int_B q_p(x,z)\,dz \ge \delta\,\mathrm{Leb}(B \cap K)$ for all $x \in C$ and $B \in \mathcal{B}_{np}$: C is small with $m = p$ and minorizing measure $\nu_p(\cdot) = \delta\,\mathrm{Leb}(\cdot \cap K)$.

The general framework is Meyn and Tweedie (1993, Proposition 16.1.3): non-degenerate noise (**A1**, **A3**) with bounded coefficients (**A2**) ensures that the chain can be neither trapped nor periodic; relative to Chen and Chen (2000), the regime-specific constant volatility (**A3**) makes non-degeneracy uniform within each regime. □

Theorem A1 (Geometric ergodicity and fourth moments). *Under assumptions **A1**–**A4**, the Markov chain $\{X_t\}$ defined by (8) is geometrically ergodic: it has a unique invariant probability measure π, and there exist $R < \infty$ and $\rho \in (0,1)$ such that, with $g(x) := 1 + x'Px$ for the matrix P of **A4**,*

$$\|P^t(x,\cdot) - \pi\|_{TV} \;\le\; R\,g(x)\,\rho^t \qquad \text{for all } x \in \mathbb{R}^{np},\ t \ge 1.$$

Moreover $\int \|x\|^4\,\pi(dx) < \infty$; in particular the VAR process has finite fourth moments under stationarity.

Remark A.1 (Common Lyapunov function). **A4** is stronger than individual stability $\rho(A_i) < 1$ for all i, which is necessary but not sufficient for a common P: all mean regimes must contract the same energy metric $g(x) = 1 + x'Px$, which is what prevents instability from switching; for given parameter values it can be verified by LMI optimization (Liberzon, 2003, Chapter 2). A sufficient primitive condition is $\rho(A_i) < \overline{\lambda} < 1$ for all i and $\max_{i,j}\|A_i - A_j\| < \varepsilon$, with $\varepsilon > 0$ depending only on $\overline{\lambda}$, np and $\max_i\|A_i\|$ (through the conditioning of the Lyapunov matrix of any single regime), not on the sample size: regimes with sufficiently similar dynamics cannot diverge by switching. The condition is sufficient, not necessary: for the univariate TAR(1), Chan et al. (1985) characterize ergodicity exactly, and the characterization allows a locally explosive regime; no such characterization is available for the multivariate model, and the drift argument below requires the common Lyapunov matrix.

Remark A.2 (Variance regimes and stability). **A4** restricts only the k_μ mean companion matrices: the variance-regime matrices Σ_ℓ enter (8) only through the bounded noise loading $H_{\ell^*(X_t)}$ and, in the proof below, only through the additive drift constants.

Remark A.3 (Relation to the literature). The proof combines ν-irreducibility and aperiodicity from the everywhere-positive innovation density, a Foster–Lyapunov drift condition for g (Meyn and Tweedie, 1993, Ch. 15), and the equivalence of drift and geometric ergodicity with moment bounds for the invariant law (Tong, 1990; Chen and Chen, 2000); the fourth moments under arbitrary regime switching follow by lifting **A4** to the Kronecker-squared companion matrices.

Theorem A1 is proved by verifying the conditions of the following lemma.

Lemma 2. *Suppose that $(\mathbf{x}_t)$ is an aperiodic, ν-irreducible Markov chain, where ν is the Lebesgue measure. If there exists a small set A, a non-negative measurable function g and constants $\rho \in (0,1)$ and $b > 0$ such that:*

(i) *g is bounded away from zero and infinity on A;*

(ii) $\sup_{\mathbf{x}\in A} \mathbb{E}\{g(\mathbf{x}_t)|\mathbf{x}_{t-1} = \mathbf{x}\} < \infty$;

(iii) $\mathbb{E}\{g(\mathbf{x}_t)|\mathbf{x}_{t-1} = \mathbf{x}\} < \rho g(\mathbf{x}) - b, \mathbf{x} \in A^c$;

then $(\mathbf{x}_t)$ is geometrically ergodic and

$$\int_{\mathbb{R}^{np}} g(\mathbf{x})\pi(d\mathbf{x}) < \infty \tag{A-1}$$

where π is an invariant probability measure on $\mathbb{R}^{np}$ with respect to the Markov chain $(\mathbf{x}_t)$.

Proof of Lemma 2. The geometric ergodicity follows from the drift criterion of Meyn and Tweedie (1993, Theorem 15.0.1 and Ch. 15–16) as presented in Tong (1990, Appendix 1), and (A-1) follows from Feigin and Tweedie (1985).

Proof of Theorem A1. By Lemma 1 the chain is ν-irreducible and aperiodic and compact sets are small, so it remains to exhibit Lyapunov functions satisfying (i)–(iii) of Lemma 2: a quadratic one, which gives geometric ergodicity and the weighted total-variation bound, and a quartic one, which gives the fourth moment. Throughout, $i = i^*(x)$ and $\ell = \ell^*(x)$ are the regimes active at the state x, $\bar{A} := \max_i \|A_i\|$ and $\bar{b} := \max_i \|b_i\|$ (finite by **A2**), $\|H_\ell\|_{\text{op}} \le C_\Sigma^{1/2}$ (**A3**), and $E\|H_\ell\varepsilon_{t+1}\|^r \le C_\Sigma^{r/2} E\|\varepsilon_t\|^r < \infty$ for $r \le 4$ (**A1**), uniformly in ℓ; the variance regime enters the argument only through these constants.

Step 1: quadratic Lyapunov function. Let $P \succ 0$ satisfy $A_i' P A_i \preceq \lambda_i^2 P$ for all i, with $\lambda := \max_i \lambda_i < 1$ (**A4**), and set $g(x) := 1 + x'Px$. Then $g \ge 1$ is measurable and $c_1(1 + \|x\|^2) \le g(x) \le c_2(1 + \|x\|^2)$ with $c_1 := \min(1, \lambda_{\min}(P))$, $c_2 := \max(1, \lambda_{\max}(P))$, through which all drift and moment statements transfer between g and the Euclidean norm. From $X_{t+1} = A_i x + b_i + H_\ell \varepsilon_{t+1}$, $E\varepsilon_{t+1} = 0$ and $E\varepsilon_{t+1}\varepsilon_{t+1}' = I$ (**A1**),

$$\begin{aligned} E[g(X_{t+1}) \mid X_t = x] &= 1 + (A_i x + b_i)' P (A_i x + b_i) + \operatorname{tr}(H_\ell' P H_\ell) \\ &\le 1 + \lambda^2 x'Px + 2\bar{A}\|P\|\bar{b}\,\|x\| + \lambda_{\max}(P)\bar{b}^2 + C_1 \\ &= \lambda^2 g(x) + 2\bar{A}\|P\|\bar{b}\,\|x\| + C_1', \end{aligned}$$

where $\operatorname{tr}(H_\ell' P H_\ell) = \operatorname{tr}(P_{11}\Sigma_\ell) \le \lambda_{\max}(P)\, n\, C_\Sigma =: C_1$ by **A3**, uniformly in ℓ, and $C_1' := (1-\lambda^2) + \lambda_{\max}(P)\bar{b}^2 + C_1$.

Step 2: geometric ergodicity. Fix $\rho \in (\lambda^2, 1)$ and $b > 0$. Since $g(x) \ge 1 + \lambda_{\min}(P)\|x\|^2$ grows quadratically while the cross term in Step 1 grows linearly, there is $R < \infty$ with $(\rho - \lambda^2)\lambda_{\min}(P)R^2 \ge 2\bar{A}\|P\|\bar{b}R + C_1' + b$, so that $E[g(X_{t+1}) \mid X_t = x] \le \rho\, g(x) - b$ for $\|x\| > R$: condition (iii) of Lemma 2 on the complement of $A := \{x : \|x\| \le R\}$, which is small by Lemma 1. On A, $1 \le g(x) \le 1 + \lambda_{\max}(P)R^2$ and $\sup_{x\in A} E[g(X_{t+1}) \mid X_t = x] \le 1 + \lambda_{\max}(P)\big[(\bar{A}R + \bar{b})^2 + nC_\Sigma\big] < \infty$: conditions (i) and (ii). Lemma 2 yields geometric ergodicity and $\int g\, d\pi < \infty$, hence $\int \|x\|^2 \pi(dx) < \infty$. The drift inequality is condition (V4) of Meyn and Tweedie (1993,

Ch. 15) with $V = g$, so Theorem 15.0.1 there gives the g-weighted total-variation bound of the statement, with R, ρ depending only on the drift constants.

Step 3: quartic Lyapunov function. Repeat the drift argument on the lifted system $z := x \otimes x \in \mathbb{R}^{(np)^2}$ with $\tilde{A}_i := A_i \otimes A_i$. By the mixed-product property and because the Kronecker product of positive semidefinite matrices is positive semidefinite, $\tilde{A}_i'(P \otimes P)\tilde{A}_i = (A_i' P A_i) \otimes (A_i' P A_i) \preceq \lambda_i^4 (P \otimes P)$, so $W := P \otimes P \succ 0$ satisfies $\tilde{A}_i' W \tilde{A}_i \preceq \lambda^{(4)} W$ for all i, with $\lambda^{(4)} := \max_i \lambda_i^4 < 1$. Define

$$G(x) := 1 + z' W z, \qquad z = x \otimes x,$$

so that $1 + \lambda_{\min}(W)\|x\|^4 \le G(x) \le 1 + \lambda_{\max}(W)\|x\|^4$, since $\|z\| = \|x\|^2$. Let $y := A_i x + b_i$ and $\eta := H_\ell \varepsilon_{t+1}$, so that $X_{t+1} = y + \eta$ and

$$z_{t+1} := X_{t+1} \otimes X_{t+1} = \tilde{A}_i z + R_i(x) + \mathcal{N},$$
$$R_i(x) := (A_i x) \otimes b_i + b_i \otimes (A_i x) + b_i \otimes b_i, \qquad \mathcal{N} := y \otimes \eta + \eta \otimes y + \eta \otimes \eta.$$

Expand $z_{t+1}' W z_{t+1}$ and take conditional expectations given $X_t = x$. Terms linear in ε_{t+1} vanish (**A1**); the remaining terms are:

(a) the leading term, $z' \tilde{A}_i' W \tilde{A}_i z \le \lambda^{(4)} z' W z$ by the lifted inequality;

(b) the intercept terms, $2(\tilde{A}_i z)' W R_i(x) + R_i(x)' W R_i(x) \le c_1 \|z\|^{3/2} + c_2 \|z\|$, with c_1, c_2 depending on $\|W\|, \bar{A}, \bar{b}$ only, since $\|R_i(x)\| \le 2\bar{A}\bar{b}\|x\| + \bar{b}^2 \le C(1 + \|z\|^{1/2})$;

(c) the terms of second and third order in η (the quadratic forms in $y \otimes \eta$ and $\eta \otimes y$, and the cross terms of $\tilde{A}_i z + R_i(x)$, $y \otimes \eta$, $\eta \otimes y$ with $\eta \otimes \eta$), each bounded by $\|W\|$ times a product of $\|y\|$ or $\|\tilde{A}_i z + R_i(x)\|$ with $E\|\eta\|^2$ or $E\|\eta\|^3$; since $\|y\| \le \bar{A}\|z\|^{1/2} + \bar{b}$ and $\|\tilde{A}_i z + R_i(x)\| \le C(1 + \|z\|)$, their sum is at most $C_2(1 + \|z\|)$;

(d) the quartic noise term, $E[(\eta \otimes \eta)' W (\eta \otimes \eta)] \le \|W\| \, E\|\eta\|^4 \le \|W\| \, C_\Sigma^2 \, m_4 =: C_4$.

By Young's inequality, for any $\epsilon > 0$ there is $C_\epsilon < \infty$ with $c_1\|z\|^{3/2} + (c_2 + C_2)\|z\| \le \epsilon \, \lambda_{\min}(W)\|z\|^2 + C_\epsilon \le \epsilon \, z' W z + C_\epsilon$. Choosing ϵ so that $\tilde{\lambda}^2 := \lambda^{(4)} + \epsilon < 1$ and collecting (a)–(d),

$$E[G(X_{t+1}) \mid X_t = x] \;\le\; \tilde{\lambda}^2 \, G(x) + C, \qquad C := C_\epsilon + C_2 + C_4 + 1 - \tilde{\lambda}^2 < \infty.$$

Step 4: fourth moment. The display is the drift inequality of Step 1 with (g, λ^2) replaced by $(G, \tilde{\lambda}^2)$ and no linear cross term, and G dominates $\lambda_{\min}(W)\|x\|^4$, so Step 2 applies verbatim: for $\rho \in (\tilde{\lambda}^2, 1)$ and $b > 0$ choose R with $(\rho - \tilde{\lambda}^2)\lambda_{\min}(W) R^4 \ge C + b$, so that $E[G(X_{t+1}) \mid X_t = x] \le \rho \, G(x) - b$ for $\|x\| > R$, while on the small set $A := \{x : \|x\| \le R\}$, $1 \le G \le 1 + \lambda_{\max}(W) R^4$ and $\sup_{x \in A} E[G(X_{t+1}) \mid X_t = x] \le 1 + \lambda_{\max}(W)(\bar{A}R + \bar{b})^4 + C_2(1 + R^2) + C_4 < \infty$. Lemma 2 gives $\int G \, d\pi < \infty$, hence $\int \|x\|^4 \pi(dx) < \infty$ and $E\|y_t\|^4 < \infty$ under stationarity.

The contraction constants $\lambda, \lambda^{(4)}$ and the compact sets are determined by the mean companion matrices $\{A_i\}_{i=1}^{k_\mu}$ alone (**A4**); the variance regimes enter only through the additive constants C_1, C_2, C_4, bounded

uniformly in ℓ by **A3**, mirroring the structure of (1): stability is governed by the mean regimes, the variance regimes affect only the noise loading. □

A.1.2 Proof of Theorem 3(i)

Proof. Model (10) is the special case of (8) with $k_\mu = k_\sigma = k$, a common state and delay, and $\ell^*(\cdot) = i^*(\cdot)$, so Theorem A1 applies verbatim with **A4** imposed on the k common companion matrices. □

A.2 Auxiliary Lemmas

A.2.1 Lemma 3

In model (10) the density regularity that Chan (1993, Conditions 2–3) and Samia and Chan (2011, Assumption 3) must assume — bounded marginal and pairwise densities of the threshold variable, full support of the state — holds under A1–A3 and B1. Define $\sigma_* := \min_{1\le\ell\le k_\sigma} \lambda_{\min}(\Sigma_\ell^0) > 0$ (A3; finitely many regimes); this is a property of the true parameter, not an additional assumption[1].

Lemma 3. *Under the Gaussian innovation assumption in model (1) (and its special case (10)), A1–A4 and B1, with π the stationary law of X_t from Theorem A1:*

(i) The conditional law of $q_t = e_s' y_t$ given $\mathcal{F}_{t-1}$ has a density bounded by $C_* := (2\pi\sigma_*)^{-1/2}$, uniformly over histories.

(ii) The stationary marginal density f_π of q_{t-d} exists, is everywhere positive, and is bounded by C_*; and for every $t \neq s$ the pair (q_{t-d}, q_{s-d}) has a joint density bounded by C_*^2, uniformly over (t, s).

(iii) π has an everywhere-positive Lebesgue density on $\mathbb{R}^{np}$. Consequently, for every nonzero matrix $C \in \mathbb{R}^{n\times(np+1)}$ whose restriction to the non-constant coordinates of x_t is nonzero, and every event $E \in \sigma(X_{t-1})$ with $P(E) > 0$, we have $P(Cx_t \neq 0 \mid E) > 0$.

Proof

(i) Given $\mathcal{F}_{t-1}$, the active regime $i(t) = i^*(X_{t-1})$ is known and $q_t \mid \mathcal{F}_{t-1} \sim N\big(e_s' B_{i(t)} x_t,\ e_s' \Sigma_{i(t)} e_s\big)$ with variance $e_s'\Sigma_{i(t)}e_s \ge \sigma_*$, so its density is bounded by C_* uniformly over histories.

(ii) By the invariance equation, $f_\pi(u) = E_\pi\big[f_{q_t|\mathcal{F}_{t-1}}(u)\big] \le C_*$, and $f_\pi(u) > 0$ everywhere because each conditional Gaussian density is everywhere positive. For $s < t$, iterating expectations,

$$f_{(q_{s-d},\, q_{t-d})}(u, v) = E_\pi\Big[f_{q_{s-d}|\mathcal{F}_{s-d-1}}(u)\; E\big[f_{q_{t-d}|\mathcal{F}_{t-d-1}}(v) \,\big|\, \mathcal{F}_{s-d}\big]\Big] \le C_*^2.$$

(iii) By Lemma 1, $P^p(x, \cdot)$ has an everywhere-positive density $q_p(x, \cdot)$ for every x; hence $\pi(B) = \int P^p(x, B)\,\pi(dx) > 0$ for every Borel B of positive Lebesgue measure and, since $\pi = \pi P^p$, π has the everywhere-positive density $f_X(z) = \int q_p(x, z)\,\pi(dx)$. For the last claim, write $Cx_t = c_0 + C_1 X_{t-1}$: if $C_1 \neq 0$, $\{c_0 + C_1 X_{t-1} = 0\}$ is a proper affine subset of $\mathbb{R}^{np}$, hence π-null and null under any conditioning on positive-probability $\sigma(X_{t-1})$-events; if $C_1 = 0$ and $c_0 \neq 0$, the event $\{Cx_t \neq 0\}$ is sure. □

[1] Where statements uniform over the parameter space are needed, as in Lemma 7, the compactness of $\mathcal{S}$ in **B1** supplies the uniform counterpart.

Remark. Lemma 3(ii) supplies, without assumption, the uniformly bounded pairwise densities that Samia and Chan (2011) require in Assumption 3 and that drive the tightness bound in Step 3 below; Lemma 3(iii) is used in the identification proof of Appendix A.3 and in the remark following the mean clause of **B2**. In the decoupled model, parts (i)–(ii) apply verbatim to any threshold variable that is an element of y_t at some lag; for general measurable transformations of X_t the corresponding regularity is maintained through the density clauses of **B2** and **B8**.

A.2.2 Lemma 4: discharging the mixing condition

Assumption **B5** is the one condition of the B-block that restricts dependence rather than moments or densities, and the hardest to verify: geometric ergodicity delivers β-mixing but not ρ-mixing (Bradley, 2005). Samia and Chan (2011) verify their Assumption 7 by hand in a univariate example (condition (C7) of their online supplement), using two features of the scalar case – an envelope depending only on the threshold variable and future noise, and a stationary density bounded below on the band – neither of which survives in the multivariate model. Lemma 4 takes a different route: the proofs use only two consequences of ρ-mixing for functionals supported on the shrinking threshold band (the covariance bounds behind the Claim in Step 5 of the proof of Theorem 3(iii), and the dependence-removal ingredient (c) of Step 4 of the proof of Theorem 3(iv)), and for such functionals a *density bridge* substitutes for mixing: conditioning $d+1$ steps before the band time, the bounded one-step conditional density of Lemma 3(i) turns the functional's conditional expectation into a function of the state bounded by C_* times the band width, which the weighted total-variation bound of Theorem A1 controls at a geometric rate. The price is a strengthening of **B6** from second to $(2+\delta)$-th conditional moments of an extended envelope:

B6* There exist $\tau > 0$, $\delta > 0$ and $M < \infty$ such that, with $g(x) = 1 + x'Px$ the weight function of Theorem A1,

$$E\Big[\big(\Lambda_t + g(X_t)\big)^{2+\delta} \,\Big|\, q_{t-d} = v\Big] \le M \qquad \text{for a.e. } v \in [\gamma^0 - \tau, \gamma^0 + \tau].$$

The same bound is imposed around each threshold, mean and variance alike (conditionally on $q^{\mu}_{t-d_\mu} = v$ near $\gamma^{\mu,0}$ and on $q^{\sigma}_{t-d_\sigma} = v$ near $\gamma^{\sigma,0}$); both are conditional *moment* requirements of the same nature as **B6**.

Lemma 4. *Let Assumptions A1–A4, B1 and B6* hold, and let $R < \infty$, $\rho \in (0,1)$ be the constants of Theorem A1, so that $\|P^j(x,\cdot) - \pi\|_{TV} \le R\, g(x)\, \rho^j$ for all x and $j \ge 1$. Fix $\Delta \le \tau$ and write $\mathbb{I}_t(I) := \mathbb{I}\{q_{t-d} \in I\}$ for intervals $I \subseteq [\gamma^0 - \Delta, \gamma^0 + \Delta]$. Then:*

(i) (Covariance bounds.) There exist $H < \infty$ and a summable sequence $\{\bar\rho(j)\}_{j\ge1}$ with $\bar\rho(j) \le C\rho^{j\delta/(2+\delta)}$ such that, for every interval I as above, every pair $s < t$, and every pair of measurable functions $m_t = m(x_t, y_t)$, $m_s = m'(x_s, y_s)$ with $|m|, |m'| \le C_m \Lambda$,

$$\big|\operatorname{Cov}\big(m_t\, \mathbb{I}_t(I),\ m_s\, \mathbb{I}_s(I)\big)\big| \ \le\ C_m^2\, H\, \bar\rho(t-s)\, P_\pi\big(q_{t-d} \in I\big).$$

In particular, the variance bounds $\operatorname{var}\big(\sum_{t\le T} m_t \mathbb{I}_t(I)\big) \le TH' P_\pi(q \in I)$ required in the Claim of Step 5 of the proof of Theorem 3(iii) hold.

(ii) (Averaging bound.) Let $\varepsilon = 1/T$, let $I_\varepsilon := [\gamma^0, \gamma^0 + \kappa\varepsilon)$, let $J_t := \xi_t^+ \, \mathbb{I}_t(I_\varepsilon)$, and let F be bounded and measurable. Then for every $k \geq 1$, every $w \in \mathbb{R}$, and every sub-σ-field $\mathcal{G} \subseteq \mathcal{F}_t$, writing $h_k := F(w + J_{t+k}) - F(w)$,

$$E\left|E[h_k \mid \mathcal{G}] - E[h_k]\right| \;\leq\; 4\,\|F\|_\infty\, C_*\,\kappa\varepsilon\,\Big(\mathbb{I}\{k \leq d+1\} + R\,\rho^{\,k-d-1} E_\pi[g]\,\mathbb{I}\{k > d+1\}\Big),$$

so that, summed over a window of $m_\varepsilon = [T\tau_\varepsilon]$ terms and normalized by τ_ε as in Step 4 of the proof of Theorem 3(iv), the total is $O(\varepsilon/\tau_\varepsilon) = O(T^{-2/3})$ for the window choice $\tau_\varepsilon = T^{-1/3}$ – the analogue of display (A49) of Samia and Chan (2011), with the mixing coefficient replaced by the geometric total-variation rate.

Proof. **(The bridge.)** For any random variable of the form $\eta_t = m(x_t, y_t)\,\mathbb{I}_t(I)$ with $|m| \leq L$ define $b_\eta(x) := E[\eta_t \mid X_{t-d-1} = x]$. Conditioning first on q_{t-d},

$$b_\eta(x) = \int_I E\big[m(x_t, y_t) \,\big|\, X_{t-d-1} = x,\; q_{t-d} = v\big]\, f_{q_{t-d}|X_{t-d-1}=x}(v)\, dv, \qquad |b_\eta(x)| \leq L\,C_*\,|I|,$$

by Lemma 3(i). Thus b_η is *bounded*, and for any $j \geq 1$,

$$\big|P^j b_\eta(x) - \pi b_\eta\big| \leq \|b_\eta\|_\infty \,\min\{2,\; R\,g(x)\,\rho^j\} \leq L\,C_*\,|I|\,\min\{2,\; R\,g(x)\,\rho^j\}. \tag{L7.1}$$

(i). Write $j := t - s$ and $\phi_t := m_t \mathbb{I}_t(I)$. For $j \leq d+1$ the bound is immediate from Cauchy–Schwarz and B6*: $E[\phi_t^2] \leq C_m^2 \int_I E[\Lambda_t^2 \mid q = v] f_\pi(v)\, dv \leq C_m^2 M C_* |I|$, and similarly for ϕ_s, so $|\operatorname{Cov}| \leq C\,|I|$, which is of the required form since there are finitely many such lags. For $j > d+1$, truncate: $m_t = m_t^L + m_t^{>L}$ with $m_t^L := m_t \mathbb{I}\{\Lambda_t \leq L\}$. By the Markov property and $\mathcal{F}_s$-measurability of ϕ_s,

$$\operatorname{Cov}\big(m_t^L \mathbb{I}_t(I), \phi_s\big) = E\Big[\big(\phi_s - E\phi_s\big)\,\big(P^{\,j-d-1} b_L(X_s) - \pi b_L\big)\Big],$$

with b_L the bridge of $m_t^L \mathbb{I}_t(I)$. By (L7.1) and Cauchy–Schwarz with B6* ($E[\phi_s^2\, g(X_s)^2]^{1/2} \leq C_m E[(\Lambda_s + g(X_s))^4\, \mathbb{I}_s(I)]^{1/2} \leq C|I|^{1/2}$ when $\delta \geq 2$, and by Hölder with exponent $(2+\delta)/2$ in general),

$$\big|\operatorname{Cov}\big(m_t^L \mathbb{I}_t(I), \phi_s\big)\big| \leq C_m L\,C_*\,|I|\,R\,\rho^{\,j-d-1}\,E\big[\big(|\phi_s| + E|\phi_s|\big)\,\big(1 + g(X_s)\big)\big] \leq C\,L\,\rho^{\,j-d-1}\,|I|.$$

For the tail, by Cauchy–Schwarz, B6* and Markov's inequality,

$$\big|\operatorname{Cov}\big(m_t^{>L} \mathbb{I}_t(I), \phi_s\big)\big| \leq \Big(E\big[m_t^2\,\mathbb{I}\{\Lambda_t > L\}\,\mathbb{I}_t(I)\big]\Big)^{1/2}\big(E[\phi_s^2]\big)^{1/2} \leq C\,L^{-\delta/2}\,|I|.$$

Optimizing $L = \rho^{-2(j-d-1)/(2+\delta)}$ gives the stated $\bar\rho(j) \leq C\rho^{(j-d-1)\delta/(2+\delta)}$; summability yields the variance bound by the usual double sum. Finally $P_\pi(q \in I) \geq c\,|I|$ on the band (the density clause of **B2** and Lemma 3(ii), after shrinking Δ), so $|I|$ may be replaced by $P_\pi(q \in I)$ throughout.

(ii). Here $|h_k| \leq 2\|F\|_\infty\, \mathbb{I}_{t+k}(I_\varepsilon)$, so the bridge $b(x) := E[h_k \mid X_{t+k-d-1} = x]$ satisfies $\|b\|_\infty \leq 2\|F\|_\infty C_* \kappa\varepsilon$ with *no truncation*. For $k > d+1$, by the tower property, Jensen's inequality ($\mathcal{G} \subseteq \mathcal{F}_t$), the Markov property and (L7.1),

$$E\left|E[h_k \mid \mathcal{G}] - Eh_k\right| \leq E\left|P^{\,k-d-1} b(X_t) - \pi b\right| \leq 2\|F\|_\infty C_* \kappa\varepsilon\, R\,\rho^{\,k-d-1}\,E_\pi[g].$$

For $k \le d+1$, bound both terms separately: $E|E[h_k \mid \mathcal{G}]| \le E|h_k| \le 2\|F\|_\infty P_\pi(q \in I_\varepsilon) \le 2\|F\|_\infty C_* \kappa\varepsilon$, and likewise $|Eh_k|$. Summing over $k = 0, \dots, m_\varepsilon - 1$ and dividing by τ_ε gives $O\big(\kappa\varepsilon(d+1+\sum_k \rho^k)/\tau_\varepsilon\big) = O(\varepsilon/\tau_\varepsilon)$. □

Remark (scope). Part (i) supplies the variance bounds of the Claim in Step 5 of the proof of Theorem 3(iii) (with $m_t \in \{1, \Lambda_t, \xi_t^{\pm}\}$, all dominated by $C\Lambda_t$), and part (ii) ingredient (c) of Step 4 of the proof of Theorem 3(iv), where the test functions have compact support and the increments h_k are band-supported. Consequently, *Theorem 3(iii)–(iv) holds with Assumption B5 replaced by B6⋆*; the same argument discharges **B5** for Theorem 2 whenever both threshold variables have one-step conditional densities bounded uniformly over histories (automatic for elements of y at fixed lags, by Lemma 3(i)), with the bridge taken $d_\mu + 1$ (respectively $d_\sigma + 1$) steps before the band time; for general measurable transformations of the state, **B5** is retained. Lemma 4 does not assert that the localized process is ρ-mixing (Bradley, 2005), only that the two consequences used in the proofs follow from Theorem A1 under B6⋆.

A.3 Consistency

A.3.1 Proof of Theorem 3(ii)

The proof follows the standard strategy for extremum estimators (Newey and McFadden, 1994): uniform convergence of the sample criterion, uniqueness of the population maximizer, and the argmax theorem, using the notation of Section 2 and the ergodicity and moment results of Theorem 3(i) and Appendix A.1.

Step 0: population log-likelihood. Let $\ell(\theta) = \mathbb{E}_\pi[\ell_t(\theta)]$, where

$$\ell_t(\theta) = \sum_{i=1}^{k} \left[-\frac{1}{2} \ln\det \Sigma_i - \frac{1}{2} \varepsilon_{it}' \Sigma_i^{-1} \varepsilon_{it} \right] \mathbb{I}(q_{t-d} \in \mathcal{R}_i(\gamma))$$

is the time-t contribution. By the density clause of **B2**, $f_\pi(\gamma_i^0) > 0$ at all thresholds, so each regime has positive stationary probability $\pi_i := P_\pi(q_{t-d} \in R_i(\gamma^0)) > 0$, $i = 1, \dots, k$.

Step 1: uniform LLN.

Lemma 5. Uniform law of large numbers (ULLN) *Assume **A1–A4** and **B1**, and in addition:*

(M1) *The stationary process $\{y_t\}$ is geometrically β-mixing (this follows from geometric ergodicity, Appendix A.1).*

(M2) $\mathbb{E}_\pi\|y_t\|^4 < \infty$ *(this follows from Theorem 3(i))*

Then

$$\sup_{\theta \in \Theta} \big| L_T(\theta) - \ell(\theta) \big| \xrightarrow{a.s.} 0 \quad (T \to \infty). \tag{A-2}$$

Proof sketch For fixed θ, ergodicity gives $L_T(\theta) \to \ell(\theta)$ a.s.; compactness of Θ, the fourth-moment bound and the uniform positive-definiteness restriction on Σ_i yield an integrable envelope for $\ell_t(\theta)$; uniformity over (B, Σ) follows from stochastic equicontinuity, and over the threshold coordinate from the finite-grid approximation for threshold models together with a ULLN for dependent sequences (Andrews, 1988; Newey and McFadden, 1994).

Step 2: identification - unique maximizer.

Lemma 6. (Unique maximizer) *Under **A1–A4**, **B2–B4**, θ_0 uniquely maximizes $\ell(\theta)$:*

$$\theta_0 = \arg\max_{\theta\in\Theta} \ell(\theta) \tag{A-3}$$

Proof of Lemma 6.

(i) **Information inequality.** Let $f(y_t \mid \mathcal{F}_{t-1};\theta)$ be the conditional density implied by parameter θ. By Jensen's inequality (information inequality),

$$\ell(\theta) - \ell(\theta_0) = \mathbb{E}_\pi\left[\ln \frac{f(y_t \mid \mathcal{F}_{t-1};\theta)}{f(y_t \mid \mathcal{F}_{t-1};\theta_0)}\right] \le 0,$$

with equality iff $f(\cdot;\theta) = f(\cdot;\theta_0)$ almost surely.

(ii) **Parameter identification.** We now show that equality can hold only at $\theta = \theta_0$.

Case 1: $\gamma = \gamma^0$, $(B_i, \Sigma_i) \neq (B_i^0, \Sigma_i^0)$ for some regime i. On $\{q_{t-d} \in \mathcal{R}_i^0\}$, which has positive stationary probability by the density clause of **B2**, equality of the conditional densities requires $B_i x_t = B_i^0 x_t$ and $\Sigma_i = \Sigma_i^0$ for π-almost every state. If $B_i \neq B_i^0$, the set $\{(B_i - B_i^0)x_t = 0\}$ is contained in a lower-dimensional affine subspace, hence is Lebesgue-null and therefore π-null by Lemma 3(iii); if $B_i = B_i^0$ but $\Sigma_i \neq \Sigma_i^0$, the Gaussian laws differ everywhere. In either case the densities differ on a set of positive probability, so $\ell(\theta) < \ell(\theta_0)$.

Case 2: $\gamma \neq \gamma^0$. Superpose the two partitions: the candidate thresholds $\{\gamma_1, \ldots, \gamma_{k-1}\}$ and the true thresholds $\{\gamma_1^0, \ldots, \gamma_{k-1}^0\}$ jointly partition $\mathbb{R}$ into finitely many intervals, on each of which the true regime index $\tau(v)$ and the candidate regime index $c(v)$ are constant, and every such interval of positive length has positive probability (Lemma 3(iii)). If $\ell(\theta) = \ell(\theta_0)$, then on each positive-length interval I the candidate conditional law coincides with the true one for π-almost every state with $q_{t-d} \in I$, which by Case 1 forces the parameter identity $(B_{c(I)}, \Sigma_{c(I)}) = (B_{\tau(I)}^0, \Sigma_{\tau(I)}^0)$. Since $\gamma \neq \gamma^0$ and both are ordered $(k-1)$-vectors, some true threshold γ_j^0 is not a candidate threshold and lies in the interior of some candidate regime $\mathcal{R}_{c^*}(\gamma)$; for $\varepsilon > 0$ smaller than the distance from γ_j^0 to every other threshold of either partition, $(\gamma_j^0 - \varepsilon, \gamma_j^0)$ and $(\gamma_j^0, \gamma_j^0 + \varepsilon)$ are positive-length superposition intervals inside the same candidate regime c^* with true indices j and $j+1$, so $(B_{c^*}, \Sigma_{c^*}) = (B_j^0, \Sigma_j^0) = (B_{j+1}^0, \Sigma_{j+1}^0)$, contradicting **B3**.

Combining the two cases shows that θ_0 is the unique maximizer of $\ell(\theta)$. (Unlike Chan (1993), which assumes constant variance across regimes, regime-specific Σ_i provides additional identification power; see Remark 19 and Section 3.2.)

Step 3: consistency via argmax theorem. By Lemma 5, $L_T \to \ell$ uniformly on Θ; by Lemma 6, θ_0 is the unique maximizer of ℓ on the compact Θ (**B1**); Theorem 2.1 of Newey and McFadden (1994) gives $\hat{\theta}_T = \arg\max_{\theta\in\Theta} L_T(\theta) \to_{a.s.} \theta_0$. □

A.3.2 Proof of Theorem 1

The proof parallels that of Theorem 3(ii), now applied to the product parameter space $\Gamma^\mu \times \Gamma^\sigma$, and to the profiled criterion of the decoupled model.

Lemma T5-F (uniform validity of the feasible profile). *Write $L_T^{\mathrm{F}}(\gamma^\mu, \gamma^\sigma)$ for the one-step feasible criterion of the implementation Remark of Section 2.2 (the object denoted L_T^{p} there) reserving L_T^{p} in this appendix for the exact profile $\sup_\vartheta L_T(\gamma^\mu, \gamma^\sigma, \vartheta)$, with population counterparts $\tilde{\ell}$ (the population criterion evaluated at the population one-step maps) and ℓ^{p}. Assume **A1–A3**, **A4**, **B1–B4**, and:*

B9 *(Uniform candidate feasibility.)* $\inf_{\gamma^\mu \in \Gamma^\mu} \min_{1 \le i \le k_\mu} P_\pi\big(q^\mu_{t-d_\mu} \in \mathcal{R}_i^\mu(\gamma^\mu)\big) > 0$ *and* $\inf_{\gamma^\sigma \in \Gamma^\sigma} \min_{1 \le \ell \le k_\sigma} P_\pi\big(q^\sigma_{t-d_\sigma} \in \mathcal{R}_\ell^\sigma(\gamma^\sigma)\big) > 0$.

Then

(i) $\sup_{(\gamma^\mu, \gamma^\sigma) \in \Gamma^\mu \times \Gamma^\sigma} \big| L_T^{\mathrm{F}} - \tilde{\ell} \big| \to 0$ *a.s.;*

(ii) $\tilde{\ell} \le \ell^{\mathrm{p}}$ *on* $\Gamma^\mu \times \Gamma^\sigma$*, provided the parameter blocks $\mathcal{B}, \mathcal{S}$ of Θ contain the population pipeline values $\{\widetilde{B}_i(\gamma^\mu, \gamma^\sigma), \widetilde{\Sigma}_\ell(\gamma^\mu, \gamma^\sigma)\}$; a finite enlargement of the bounds in **B1** always suffices, since these values form a bounded family, being continuous on the compact $\Gamma^\mu \times \Gamma^\sigma$; and*

(iii) $\tilde{\ell}(\gamma^{\mu,0}, \gamma^{\sigma,0}) = \ell^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0}) = \ell(\theta_0)$.

Remark. B9 is discussed in Remark 4 (Section 2.3); its content is not repeated here.

Proof. (i) Every ingredient of the feasible pipeline is a function of finitely many cell sample moments $T^{-1}\sum_t g(x_t, y_t)\, \mathbb{I}\{q^\mu_{t-d_\mu} \le a\}\mathbb{I}\{q^\sigma_{t-d_\sigma} \le b\}$, with g one of finitely many quadratic maps. For each g the family indexed by (a, b) is monotone in each index with integrable envelope (Theorem 3), and its pointwise ergodic limits are continuous in (a, b) because the states have absolutely continuous laws (Lemma 3); uniform a.s. convergence over the compact $\Gamma^\mu \times \Gamma^\sigma$ follows by the monotonicity argument underlying Lemma 5. By **B9**, the population mean- and variance-regime probabilities are bounded below by some $\underline{\pi} > 0$ uniformly over candidates. The population mean-regime second-moment matrices are uniformly nonsingular: for $a \neq 0$, $a' E_\pi[x_t x_t' \mathbb{I}\{q^\mu \in \mathcal{R}_i^\mu(\gamma^\mu)\}]a = E_\pi[(a'x_t)^2 \mathbb{I}\{\cdot\}]$ can vanish only if $a'x_t = 0$ a.s. on an event of probability at least $\underline{\pi}$, impossible because the stationary law of X_t has an everywhere-positive density (Lemma 3(iii)) and x_t contains the intercept; $\gamma^\mu \mapsto \lambda_{\min}\big(E_\pi[x_t x_t' \mathbb{I}\{\cdot\}]\big)$ is continuous by dominated convergence (atomless state laws), hence bounded below on the compact Γ^μ. The population pooled residual covariances dominate the innovation covariances, $\tilde{\Sigma}_\ell(\gamma^\mu, \gamma^\sigma) \succeq \min_{\ell'} \lambda_{\min}(\Sigma^0_{\ell'})\, I_n \succ 0$, uniformly. On this domain (regime probabilities $\ge \underline{\pi}$, moment matrices with eigenvalues in a fixed compact of $(0, \infty)$) the pipeline maps (cell OLS, pooled covariance, one-step GLS, concentrated criterion) are jointly Lipschitz in the cell moments, so the sample pipeline is, almost surely for all large T, well defined uniformly over $\Gamma^\mu \times \Gamma^\sigma$ (sample regime shares $\ge \underline{\pi}/2$, sample moment matrices within the compact eigenvalue domain), and uniform convergence is preserved under the composition. (ii) The plug-in evaluates ℓ at the population pipeline parameters, the profile at the supremum over $\mathcal{B} \times \mathcal{S}$; the inequality follows whenever the former lie in the latter, which is the stated qualification; the pipeline family is bounded because, by **B9** and the uniform moment bounds of part (i), $(\gamma^\mu, \gamma^\sigma) \mapsto (\widetilde{B}_i, \widetilde{\Sigma}_\ell)$

is continuous on the compact threshold space (dominated convergence, atomless state laws). (iii) At the true thresholds the cells are correctly specified, so the population within-mean-regime OLS estimand is B_i^0 regardless of the variance-regime heteroskedasticity, the pooled population residual covariance is Σ_ℓ^0, the population GLS estimand is again B_i^0, and the criterion value is $\ell(\theta_0)$. □

Step 1. ULLN. It is convenient to work first with the full Gaussian criterion (2). For $\vartheta := (B, \Sigma) \in \mathcal{B} \times \mathcal{S}$ and $\theta := (\gamma^\mu, \gamma^\sigma, \vartheta) \in \Theta$, the per-observation contribution is

$$\ell_t(\theta) = \sum_{i=1}^{k_\mu} \sum_{\ell=1}^{k_\sigma} \Big(-\tfrac{1}{2} \log\det \Sigma_\ell - \tfrac{1}{2}\, \varepsilon_{it}(B_i)' \Sigma_\ell^{-1} \varepsilon_{it}(B_i) \Big) \mathbb{I}\{q^\mu_{t-d_\mu} \in \mathcal{R}_i^\mu(\gamma^\mu)\}\, \mathbb{I}\{q^\sigma_{t-d_\sigma} \in \mathcal{R}_\ell^\sigma(\gamma^\sigma)\},$$

with $\varepsilon_{it}(B_i) = y_t - (x_t' \otimes I_n)\,\mathrm{vec}(B_i)$, $L_T(\theta) = T^{-1} \sum_{t=1}^T \ell_t(\theta)$ and $\ell(\theta) = E_\pi[\ell_t(\theta)]$. We show

$$\sup_{\theta \in \Theta} \big| L_T(\theta) - \ell(\theta) \big| \xrightarrow{a.s.} 0. \tag{A.23}$$

By **B1**, $\mathcal{B} \times \mathcal{S}$ is compact and $0 < \underline{\sigma} \le \lambda_{\min}(\Sigma_\ell) \le \lambda_{\max}(\Sigma_\ell) \le \bar{\sigma} < \infty$ uniformly over $\Sigma_\ell \in \mathcal{S}_\ell$, so $|\log\det \Sigma_\ell|$ and $\|\Sigma_\ell^{-1}\|$ are uniformly bounded and $\sup_{B_i \in \mathcal{B}_i} \|\varepsilon_{it}(B_i)\| \le C(1 + \|y_t\| + \|x_t\|)$; hence

$$\sup_{\theta \in \Theta} |\ell_t(\theta)| \le M_t := C\big(1 + \|y_t\|^2 + \|x_t\|^2\big), \qquad E_\pi M_t < \infty \tag{A.24}$$

by Theorem A1: $\{M_t\}$ is an integrable envelope. For the threshold coordinates, let $\Delta^\mu(\gamma^\mu, \tilde{\gamma}^\mu) := \bigcup_{j=1}^{k_\mu - 1} [\gamma_j^\mu \wedge \tilde{\gamma}_j^\mu,\ \gamma_j^\mu \vee \tilde{\gamma}_j^\mu)$, and $\Delta^\sigma(\gamma^\sigma, \tilde{\gamma}^\sigma)$ analogously. Two parameter points sharing the same ϑ induce different cell assignments only when $q^\mu_{t-d_\mu} \in \Delta^\mu$ or $q^\sigma_{t-d_\sigma} \in \Delta^\sigma$, so

$$|\ell_t(\theta) - \ell_t(\tilde{\theta})| \le 2M_t \Big(\mathbb{I}\{q^\mu_{t-d_\mu} \in \Delta^\mu(\gamma^\mu, \tilde{\gamma}^\mu)\} + \mathbb{I}\{q^\sigma_{t-d_\sigma} \in \Delta^\sigma(\gamma^\sigma, \tilde{\gamma}^\sigma)\} \Big),$$

and since the stationary densities of $q^\mu_{t-d_\mu}$ and $q^\sigma_{t-d_\sigma}$ are bounded on the compact threshold spaces (the density clauses of **B2**), $P_\pi(q^\mu_{t-d_\mu} \in \Delta^\mu) \le C_\mu \|\gamma^\mu - \tilde{\gamma}^\mu\|$ and $P_\pi(q^\sigma_{t-d_\sigma} \in \Delta^\sigma) \le C_\sigma \|\gamma^\sigma - \tilde{\gamma}^\sigma\|$. With (A.24) this gives stochastic equicontinuity in the threshold coordinates; uniformity in (B, Σ) follows from compactness and continuity of $\ell_t(\theta)$ with the same envelope. Since $\{X_t\}$ is geometrically ergodic (Theorem A1), hence geometrically β-mixing, the ULLN for dependent processes applies (Andrews, 1988; Newey and McFadden, 1994), proving (A.23). Finally, with the exact profile criteria $L_T^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) := \sup_{\vartheta \in \mathcal{B} \times \mathcal{S}} L_T(\gamma^\mu, \gamma^\sigma, \vartheta)$ and $\ell^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) := \sup_{\vartheta \in \mathcal{B} \times \mathcal{S}} \ell(\gamma^\mu, \gamma^\sigma, \vartheta)$,

$$\sup_{(\gamma^\mu, \gamma^\sigma) \in \Gamma^\mu \times \Gamma^\sigma} \big| L_T^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) - \ell^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) \big| \le \sup_{\theta \in \Theta} \big| L_T(\theta) - \ell(\theta) \big| \xrightarrow{a.s.} 0, \tag{A.27}$$

the ULLN for the profiled criterion.

Step 2. Unique maximizer. We show that θ_0 uniquely maximizes $\ell(\theta) := E_\pi[\ell_t(\theta)]$ on Θ and then pass to the profiled criterion, following the two-case superposition pattern of Lemma 6 in each threshold dimension.

(i) **Information inequality.** For any $\theta \in \Theta$, letting $f(y_t \mid \mathcal{F}_{t-1}; \theta)$ denote the conditional Gaussian

density implied by θ,

$$\ell(\theta) - \ell(\theta_0) = \mathbb{E}_\pi\Big[\ln \frac{f(y_t \mid \mathcal{F}_{t-1}; \theta)}{f(y_t \mid \mathcal{F}_{t-1}; \theta_0)}\Big] \leq 0,$$

with equality if and only if $f(\cdot\,; \theta) = f(\cdot\,; \theta_0)$ almost surely under π.

(ii) **Any $\theta \neq \theta_0$ produces a density mismatch on a set of positive stationary probability.** If $\gamma^\mu \neq \gamma^{\mu,0}$, superpose the candidate and true mean partitions as in Case 2 of Lemma 6: some true mean threshold is interior to a candidate mean regime, so a.s. equality of the conditional densities would force a single candidate B_i to satisfy $B_i x_t = B_j^0 x_t$ and $B_i x_t = B_{j'}^0 x_t$ for distinct true regimes $j \neq j'$, each on a set of positive stationary probability (the density clause of **B2**); hence $(B_j^0 - B_{j'}^0)x_t = 0$ on a set of positive probability, impossible by **B3** and Lemma 3(iii) (a nonzero linear functional of x_t vanishes only on a null set, the knife-edge intercept/threshold-coordinate case being excluded by the mean clause of **B2**). If $\gamma^\sigma \neq \gamma^{\sigma,0}$ (with γ^μ arbitrary), the same superposition on the variance side forces a single candidate Σ_ℓ to equal two distinct true matrices $\Sigma_m^0 \neq \Sigma_{m'}^0$ (**B3**), impossible and requiring no condition on x_t, since covariances are matched matrix-to-matrix. If both thresholds are correct but $(B_i, \Sigma_\ell) \neq (B_i^0, \Sigma_\ell^0)$ on some cell of positive probability (the cell clause of **B2**), the conditional Gaussian laws differ on that cell by Lemma 3(iii) (mean block) or directly (covariance block). In all cases $\ell(\theta) < \ell(\theta_0)$.

(iii) **Moving to the profile.** $\ell(\gamma^\mu, \gamma^\sigma, \vartheta)$ is continuous in $\vartheta = (B, \Sigma)$ on the compact $\mathcal{B} \times \mathcal{S}$ (dominated convergence with the envelope (A.24)), so the supremum defining $\ell^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) = \sup_\vartheta \ell(\gamma^\mu, \gamma^\sigma, \vartheta)$ is attained. For $(\gamma^\mu, \gamma^\sigma) \neq (\gamma^{\mu,0}, \gamma^{\sigma,0})$, every attaining ϑ yields a parameter $\theta \neq \theta_0$, so by (i)–(ii), $\ell^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) < \ell(\theta_0) = \ell^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0})$. Hence $(\gamma^{\mu,0}, \gamma^{\sigma,0})$ is the unique maximizer of ℓ^{p} on $\Gamma^\mu \times \Gamma^\sigma$.

Step 3. Consistency via the argmax theorem. By (A.27) the profiled criterion converges uniformly, and by Step 2 $(\gamma^{\mu,0}, \gamma^{\sigma,0})$ is the unique maximizer of ℓ^{p} on the compact $\Gamma^\mu \times \Gamma^\sigma$; Theorem 2.1 of Newey and McFadden (1994) gives $(\widehat{\gamma}^\mu, \widehat{\gamma}^\sigma) = \arg\max_{\Gamma^\mu \times \Gamma^\sigma} L_T^{\mathrm{p}} \xrightarrow{\text{a.s.}} (\gamma^{\mu,0}, \gamma^{\sigma,0})$.

Step 4: the feasible estimator. Steps 1–3 establish consistency of the exact-profile maximizer; the estimator of Section 2.2 maximizes the feasible criterion L_T^{F}, and is denoted $(\widehat{\gamma}_{\mathrm{F}}^\mu, \widehat{\gamma}_{\mathrm{F}}^\sigma)$ within this step. Fix $\delta > 0$. By Step 2 and the continuity of ℓ^{p} on the compact $\Gamma^\mu \times \Gamma^\sigma$ (dominated convergence, atomless state laws), there is $c = c(\delta) > 0$ with $\ell^{\mathrm{p}}(\gamma^\mu, \gamma^\sigma) \leq \ell^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0}) - c$ whenever $\|(\gamma^\mu, \gamma^\sigma) - (\gamma^{\mu,0}, \gamma^{\sigma,0})\| \geq \delta$. Almost surely, for all large T, Lemma T5-F(i) gives

$$\tilde{\ell}(\widehat{\gamma}_{\mathrm{F}}^\mu, \widehat{\gamma}_{\mathrm{F}}^\sigma) \;\geq\; L_T^{\mathrm{F}}(\widehat{\gamma}_{\mathrm{F}}^\mu, \widehat{\gamma}_{\mathrm{F}}^\sigma) - c/3 \;\geq\; L_T^{\mathrm{F}}(\gamma^{\mu,0}, \gamma^{\sigma,0}) - c/3 \;\geq\; \tilde{\ell}(\gamma^{\mu,0}, \gamma^{\sigma,0}) - 2c/3 \;=\; \ell^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0}) - 2c/3,$$

by the definition of the feasible maximizer (second inequality) and Lemma T5-F(iii) (equality). Since $\tilde{\ell} \leq \ell^{\mathrm{p}}$ everywhere (Lemma T5-F(ii), with the bounds of **B1** enlarged if necessary), the feasible maximizer satisfies $\ell^{\mathrm{p}}(\widehat{\gamma}_{\mathrm{F}}^\mu, \widehat{\gamma}_{\mathrm{F}}^\sigma) \geq \ell^{\mathrm{p}}(\gamma^{\mu,0}, \gamma^{\sigma,0}) - 2c/3$ and therefore lies within δ of the truth for all large T; as δ was arbitrary, $(\widehat{\gamma}_{\mathrm{F}}^\mu, \widehat{\gamma}_{\mathrm{F}}^\sigma) \to (\gamma^{\mu,0}, \gamma^{\sigma,0})$ a.s. This is the estimator to which Theorem 1 refers. Consistency of the regime parameters follows as in Steps 1–3, evaluating the (uniformly convergent) feasible parameter maps at the consistent thresholds. □

A.3.3 Proof of Proposition 1

Throughout, $q^\mu \equiv q^\sigma =: q$ with a common delay, and the data follow (10) with k regimes, true threshold vector γ^0, and $B_i^0 \neq B_{i+1}^0$, $\Sigma_i^0 \neq \Sigma_{i+1}^0$ for all i. With a shared state, a candidate pair $(\gamma^\mu, \gamma^\sigma)$ generates the superposed partition of $\mathbb{R}$ by both threshold vectors; the feasible joint cells are the cells of that partition (a staircase), and the cell clause of **B2** is imposed on those cells only, as in Remark 9. **B9** is unaffected: it restricts only the marginal probabilities $P_\pi(q \in \mathcal{R}_i^\mu(\gamma^\mu))$ and $P_\pi(q \in \mathcal{R}_\ell^\sigma(\gamma^\sigma))$, bounded below uniformly over the compact threshold spaces whenever q has a positive density, and the joint cells that the staircase leaves empty (for instance $\{q \le \gamma^\mu\} \cap \{q > \gamma^\sigma\}$ when $\gamma^\mu \le \gamma^\sigma$) are harmless for the feasible pipeline, which aggregates within a mean (respectively variance) regime (Remark 4).

Step 1 (consistency). The uniform convergence of Lemma T5-F(i) and the population inequality (ii) hold verbatim: nothing in their proofs uses the distinctness of the two state variables. For (iii) and the identification step, note that the true pair (γ^0, γ^0) lies on the diagonal of the candidate space, where the two partitions coincide and the population pipeline returns (B_i^0, Σ_i^0). Uniqueness of the population maximizer follows from the superposition argument of Lemma 6 applied to the common refinement of the candidate pair and γ^0: if $\gamma^\mu \neq \gamma^0$, some true threshold is interior to a candidate mean regime and one candidate coefficient matrix must equal two distinct true ones; if $\gamma^\sigma \neq \gamma^0$, some true threshold is interior to a candidate variance regime and one candidate covariance must equal two distinct true ones; either contradicts **B3**. The squeeze of Step 4 in the proof of Theorem 1 then gives $(\hat\gamma^\mu, \hat\gamma^\sigma) \to (\gamma^0, \gamma^0)$ a.s.

Step 2 (rates). Localize both candidates at scale $1/T$ around γ^0: $(\gamma^\mu, \gamma^\sigma) = (\gamma^0 + h_1/T, \gamma^0 + h_2/T)$. A boundary observation at position v (in $1/T$ units from γ^0) is reassigned in the mean only, in the variance only, or in both, according to the position of v relative to h_1 and h_2; its expected loss is, respectively, the mean-only divergence $\frac{1}{2}E[\kappa \mid \cdot] > 0$ (the mean clause of **B2** with **B7**), the variance-only divergence $\frac{1}{2}\sum_j(\lambda_j - \ln\lambda_j - 1) > 0$ (**B3**), or the joint divergence (J3) > 0. All three being strictly positive, the uniform negative drift of the local criterion over the box $\{(h_1, h_2)\}$ holds exactly as in Step 5 of the proof of Theorem 3(iii), with the single band counting of that proof in place of the two disjoint bands of Theorem 2(i); hence $T(\hat\gamma^\mu - \gamma^0) = O_p(1)$ and $T(\hat\gamma^\sigma - \gamma^0) = O_p(1)$. The number of misclassified observations is therefore $O_p(1)$, and the feasible-GLS regime-parameter estimates differ from the joint estimator's by $o_p(T^{-1/2})$: this is (ii).

Step 3 (selection). Both maximized criteria are evaluated within $O_p(1/T)$ of (γ^0, γ^0) by Steps 1–2, so $T(\hat L_T^{\mathrm{F}} - \hat L_T^{\mathrm{J}})$ is the difference between the maximum of the local criterion over the box and its maximum over the diagonal $h_1 = h_2$; both are finite sums of the $O_p(1)$ boundary losses, hence the difference is $O_p(1)$. Under a shared-state truth with distinct thresholds $\gamma^{\mu,0} \neq \gamma^{\sigma,0}$, the restricted estimator misclassifies a positive fraction of the sample whatever its threshold, so $\hat L_T^{\mathrm{J}}$ is bounded away from $\hat L_T^{\mathrm{F}}$ by the identification gap of Theorem 1 and the difference is of order T. Consistency of the rule "select the restriction if $T(\hat L_T^{\mathrm{F}} - \hat L_T^{\mathrm{J}}) < c_T$" for $c_T \to \infty$, $c_T = o(T)$, follows; this is the threshold analogue of the break-number selection argument of Bai and Perron (1998). □

A.4 Rates and Limit Laws: the Single-threshold Case

A.4.1 Lemma 7

Lemma 7 is the analogue, for our two models, of Lemmas 5 and 6 of Samia and Chan (2011) (and of displays (4.11)–(4.12) in Chan, 1993): part (i) says that the fixed-threshold estimators are insensitive, at the $o_p(T^{-1/2})$ scale, to $O(1/T)$ perturbations of the threshold; part (ii) that the normalized profile-likelihood increments may be computed as if the regime parameters were known. Part (i) is the converse direction of Step 6 of Appendix A.4 (there: threshold error does not disturb the parameter CLT; here: parameter estimation does not disturb the threshold local process); both parts feed the proofs of Theorems 3(iv) and 2(ii)–(iii).

For the joint model, write $\varphi := \big(\mathrm{vec}(B_1)', \mathrm{vec}(B_2)', \mathrm{vech}(\Sigma_1)', \mathrm{vech}(\Sigma_2)'\big)'$, let $\hat\varphi(\gamma)$ collect the fixed-γ maximum likelihood estimators, and let $l(\varphi,\gamma) := T\,L_T(\varphi,\gamma)$ denote the unnormalized log-likelihood, so that $T\,L_T^{\mathrm{p}}(\gamma) = l(\hat\varphi(\gamma),\gamma)$. For the decoupled model, $\hat\varphi(\gamma^\mu,\gamma^\sigma)$ collects the feasible profile-GLS estimators of Section 2.2 and the statements hold with the interval $|\gamma-\gamma^0| \le K/T$ replaced by the box $\|(\gamma^\mu,\gamma^\sigma) - (\gamma^{\mu,0},\gamma^{\sigma,0})\|_\infty \le K/T$.

Lemma 7. *Let the assumptions of Theorem 3(iv) (respectively Theorem 2(ii)–(iii)) hold. Then for every $K>0$:*

(i) $\sup_{|\gamma-\gamma^0|\le K/T} \big\|\hat\varphi(\gamma) - \hat\varphi(\gamma^0)\big\| = O_p\big(T^{-1}\big) = o_p\big(T^{-1/2}\big)$;

(ii) with $\mathcal{L}_T(\kappa)$ the normalized profile increment of Theorem 3(iv) and $\mathcal{L}_T^0(\kappa) := l(\varphi_0,\gamma^0+\kappa/T) - l(\varphi_0,\gamma^0)$ the increment at the true regime parameters,

$$\sup_{|\kappa|\le K} \big|\mathcal{L}_T(\kappa) - \mathcal{L}_T^0(\kappa)\big| = o_p(1).$$

The decoupled analogues hold with (κ) replaced by (h_μ,h_σ) ranging over $[-K,K]^2$.

Proof.

(i) Consider the joint model. The fixed-γ Gaussian MLE is regime-by-regime OLS plus the residual covariance. With $\mathbb{I}_{i,\gamma}(t) := \mathbb{I}\{q_{t-d} \in \mathcal{R}_i(\gamma)\}$,

$$\hat S_i(\gamma) := \sum_t x_t x_t'\,\mathbb{I}_{i,\gamma}(t), \quad \hat h_i(\gamma) := \sum_t x_t y_t'\,\mathbb{I}_{i,\gamma}(t), \quad \hat B_i(\gamma)' = \hat S_i(\gamma)^{-1}\hat h_i(\gamma),$$
$$\hat\Sigma_i(\gamma) = T_i(\gamma)^{-1}\sum_t \big(y_t - \hat B_i(\gamma)x_t\big)\big(y_t - \hat B_i(\gamma)x_t\big)'\,\mathbb{I}_{i,\gamma}(t), \quad T_i(\gamma) := \sum_t \mathbb{I}_{i,\gamma}(t).$$

Over $|\gamma-\gamma^0| \le K/T$ the indicators $\mathbb{I}_{i,\gamma}(t)$ and $\mathbb{I}_{i,\gamma^0}(t)$ differ only for observations in the band $\mathcal{B}_T := \big[\gamma^0 - K/T,\ \gamma^0 + K/T\big)$. By Lemma 3(ii), $P_\pi(q_{t-d} \in \mathcal{B}_T) \le C_*|\mathcal{B}_T| = 2KC_*/T$, so $N_T := \#\{t : q_{t-d} \in \mathcal{B}_T\}$ has $E\,N_T \le 2KC_*$ and is $O_p(1)$ by Markov's inequality; by B6 each band observation contributes moment terms $\|x_t x_t'\|, \|x_t y_t'\| \le \Lambda_t$ with $E\big[\Lambda_t\,\mathbb{I}\{q_{t-d} \in \mathcal{B}_T\}\big] \le M^{1/2}C_*\,|\mathcal{B}_T|\,T \le 2KC_*M^{1/2}$. Since the indicator differences are monotone in γ on each side of γ^0, the supremum of the moment perturbations over the interval is dominated by the total over the band: $\sup_{|\gamma-\gamma^0|\le K/T}\|\hat S_i(\gamma) - \hat S_i(\gamma^0)\| = O_p(1)$,

$\sup_{|\gamma-\gamma^0|\le K/T} \|\hat{h}_i(\gamma) - \hat{h}_i(\gamma^0)\| = O_p(1)$, and $|T_i(\gamma) - T_i(\gamma^0)| \le N_T = O_p(1)$. By the ergodic theorem, $T^{-1}\hat{S}_i(\gamma^0) \to_{a.s.} S_i = E_\pi[x_t x_t' \mathbb{1}\{q \in \mathcal{R}_i^0\}]$, positive definite because $\pi_i > 0$ (the density clauses of **B2**) and x_t is not supported on a proper affine subspace conditionally on $\{q_{t-d} \in \mathcal{R}_i^0\}$ (Lemma 3(iii)). On the event $\{\lambda_{\min}(T^{-1}\hat{S}_i(\gamma^0)) \ge \lambda_{\min}(S_i)/2\}$, of probability tending to one, $(\hat{S}, \hat{h}) \mapsto \hat{S}^{-1}\hat{h}$ is Lipschitz in a neighborhood, and the $O_p(1)$ unnormalized perturbations give $\sup_{|\gamma-\gamma^0|\le K/T} \|\hat{B}_i(\gamma) - \hat{B}_i(\gamma^0)\| = O_p(T^{-1})$.

The same argument applied to the residual cross-product sums (whose perturbation combines $O_p(1)$ band terms with the $O_p(T^{-1})$ coefficient perturbation multiplied through $O_p(T)$ sums, an $O_p(1)$ total) and to $T_i(\gamma)$ gives $\sup \|\hat{\Sigma}_i(\gamma) - \hat{\Sigma}_i(\gamma^0)\| = O_p(T^{-1})$.

For the decoupled model, the feasible profile-GLS estimator of Section 2.2 is the composition of three maps of the cellwise moments $\hat{S}_{i\ell}, \hat{h}_{i\ell}, t_{i\ell}$ (preliminary OLS, pooled preliminary covariance, feasible GLS), each smooth with nonsingular Jacobian limits ($E_\pi[x_t x_t' \mathbb{1}_{i\ell}] \succ 0$ by the cell clause of **B2** and Lemma 3(iii); $\Sigma_\ell^0 \succ 0$ by B1). An $O(1/T)$ move of either threshold perturbs each unnormalized cell moment by $O_p(1)$, by the band-counting argument applied to the μ- and σ-bands separately (B6 and Lemma 3(ii)/B8); chaining the Lipschitz maps gives the same $O_p(T^{-1})$ bound uniformly over the box.

(ii) Since $\hat{\varphi}(\gamma)$ maximizes $l(\cdot, \gamma)$, the profiling error $l(\hat{\varphi}(\gamma), \gamma) - l(\varphi_0, \gamma)$ has no linear term and is purely quadratic in $a_\gamma := \hat{\varphi}(\gamma) - \varphi_0$. Each such quadratic form is $O_p(1)$, but the lemma concerns their *difference* across γ_κ and γ^0, where the leading terms cancel and three channels remain: (a) the movement of the estimate, $O_p(T^{-1})$ by part (i); (b) the Hessian change from reallocating $O_p(1)$ band observations; (c) the Hessian change along the moved evaluation path. Fix κ, write $\gamma_\kappa := \gamma^0 + \kappa/T$ and $H(\varphi, \gamma) := -\nabla_\varphi^2 l(\varphi, \gamma)$. For T large, $\hat{\varphi}(\gamma)$ lies in the interior of the compact $\mathcal{B} \times \mathcal{S}$ with probability approaching one (B1, B4, Theorem 3(ii)), so $\nabla_\varphi l(\hat{\varphi}(\gamma), \gamma) = 0$ and a second-order Taylor expansion with integral remainder gives

$$l\big(\hat{\varphi}(\gamma), \gamma\big) - l\big(\varphi_0, \gamma\big) = \tfrac{1}{2}\, a_\gamma'\, \bar{H}_\gamma\, a_\gamma, \qquad \bar{H}_\gamma := \int_0^1 2(1-u)\, H\big(\varphi_0 + u\, a_\gamma,\ \gamma\big)\, du.$$

Hence

$$\mathcal{L}_T(\kappa) - \mathcal{L}_T^0(\kappa) = \tfrac{1}{2}\, a_{\gamma_\kappa}' \bar{H}_{\gamma_\kappa} a_{\gamma_\kappa} - \tfrac{1}{2}\, a_{\gamma^0}' \bar{H}_{\gamma^0} a_{\gamma^0}. \tag{A.32}$$

Write $a_{\gamma_\kappa} = a_{\gamma^0} + r_\kappa$ with $\sup_{|\kappa|\le K} \|r_\kappa\| = O_p(T^{-1})$ by part (i), and $a_{\gamma^0} = O_p(T^{-1/2})$ by standard maximum-likelihood asymptotics for the fixed-threshold model at γ^0 (a fixed-γ^0 statement, logically prior to Theorem 2): the score at (φ_0, γ^0) is a stationary ergodic martingale difference with finite second moments (Theorem A1), the normalized Hessian converges to the negative-definite information matrix (the ULLN of Lemma 5 with Lemma 3(iii)), and a standard sphere argument (cf. Samia and Chan, 2011, proof of Lemma 5, displays (A29)–(A37) therein) yields the rate. Decompose $\bar{H}_{\gamma_\kappa} - \bar{H}_{\gamma^0} = D_\kappa^{ind} + D_\kappa^{sm}$, where D_κ^{ind} collects the terms generated by indicator reallocation and D_κ^{sm} the effect of the changed evaluation path in φ. Each Hessian summand is dominated by $C\Lambda_t$ on the compact parameter set, so $\sup_{|\kappa|\le K} \|D_\kappa^{ind}\| \le C \sum_t \Lambda_t \mathbb{1}\{q_{t-d} \in \mathcal{B}_T\} = O_p(1)$ as in part (i); and the Gaussian log-likelihood has third φ-derivatives dominated by an integrable envelope on $\mathcal{B} \times \mathcal{S}$, so $\varphi \mapsto H(\varphi, \gamma^0)$ is Lipschitz with random

constant $O_p(T)$ on the relevant neighborhood, giving $\sup_{|\kappa|\le K}\|D_\kappa^{sm}\| = O_p(T)\cdot O_p(T^{-1/2}) = O_p(T^{1/2})$. Expanding (A.32),

$$2\big|\mathcal{L}_T(\kappa)-\mathcal{L}_T^0(\kappa)\big| \;\le\; 2\,\|a_{\gamma^0}\|\,\|\bar H_{\gamma^0}\|\,\|r_\kappa\| + \|r_\kappa\|^2\|\bar H_{\gamma^0}\| + \big(\|a_{\gamma^0}\|+\|r_\kappa\|\big)^2\big(\|D_\kappa^{ind}\|+\|D_\kappa^{sm}\|\big),$$

and with $\|\bar H_{\gamma^0}\| = O_p(T)$ (ULLN on the compact set) the three terms are $O_p(T^{-1/2})\cdot O_p(T)\cdot O_p(T^{-1}) = O_p(T^{-1/2})$, $O_p(T^{-2})O_p(T) = O_p(T^{-1})$, and $O_p(T^{-1})\big(O_p(1)+O_p(T^{1/2})\big) = O_p(T^{-1/2})$, all $o_p(1)$ uniformly over $|\kappa|\le K$ since every bound above is uniform. The decoupled version is identical over the box, with the feasible-GLS profile in place of the ML profile: the role of a_{γ^0} is played by the known-thresholds infeasible GLS estimator at $(\gamma^{\mu,0},\gamma^{\sigma,0})$, whose $\sqrt{T}$-rate follows from the same fixed-threshold argument, and the feasible–infeasible gap is $o_p(T^{-1/2})$ by the score expansion of Appendix A.5, Steps 3–4, so the feasible and infeasible profiled criteria differ by $o_p(1)$ at this scale. □

A.4.2 Proof of Theorem 3(iii)

Throughout, $\hat\gamma$ denotes the smallest maximizer (Section 2.5), and by Theorem 3(ii) all statements may be, and are, restricted to an arbitrarily small neighborhood of θ_0, exactly as in Proposition 1 of Chan (1993) and Theorem 2 of Samia and Chan (2011). The proof invokes Lemma 3(ii) and, as an alternative to **B5**, Lemma 4(i) of Appendix A.2.

Part (i) - super consistency of the threshold

Steps 1–4: localization and reduction. For $h\in\mathbb{R}$ define the rescaled local process $\mathcal{L}_T(h) := T\{L_T^{\mathrm{p}}(\gamma^0 + h/T) - L_T^{\mathrm{p}}(\gamma^0)\}$, with L_T^{p} the profile criterion of Section 2.5. Perturbing γ^0 to γ^0+h/T reassigns only observations with q_{t-d} in a band of width $|h|/T$ at γ^0; their number $N_T(h/T) := \sum_{t=1}^T \mathbb{I}\{\gamma^0 \le q_{t-d} < \gamma^0+h/T\}$ has expectation $T\int_{\gamma^0}^{\gamma^0+h/T} f_\pi(u)\,du \to f_\pi(\gamma^0)\,h$ (the density clause of **B2** and Lemma 3(ii)), hence is $O_p(1)$ by Markov's inequality (Poisson in the limit, Theorem 3(iv)); this is what motivates the T-rescaling. Each reassigned observation is evaluated under the adjacent regime's conditional Gaussian law, which differs from the true one with positive conditional probability at γ^0 (the mean clause of **B2** and **B3**) and in a neighborhood of it (**B7**), so its expected contribution is minus a Kullback–Leibler divergence between distinct Gaussian laws, strictly negative. Observations outside the band enter only through the profiled estimates $(\hat B(\gamma),\hat\Sigma(\gamma))$, which satisfy the inner first-order conditions and move by $O_p(T^{-1})$ across the band (Lemma 7(i)), so that by Lemma 7(ii) the normalized profile increment coincides, uniformly over $|h|\le K$, with the increment at the true regime parameters up to $o_p(1)$. The argmax of $\mathcal{L}_T$ is therefore $O_p(1)$ once the switched-observation sum is shown to be uniformly negative outside a K/T-neighborhood, uniformly over the regime parameters; this is Step 5.

Step 5: Conclusion for the threshold rate. By Theorem 3(ii), with probability approaching one $\hat\theta_T$ lies in $\Omega_1(\bar\Delta) := \{\theta : \|B_i - B_i^0\|\le\bar\Delta,\ \|\Sigma_i-\Sigma_i^0\|\le\bar\Delta,\ i=1,2;\ |\gamma-\gamma^0|\le\Delta\}$ for any fixed $\bar\Delta,\Delta>0$. We show: for every $\varepsilon>0$ there exist $K>0$ and small $\bar\Delta,\Delta$ such that, with probability greater than $1-\varepsilon$,

$$\theta\in\Omega_1(\bar\Delta)\ \text{ and }\ K/T<\gamma-\gamma^0\le\Delta \quad\Longrightarrow\quad L_T(\varphi,\gamma)-L_T(\varphi,\gamma^0)<0,$$

with the mirror statement for $\gamma<\gamma^0$; since $L_T^{\mathrm{p}}(\hat\gamma)\ge L_T^{\mathrm{p}}(\gamma^0)\ge L_T(\hat\varphi(\hat\gamma),\gamma^0)$, this forces $|\hat\gamma-\gamma^0|\le K/T$ with

probability greater than $1-\varepsilon$, i.e. $T(\hat{\gamma}-\gamma^0)=O_p(1)$.

Fix $\varphi \in \Omega_1(\bar{\Delta})$ and $r := \gamma - \gamma^0 \in (K/T, \Delta]$. For fixed φ, $L_T(\varphi, \gamma^0 + r)$ and $L_T(\varphi, \gamma^0)$ assign every observation to the same regime except those with $q_{t-d} \in [\gamma^0, \gamma^0 + r)$, whose contribution switches from the regime-2 to the regime-1 conditional Gaussian log-density. Writing $m_i(y_t; \varphi)$ for the regime-i log-density at candidate parameters and $M_t(\varphi) := m_1(y_t; \varphi) - m_2(y_t; \varphi)$,

$$T\big[L_T(\varphi, \gamma^0 + r) - L_T(\varphi, \gamma^0)\big] = \sum_{t=1}^{T} M_t(\varphi)\, \mathbb{I}\{\gamma^0 \le q_{t-d} < \gamma^0 + r\} = \sum_{t=1}^{T} \big[\xi_t^+ + \{M_t(\varphi) - M_t(\varphi_0)\}\big]\, \mathbb{I}_t(r),$$

where $\mathbb{I}_t(r) := \mathbb{I}\{\gamma^0 \le q_{t-d} < \gamma^0 + r\}$ and, on the compact $\mathcal{B} \times \mathcal{S}$, $|M_t(\varphi) - M_t(\varphi_0)| \le C\, \Lambda_t\, \bar{\Delta}$ by the Gaussian log-density's Lipschitz property in φ with envelope Λ_t. Let $Q(r) := P_\pi(\gamma^0 \le q_{t-d} < \gamma^0 + r)$. Dividing by $TQ(r)$,

$$\frac{T\big[L_T(\varphi, \gamma^0 + r) - L_T(\varphi, \gamma^0)\big]}{TQ(r)} \;\le\; \frac{\sum_t \xi_t^+ \mathbb{I}_t(r)}{TQ(r)} \;+\; C\bar{\Delta}\, \frac{\sum_t \Lambda_t \mathbb{I}_t(r)}{TQ(r)}. \tag{A.20}$$

Claim (uniform ratio control; the analogue of Claim I in Samia and Chan (2011)). For all $\varepsilon, \zeta > 0$ there exists $K > 0$ such that, for all T, with probability greater than $1-\varepsilon$, uniformly over $r \in (K/T, \Delta]$:

(i) $\big|\sum_t \mathbb{I}_t(r)/(TQ(r)) - 1\big| < \zeta$;

(ii) $\big|\sum_t \Lambda_t \mathbb{I}_t(r) - E\sum_t \Lambda_t \mathbb{I}_t(r)\big|/(TQ(r)) < \zeta$, with $E[\Lambda_t \mathbb{I}_t(r)] \le M_\Lambda Q(r)$;

(iii) $\big|\sum_t \xi_t^+ \mathbb{I}_t(r) - E\sum_t \xi_t^+ \mathbb{I}_t(r)\big|/(TQ(r)) < \zeta$.

Proof of the Claim. The ingredients are the analogues of displays (A22)–(A28) of Samia and Chan (2011, online appendix). First, after shrinking Δ if necessary, there exist $0 < m < \bar{M} < \infty$ with $mr \le Q(r) \le \bar{M}r$ for all $r \in (0, \Delta]$: the upper bound is Lemma 3(ii) ($f_\pi \le C_*$), and the lower bound follows since f_π is positive (Lemma 3(ii)) and continuous at γ^0 (the density clause of **B2**), hence bounded away from zero on $[\gamma^0, \gamma^0 + \Delta]$ for small Δ. Second, $\mathrm{var}(\mathbb{I}_t(r)) \le Q(r)$, and by **B6** together with $|\xi_t^+| \le C\Lambda_t$, $E[\Lambda_t^2\, \mathbb{I}_t(r)] \le MQ(r)$ and $E[(\xi_t^+)^2\, \mathbb{I}_t(r)] \le C^2 MQ(r)$, whence the same bounds hold for the variances. Third, all three summand processes are measurable functions of the localized pair in **B5**; by the covariance inequality for ρ-mixing sequences (Doukhan, 1994, p. 9) and summability of the mixing coefficients, the variances of the *sums* are bounded by $THQ(r)$ for some $H < \infty$, uniformly in $r \in (0, \Delta]$ (in model (10), Lemma 4(i) delivers the same variance bounds directly from Theorem A1, without **B5**). The uniform statements (i)–(iii) then follow by Chebyshev bounds on the dyadic grid $r_j = b^j K/T$, $j = 0, 1, \ldots,$ combined with the monotonicity of $r \mapsto \sum_t(\cdot)\mathbb{I}_t(r)$ and $r \mapsto Q(r)$ to interpolate between grid points, exactly as in Chan (1993, p. 529, displays (4.9a)–(4.9e)); K large makes all grid bounds summable to less than ε. ◊

Drift. By **B7** and **B6**, the map
$v \mapsto E[\xi_t^+ \mid q_{t-d} = v] = -E\big[\mathrm{KL}\big(N(B_2^0 x_t, \Sigma_2^0) \,\|\, N(B_1^0 x_t, \Sigma_1^0)\big) \,\big|\, q_{t-d} = v\big]$ is continuous on $[\gamma^0, \gamma^0 + \Delta]$ (shrinking Δ if necessary) and strictly negative at $v = \gamma^0$ by by the mean clause of **B2** and **B3**; hence $E[\xi_t^+ \mid q_{t-d} = v] \le -\chi < 0$ on the band, and $E\sum_t \xi_t^+ \mathbb{I}_t(r) \le -\chi\, TQ(r)$.

Combining the Claim with the drift bound in (A.20): with probability greater than $1-\varepsilon$, uniformly over

$\varphi \in \Omega_1(\bar{\Delta})$ and $r \in (K/T, \Delta]$,

$$\frac{T\big[L_T(\varphi, \gamma^0 + r) - L_T(\varphi, \gamma^0)\big]}{TQ(r)} \;\leq\; -\chi(1-\zeta) + \zeta + C\bar{\Delta}\,(M_\Lambda + \zeta) \;<\; 0$$

for ζ and $\bar{\Delta}$ small enough that $\chi(1-\zeta) > \zeta + C\bar{\Delta}(M_\Lambda + \zeta)$. The case $\gamma < \gamma^0$ is symmetric with ξ_t^- in place of ξ_t^+. This establishes the uniform negativity, hence $T(\hat{\gamma} - \gamma^0) = O_p(1)$.

Part (ii) - $\sqrt{T}$-consistency of regime parameters

Step 6: Negligible impact of threshold estimation on regime parameters. Let $s_{it}(\gamma)$ denote the regime-i score contribution for (B_i, Σ_i) at threshold γ. By the band count of Steps 1–4 and $|\widehat{\gamma} - \gamma^0| = O_p(T^{-1})$, the number of observations that switch regimes when γ moves from γ^0 to $\widehat{\gamma}$ is $O_p(1)$, i.e. $\sum_{t=1}^{T} \mathbb{I}\{q_{t-d} \text{ misclassified under } \widehat{\gamma}\} = O_p(1)$, and each score term satisfies $\mathbb{E}_\pi \|s_{it}(\gamma^0)\|^2 < \infty$ by the fourth-moment bound (Theorem A1). Therefore

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} s_{it}(\widehat{\gamma}) \mathbb{I}\{q_{t-d} \text{ misclassified under } \widehat{\gamma}\} = O_p\left(\frac{1}{\sqrt{T}}\right) = o_p(1)\,:$$

the score contribution of the misclassified observations is asymptotically negligible at the $\sqrt{T}$ scale. This mirrors Lemma 5 of Samia and Chan (2011) (see also Hansen (2000), Lemma 1): the plug-in of the estimated threshold does not affect first-order asymptotics; it is the converse direction of Lemma 7(i) in Appendix A.4.

Step 7: $\sqrt{T}$-rate of the regime-specific estimators. Given Step 6, the usual ML first-order expansion within each regime holds at $\widehat{\gamma}$:

$$0 = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} s_{it}(\gamma^0) + \mathcal{I}_i \sqrt{T}\,(\widehat{\beta}_i - \beta_i^0) + o_p(1), \qquad \mathcal{I}_i := -\mathbb{E}_\pi[\partial s_{it}(\gamma^0)/\partial \beta_i'],$$

so that $\sqrt{T}(\widehat{\beta}_i - \beta_i^0) = \mathcal{I}_i^{-1} T^{-1/2} \sum_{t=1}^{T} s_{it}(\gamma^0) + o_p(1) = O_p(1)$ by the central limit theorem for geometrically ergodic sequences, and analogously for $\widehat{\Sigma}_i$; the Gaussian limits themselves are derived in Step 8 of the proof of Theorem 3(iv). Because threshold estimation affects only $O_p(1)$ observations (Step 6), the rate is the same as with γ^0 known.

This completes the proof of Theorem 3(iii). □

A.4.3 Proof of Theorem 3(iv)

The proof follows the architecture of Theorem 3 of Samia and Chan (2011), which adapts Proposition 2 of Chan (1993); the weak-convergence machinery is the operator-convergence method of Kushner (1980, Theorem 1), implemented through the perturbed-test-function device of Kurtz (1975, p. 625), with uniqueness of the limiting martingale problem from Stroock and Varadhan (1971) (Kushner, 1984, Ch. 5, cited by Chan (1993), presents the same circle of ideas). We spell out the steps and the exact points at which each assumption enters, and cite the two papers for the parts that transfer verbatim.

Step 1: reduction to the true-parameter local process. By Lemma 7(ii), $\sup_{|\kappa| \leq K} |\mathcal{L}_T(\kappa) - \mathcal{L}_T^0(\kappa)| = o_p(1)$ for every K, so it suffices to establish the weak convergence in part (i) for $\mathcal{L}_T^0$. This is the analogue of

passing from (4.10) to (4.12) in Chan (1993) and of Lemma 6 in Samia and Chan (2011).

Step 2: exact marked-point-process representation. Throughout this proof we treat a single threshold and drop its index; for the two-sided process J of Theorem 3(iv) we write J_1 for its left branch ($h<0$) and J_2 for its right branch ($h>0$), following Chan (1993). At the true regime parameters, perturbing the threshold reassigns observations exactly and the increments telescope:

$$\begin{aligned}\mathcal{L}_T^0(\kappa) &= \sum_{t=1}^{T} \xi_t^{+}\, \mathbb{I}\{\gamma^0 \le q_{t-d} < \gamma^0 + \kappa/T\}, \qquad \kappa \ge 0,\\ \mathcal{L}_T^0(\kappa) &= \sum_{t=1}^{T} \xi_t^{-}\, \mathbb{I}\{\gamma^0 + \kappa/T \le q_{t-d} < \gamma^0\}, \qquad \kappa < 0,\end{aligned} \tag{A.33}$$

with no remainder: $\mathcal{L}_T^0$ is a marked point process with rescaled locations $T(q_{t-d}-\gamma^0)$ within $[0,\kappa)$ (resp. $[\kappa,0)$) and marks $\xi_t^{\pm}$.

Step 3: tightness. By Lemma 3(ii), for every interval $I \subseteq [\gamma^0-\Delta, \gamma^0+\Delta]$ and all $t \neq s$,

$$P\big(q_{t-d} \in I,\ q_{s-d} \in I\big) \le C_*^2\, |I|^2, \tag{A.34}$$

the analogue of display (4.13) in Chan (1993); together with the first-moment bound $E\big[|\xi_t^{\pm}|\,\mathbb{I}\{q_{t-d}\in I\}\big] \le C\,E\big[\Lambda_t \mathbb{I}\{q_{t-d}\in I\}\big] \le CM^{1/2}C_*|I|$ (B6 and Lemma 3(ii)), (A.34) yields tightness of the pair $(\mathcal{L}_T^0(-\cdot), \mathcal{L}_T^0(\cdot))$ in $D[0,\infty)^2$, by the argument in the proof of Proposition 2 of Chan (1993) (following Lemma 5.2 of Ibragimov and Has'minskii, 1981).

Step 4: finite-dimensional convergence and identification of the limit. We verify the three ingredients of the direct-averaging/martingale-problem argument of Chan (1993, proof of Proposition 2, displays (4.14)–(4.23)), as adapted by Samia and Chan (2011, proof of Theorem 3):

(a) Local band rates. Uniformly over histories, $P\big(\gamma^0 \le q_{j-d} < \gamma^0 + z\varepsilon \mid \mathcal{F}_{j-d-1}\big) \le C_*\, z\varepsilon$ by Lemma 3(i) (the analogue of Chan's (4.17)) and, in the averaged sense required by the martingale problem, $\lim_{j\to\infty} P\big(\gamma^0 \le q_{j-d} < \gamma^0 + z\varepsilon\big)/\varepsilon = f_\pi(\gamma^0)\, z + o(\varepsilon)$ by continuity and positivity of f_π at γ^0 (the density clause of B2 and Lemma 3(ii)), the analogue of Chan's (4.18). The removal of the conditioning on the remote past in these displays is where the local ρ-mixing B5 does its work, as Assumption 7 does in Samia and Chan (2011); under the alternative route flagged in the Remark to B5, the total-variation forgetting bound of Theorem A1 (Chan's Condition 1 with $g(x) = 1 + x'Px$) substitutes.

(b) Convergence of the conditional mark laws. The marks are $\xi_t^{\pm} = \Phi^{\pm}(x_t, \varepsilon_t)$ for continuous $\Phi^{\pm}$ (display (J1)). Since $\varepsilon_t \perp \mathcal{F}_{t-1}$, the conditional law of $(x_t^{(-)}, \varepsilon_t)$ given $\gamma^0 \le q_{t-d} < \gamma^0 + \delta$ is (conditional law of $x_t^{(-)}$ given the band) $\otimes N(0, I_n)$, and by B7 the first factor converges weakly, as $\delta \downarrow 0$, to the conditional law at $q_{t-d} = \gamma^0$. By the continuous mapping theorem the conditional law of ξ_t^{+} given the band converges weakly to μ^{+}, and B6 supplies the uniform integrability needed to pass conditional expectations of the truncated test functions to the limit (the analogue of Chan's (4.19) and of Assumption 6 in Samia and Chan (2011)).

(c) Averaging and the martingale problem. With (a), (b) and **B5** in hand (in model (10), Lemma 4(ii) supplies the role of **B5** directly, without the mixing condition), the operator-convergence argument of Kushner (1980, Theorem 1), implemented via the perturbed test functions of Kurtz (1975, p. 625) and applied exactly as in Chan (1993) and in the proof of Theorem 3 of Samia and Chan (2011), identifies every subsequential weak limit of $\{\mathcal{L}_T^0(\kappa),\ 0 \le \kappa \le K\}$ as the unique solution (Stroock and Varadhan, 1971) of the martingale problem with generator $Af(w) = f_\pi(\gamma^0)\, z \int \{f(w+u) - f(w)\}\, \mu^+(du)$, i.e. as the compound Poisson process J_2 with rate $f_\pi(\gamma^0)$ and jump law μ^+; the Cramér–Wold device extends this to the finite-dimensional distributions of the pair $(\mathcal{L}_T^0(-\cdot), \mathcal{L}_T^0(\cdot))$. The two components are driven by observations in the disjoint bands $[\gamma^0, \gamma^0 + K/T)$ and $[\gamma^0 - K/T, \gamma^0)$, so the bivariate martingale problem factorizes and the limit pair (J_1, J_2) has independent components; equivalently, the limiting marked point process on the two half-lines is a Poisson random measure, whose restrictions to disjoint carriers are independent (Kallenberg, 2002, Ch. 12). With Step 3, this proves the process convergence of Theorem 3(iv).

The convergence just established is recorded as a marked-point-process statement, which the proof of Theorem 2(ii)–(iii) invokes componentwise; its proof uses only the conclusions of Steps 2–4.

Lemma MPP. *Fix $K > 0$. Define the random point measures on $E_K := (0, K] \times \mathbb{R}$,*

$$N_T^+ := \sum_{t=1}^{T} \delta_{\left(T(q_{t-d} - \gamma^0),\ \xi_t^+\right)}\, \mathbb{I}\{0 < T(q_{t-d} - \gamma^0) \le K\}, \quad N_T^- := \sum_{t=1}^{T} \delta_{\left(T(\gamma^0 - q_{t-d}),\ \xi_t^-\right)}\, \mathbb{I}\{0 < T(\gamma^0 - q_{t-d}) \le K\},$$

*regarded as random elements of the space $\mathcal{N}(E_K)$ of finite point measures on E_K, endowed with the topology of weak convergence of measures and the corresponding Borel σ-field. Under **A1–A4** and **B1–B7**,*

$$\left(N_T^+,\, N_T^-\right) \;\Rightarrow\; \left(N^+,\, N^-\right),$$

where N^+ and N^- are independent Poisson random measures on E_K with intensity measures $f_\pi(\gamma^0)\,\mathrm{Leb}_{(0,K]} \otimes \mu^+$ and $f_\pi(\gamma^0)\,\mathrm{Leb}_{(0,K]} \otimes \mu^-$ respectively. Equivalently, the associated jump processes $\kappa \mapsto \int_{(0,\kappa]\times\mathbb{R}} u\, N_T^{\pm}(dv, du)$ converge jointly in $D[0,K]^2$ to the compound Poisson processes (J_2, J_1) of Theorem 3(iv) restricted to $[0, K]$.

Proof. By Steps 2–4, the pair of one-sided normalized criterion processes converges in $D[0,K]^2$ to the compound Poisson pair (J_2, J_1). The same direct-averaging/martingale-problem computation applies to the enlarged pair (count, sum) on each side: the generator displays of Step 4 depend on the state only through the arrival rates and the mark laws (ingredients (a)–(c)), and the counting component is the special case of a mark identically equal to one, so the four processes (counts and sums, both sides) converge jointly, the limit counts being Poisson with rate $f_\pi(\gamma^0)$ and the limit sums their $\mu^{\pm}$-marked compounds. A path pair (count, sum) with finitely many jumps at distinct positions determines the marked point measure (positions are the count's jump times, marks the sum's increments there), and this correspondence is continuous, in the weak topology of $\mathcal{N}(E_K)$, at every configuration with finitely many points at distinct positions, an almost-sure property of the Poisson limit. The continuous mapping theorem then yields $(N_T^+, N_T^-) \Rightarrow (N^+, N^-)$ with the stated intensities, the independence of the two sides being that of the limiting pairs of Step 4 (their points are carried by disjoint sets of observations, display (A.34)). □

Step 5: drift. By (J2) and iterated expectations, the means of the jump laws are

$$\int u\,\mu^{+}(du) = -\,E\Big[\mathrm{KL}\big(N(B_2^0 x_t, \Sigma_2^0)\,\|\,N(B_1^0 x_t, \Sigma_1^0)\big)\,\Big|\,q_{t-d}=\gamma^0\Big] \;<\; 0,$$

strictly: the mean clause of B2 gives positive conditional probability, at the threshold, that the adjacent conditional Gaussian laws differ (and B3 gives $\Sigma_1^0 \neq \Sigma_2^0$ when the means coincide), B7 carries this separation to the one-sided limit laws, and the Gaussian KL divergence (J3) vanishes only when both means and covariances coincide; symmetrically for μ^{-}. Both means are finite by B6. The random walks of jump sums therefore tend to $-\infty$ almost surely by the strong law of large numbers, so $\mathcal{J}(\kappa) := J(\kappa) \to -\infty$ as $|\kappa| \to \infty$ almost surely (we write $\mathcal{J}$ for the limit process J of Theorem 3(iv) in this appendix).

Step 6: almost-sure uniqueness of the argmax interval. By the limit-objects paragraph of Section 2.4, the nonzero parts of $\mu^{\pm}$ are absolutely continuous, so the running maxima of the thinned compound Poisson processes are almost surely tie-free, and the global maximum of $\mathcal{J}$ is attained on an almost-surely unique interval $[M_{-}, M_{+})$: the multivariate-Gaussian analogue of Remark 1 in Samia and Chan (2011).

Step 7: argmax. By Theorem 3(iii), $T(\hat{\gamma}-\gamma^0) = O_p(1)$: with probability approaching one the maximizer of the profile likelihood over Γ lies within $[\gamma^0 - K/T, \gamma^0 + K/T]$ for K large, and on that event $T(\hat{\gamma}-\gamma^0)$ is the smallest maximizer of $\kappa \mapsto \mathcal{L}_T(\kappa)$ over $[-K, K]$. Realizing the convergence of part (i) almost surely by the Skorohod representation and using the almost-sure uniqueness of Step 6, the smallest maximizer of $\mathcal{L}_T$ converges to the left endpoint M_{-} of the maximizing interval of $\mathcal{J}$, exactly as in the concluding argument of Theorem 2 of Chan (1993) and Theorem 3 of Samia and Chan (2011). Hence $T(\hat{\gamma}-\gamma^0) \Rightarrow M_{-}$, the argmax statement of Theorem 3(iv).

Step 8: regime parameters and asymptotic independence. By Lemma 7(i) and $T(\hat{\gamma}-\gamma^0) = O_p(1)$,

$$\hat{\varphi}(\hat{\gamma}) = \hat{\varphi}(\gamma^0) + o_p\big(T^{-1/2}\big), \tag{A.35}$$

the analogue of display (4.24) in Chan (1993): the two-step estimator is first-order equivalent to the known-threshold ML estimator, so it suffices to derive the limit of $\hat{\varphi}(\gamma^0)$. The known-threshold scores at θ_0, written with the true innovations $\varepsilon_t^0 := y_t - B_i^0 x_t = \Sigma_i^{0,1/2}\varepsilon_t$ on regime i,

$$s_t^{B_i} = \big(x_t \otimes I_n\big)\,\Sigma_i^{0,-1}\varepsilon_t^0\;\mathbb{I}\{q_{t-d}\in\mathcal{R}_i^0\}, \qquad s_t^{\Sigma_i} = \tfrac{1}{2}\,D_n'\,\mathrm{vec}\Big(\Sigma_i^{0,-1}\big(\varepsilon_t^0\varepsilon_t^{0\prime} - \Sigma_i^0\big)\Sigma_i^{0,-1}\Big)\,\mathbb{I}\{q_{t-d}\in\mathcal{R}_i^0\},$$

are stationary ergodic martingale differences with respect to $\{\mathcal{F}_t\}$ (the indicators and x_t are $\mathcal{F}_{t-1}$-measurable; $\varepsilon_t \perp \mathcal{F}_{t-1}$) with finite second moments (Theorem A1). By the martingale-difference central limit theorem for stationary ergodic sequences (Billingsley, 1961) and the Cramér–Wold device, the stacked normalized score converges to a centered Gaussian vector with covariance equal to the information matrix, which is block diagonal: across regimes because the indicators are disjoint, and between the B- and Σ-blocks within a regime because the cross moments involve odd Gaussian moments of ε_t, which vanish. The standard ML expansion within each regime (Step 7 of the proof of Theorem 3(iii)) then delivers the stated Gaussian limits: the B_i-block information is $E_\pi[x_t x_t' \mathbb{I}_i] \otimes \Sigma_i^{0,-1} = S_i \otimes \Sigma_i^{0,-1}$, with inverse $S_i^{-1} \otimes \Sigma_i^0$, and $\sqrt{T_i}\,\mathrm{vech}(\hat{\Sigma}_i - \Sigma_i^0) \Rightarrow N\big(0,\; 2D_n^{+}(\Sigma_i^0 \otimes \Sigma_i^0)D_n^{+\prime}\big)$ with $T_i/T \to_{a.s.} \pi_i$ gives the stated Σ-block. Finally, the asymptotic independence of

$T(\hat{\gamma}-\gamma^0)$ from the Gaussian block follows as in Chan (1993, proof of Theorem 2) and Samia and Chan (2011, Theorem 4): by (A.35) the Gaussian block is a functional of the full-sample score partial sums, while by Steps 1–7 the threshold limit is a functional of the $O_p(1)$ observations in the shrinking bands, whose contribution to the normalized score is $o_p(1)$ (Step 6 of the proof of Theorem 3(iii)); the joint limit therefore factorizes. □

Proof of Theorem 3(iv) (sketch). For $k > 2$, the true thresholds are separated by fixed positive distances (B1), so for K fixed and T large the $k-1$ boundary bands $[\gamma_j^0 - K/T,\ \gamma_j^0 + K/T)$ are disjoint and only adjacent-regime pairs are involved in each. Steps 1–4 apply to the $2(k-1)$ one-sided component processes simultaneously; the limiting marked point process on the $2(k-1)$ half-lines is a Poisson random measure whose restrictions to the disjoint carriers are independent, so the $k-1$ two-sided processes $\mathcal{J}^{(j)}$ are mutually independent, each built from the adjacent pair $(j, j+1)$ with rate $f_\pi(\gamma_j^0)$. Steps 5–8 apply coordinatewise, and the joint argmax convergence follows from the continuous mapping theorem on the product space. The result parallels Theorem 2 of Li and Ling (2012) for the least-squares estimator of multiple-regime TAR models, which establishes the same asymptotic independence of the multiple threshold estimators. □

A.5 Rates and Limit Laws: the Decoupled Case

A.5.1 Proof of Theorem 2(i)

The proof extends that of Theorem 3(iii) to the product parameter space and to the feasible profile-GLS estimator.

Part (i): super-consistency of $\widehat{\gamma}^\mu$ and $\widehat{\gamma}^\sigma$. We prove the rate jointly. For $h_\mu \in \mathbb{R}^{k_\mu - 1}$ and $h_\sigma \in \mathbb{R}^{k_\sigma - 1}$ define the local process

$$\mathcal{L}_T(h_\mu, h_\sigma) := T\Big[L_T^{\mathrm{p}}\big(\gamma^{\mu,0} + h_\mu/T,\ \gamma^{\sigma,0} + h_\sigma/T\big) - L_T^{\mathrm{p}}\big(\gamma^{\mu,0}, \gamma^{\sigma,0}\big)\Big]; \tag{A.28}$$

it suffices to show that its joint maximizer is $O_p(1)$.

Step 1: local counting argument. For fixed compact sets $K_\mu \subset \mathbb{R}^{k_\mu-1}$, $K_\sigma \subset \mathbb{R}^{k_\sigma-1}$, define the boundary bands $\Delta_T^\mu(h_\mu) := \bigcup_{j=1}^{k_\mu-1}\big[\gamma_j^{\mu,0} \wedge (\gamma_j^{\mu,0} + h_{\mu,j}/T),\ \gamma_j^{\mu,0} \vee (\gamma_j^{\mu,0} + h_{\mu,j}/T)\big)$ and, analogously, $\Delta_T^\sigma(h_\sigma)$. Only observations with $q^\mu_{t-d_\mu} \in \Delta_T^\mu(h_\mu)$ or $q^\sigma_{t-d_\sigma} \in \Delta_T^\sigma(h_\sigma)$ can change cell when the threshold pair is perturbed to $(\gamma^{\mu,0} + h_\mu/T,\ \gamma^{\sigma,0} + h_\sigma/T)$, so the number of switched observations is at most

$$N_T(h_\mu, h_\sigma) := \sum_{t=1}^{T} \mathbb{I}\{q^\mu_{t-d_\mu} \in \Delta_T^\mu(h_\mu)\} + \sum_{t=1}^{T} \mathbb{I}\{q^\sigma_{t-d_\sigma} \in \Delta_T^\sigma(h_\sigma)\},$$

which is $O_p(1)$ uniformly over $(h_\mu, h_\sigma) \in K_\mu \times K_\sigma$: by the density clauses of **B2** each band has stationary probability $O(1/T)$, so $EN_T = O(1)$ and Markov's inequality applies.

Step 2: local likelihood loss. If $h_{\mu,j} \neq 0$, the observations in the j-th mean band are assigned to the wrong pair of adjacent mean regimes, whose conditional means differ at $\gamma_j^{\mu,0}$ (the mean clause of **B2**); if $h_{\sigma,\ell} \neq 0$, those in the ℓ-th variance band are evaluated under the wrong adjacent covariance matrices (the variance clause of **B2**). Each switched observation thus incurs a non-vanishing expected Kullback–Leibler loss, so for every fixed $(h_\mu, h_\sigma) \neq (0,0)$ the local criterion is driven by the $O_p(1)$ switched observations of Step 1, each

contributing a loss of the sign required by the classical localization argument. The uniform ratio control follows as in the Claim of Step 5 of the proof of Theorem 3(iii), applied to each of the $(k_\mu - 1) + (k_\sigma - 1)$ bands separately (the band sums are combined by the triangle inequality, so no cross-band covariances are needed), with **B5** and **B6** supplying the covariance and moment bounds; for threshold variables of the class described in the remark after **B5**, Lemma 4(i) supplies these bounds without **B5**.

Step 3: control of the remainder. Observations outside the boundary bands do not change cell and affect the criterion only through the re-estimated regime parameters. By Lemma 7 (decoupled clauses), the profiled estimates move by $O_p(T^{-1})$ uniformly over the box (part (i)) and, since the score vanishes at the profiled maximizer, the induced criterion change is quadratic in this movement against an $O_p(T)$ Hessian (part (ii)); hence these terms are $o_p(1)$ after multiplication by T, uniformly over (h_μ, h_σ) in compact sets.

Step 4: localization. By Steps 1–3 the local process admits the localization mechanism of Chan (1993), Hansen (2000) and Samia and Chan (2011), applied jointly to the product threshold parameter: for every $\varepsilon > 0$ there is $M < \infty$ such that $P\big(\sup_{\|(h_\mu,h_\sigma)\|>M} \mathcal{L}_T(h_\mu, h_\sigma) \geq \mathcal{L}_T(0,0)\big) < \varepsilon$ for all large T. Since $(\widehat{\gamma}^\mu, \widehat{\gamma}^\sigma)$ maximizes L_T^{p} over $\Gamma^\mu \times \Gamma^\sigma$, the argmax localization argument gives

$$T(\widehat{\gamma}^\mu - \gamma^{\mu,0}) = O_p(1), \qquad T(\widehat{\gamma}^\sigma - \gamma^{\sigma,0}) = O_p(1). \tag{A.31}$$

This proves part (i).

Part (ii): $\sqrt{T}$-consistency of the regime parameters.

Step 1. Preliminary covariances. Let $\widehat{\Sigma}_\ell^{(0)}$ be the preliminary pooled estimator of Remark 1 (Section 2.2). At the true threshold pair, the ergodic theorem and the fourth-moment bound give $\widehat{\Sigma}_\ell^{(0)} \to_{a.s.} \Sigma_\ell^0$, $\ell = 1, \ldots, k_\sigma$, and, since $\Sigma_\ell^0 \succ 0$ by **B1**, $(\widehat{\Sigma}_\ell^{(0)})^{-1} \to_{a.s.} (\Sigma_\ell^0)^{-1}$.

Step 2. Negligible misclassification. By Part (i) and the local counting argument of its Step 1, the number of observations misclassified in either threshold dimension is $O_p(1)$; since each score term has finite second moment (Theorem A1), their contribution to the normalized score is $O_p(T^{-1/2}) = o_p(1)$.

Step 3. GLS score expansion for the mean block. Let $s_{it}^{GLS} := (x_t \otimes \Sigma_{\ell(t)}^{-1})\varepsilon_t \mathbb{I}\{q_{t-d_\mu}^\mu \in \mathcal{R}_i^\mu\}$, where $\ell(t) = \ell^*(X_{t-1})$ is evaluated at the true threshold. The feasible GLS first-order condition yields

$$0 = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} s_{it}^{GLS}(B_i^0) - M_i^{GLS} \sqrt{T}\big(\widehat{\beta}_i^{GLS} - \beta_i^0\big) + o_p(1),$$

with $M_i^{GLS} = E\big[(x_t x_t' \otimes \Sigma_{\ell(t)}^{-1})\, \mathbb{I}\{q_{t-d_\mu}^\mu \in \mathcal{R}_i^\mu\}\big]$.

Step 4. FGLS plug-in error. Since $\max_{1\leq\ell\leq k_\sigma} \|(\widehat{\Sigma}_\ell^{(0)})^{-1} - (\Sigma_\ell^0)^{-1}\| = o_p(1)$, the feasible and infeasible GLS scores differ by $T^{-1/2}\sum_{t=1}^T \big(x_t \otimes [(\widehat{\Sigma}_{\ell(t)}^{(0)})^{-1} - (\Sigma_{\ell(t)}^0)^{-1}]\big)\varepsilon_t \mathbb{I}\{q_{t-d_\mu}^\mu \in \mathcal{R}_i^\mu\} = o_p(1)$, by Cauchy–Schwarz and the fourth-moment bound: one covariance update is first-order equivalent to infeasible GLS.

Steps 5–6. Mean- and variance-block asymptotics. Solving the score expansion, $\sqrt{T}\,\mathrm{vec}(\widehat{B}_i^{GLS} - B_i^0) = (M_i^{GLS})^{-1} T^{-1/2} \sum_{t=1}^T s_{it}^{GLS}(B_i^0) + o_p(1) = O_p(1)$, and a central limit theorem for geometrically ergodic processes yields the centered Gaussian limit of the theorem. For each variance regime ℓ, $\widehat{\Sigma}_\ell$ in (5) is a pooled sample covariance of residuals that differ from the true innovations only through $o_p(1)$ parameter-estimation

error and $O_p(1)$ misclassified observations, so the sample-covariance expansion gives $\sqrt{T}\,\mathrm{vech}(\widehat{\Sigma}_\ell - \Sigma_\ell^0) = O_p(1)$, $\ell = 1, \ldots, k_\sigma$, proving Part (ii). □

A.5.2 Proof of Theorem 2(ii)-(iii)

The proof extends that of Theorem 3(iv) (Appendix A.4) to the product threshold parameter, with two new ingredients: the elimination of double-boundary observations (Step 2, using B8), which delivers the asymptotic independence of the two threshold estimators, and the fact that the variance-side marks are exact i.i.d. functionals of the innovations (Step 3), which simplifies the identification of the σ-side limit.

Step 1: reduction. By the decoupled version of Lemma 7(ii), uniformly over $(h_\mu, h_\sigma) \in [-K, K]^2$ the normalized feasible-GLS profile increment

$$\mathcal{L}_T(h_\mu, h_\sigma) := T\Big[L_T^{\mathrm{p}}\big(\gamma^{\mu,0} + h_\mu/T,\ \gamma^{\sigma,0} + h_\sigma/T\big) - L_T^{\mathrm{p}}\big(\gamma^{\mu,0}, \gamma^{\sigma,0}\big)\Big]$$

differs by $o_p(1)$ from its true-parameter counterpart $\mathcal{L}_T^0(h_\mu, h_\sigma)$, computed from the full Gaussian criterion ℓ_t of (2) in Appendix A.3 at $\varphi_0 = \{B_i^0, \Sigma_\ell^0\}$. (The one-covariance-update feasibility gap is $o_p(1)$ at this scale by the score expansion of Appendix A.5, Steps 3–4).

Step 2: exact decomposition and elimination of double switchers. At the true parameters, an observation contributes to $\mathcal{L}_T^0(h_\mu, h_\sigma)$ only if it changes cell, i.e. only if $q^\mu_{t-d_\mu}$ lies in the μ-band $\mathcal{B}_T^\mu(h_\mu)$ of width $|h_\mu|/T$ at $\gamma^{\mu,0}$, or $q^\sigma_{t-d_\sigma}$ in the analogous σ-band $\mathcal{B}_T^\sigma(h_\sigma)$, or both. Splitting by which band is hit,

$$\mathcal{L}_T^0(h_\mu, h_\sigma) = \mathcal{L}_T^\mu(h_\mu) + \mathcal{L}_T^\sigma(h_\sigma) + R_T(h_\mu, h_\sigma), \tag{A.36}$$

where $\mathcal{L}_T^\mu$ sums the mean jumps $\xi_t^{\mu,\pm}$ (display (V1), at the observation's true variance regime $\ell(t)$, unchanged for a μ-only switcher) over observations hitting only the μ-band; $\mathcal{L}_T^\sigma$ sums the variance jumps $\xi_t^{\sigma,\pm}$ (display (V2), with the observation's true mean, unchanged for a σ-only switcher) over observations hitting only the σ-band; and R_T collects the double switchers, whose individual contributions are bounded by $C\Lambda_t$. By B8,

$$P\Big(\exists\, t \le T:\ q^\mu_{t-d_\mu} \in \mathcal{B}_T^\mu(h_\mu),\ q^\sigma_{t-d_\sigma} \in \mathcal{B}_T^\sigma(h_\sigma) \text{ for some } |h_\mu|, |h_\sigma| \le K\Big) \ \le\ T \cdot \sup f_{\mu\sigma} \cdot \frac{2K}{T} \cdot \frac{2K}{T} \ =\ O\big(T^{-1}\big),$$

so with probability approaching one there are no double switchers in the box, $\sup_{\|(h_\mu,h_\sigma)\|_\infty \le K} |R_T| = 0$, and in particular $R_T = o_p(1)$ uniformly. This is the precise sense in which the two threshold problems decouple: membership in one band is an event of probability $O(1/T)$ per observation, which occurs a stable $O_p(1)$ number of times in T trials and drives the limit, whereas membership in both bands requires two coincidences for the same observation and, under **B8**, which rules out any singular concentration of (q^μ, q^σ) along a curve, has probability $O(1/T^2)$, so that not even one double event occurs in the limit. The assumption is sharp: on the diagonal $q^\mu \equiv q^\sigma$ (same variable, same delay) one band membership implies the other, the joint law is singular, the double-switcher count is $O_p(1)$ and the criterion no longer separates; that configuration is exactly the joint model of Section 2.5.

Step 3: marks. The σ-side marks $\xi_t^{\sigma,\pm}$ are, by (V2), measurable functions of ε_t and (Σ_1^0, Σ_2^0) alone; since $\{\varepsilon_t\}$ is i.i.d. and independent of the ($\mathcal{F}_{t-1}$-measurable) band indicators, they are exactly i.i.d. with law $\mu^{\sigma,\pm}$, whatever the conditional distribution of the state at the threshold: no weak-continuity condition is

needed, and ingredient (b) of Step 4 in Appendix A.4 is automatic on this side. The μ-side marks are $\xi_t^{\mu,\pm} = \Phi^{\mu,\pm}(x_t, \ell(t), \varepsilon_t)$ with $\Phi^{\mu,\pm}$ continuous in (x_t, ε_t) for each value of the discrete $\ell(t)$; by B7 the conditional law of $(x_t^{(-)}, \ell(t))$ given the μ-band converges weakly to its value at $q^\mu = \gamma^{\mu,0}$, so the conditional mark law converges to the mixture $\mu^{\mu,\pm}$ of (V3), with uniform integrability from B6: ingredient (b) on the μ side.

Step 4: joint weak convergence and independence. Ingredient (a) of Step 4 in Appendix A.4 holds at each threshold by the density clauses of **B2** (continuity and positivity of f_μ, f_σ) together with the conditional-density bounds (Lemma 3(i) when the state variables are elements of y_t; maintained through the density clauses of **B2** and B6 in general); ingredient (c) holds by **B5**, and for threshold variables of the class described in the remark after **B5**, Lemma 4(ii) applied at each band supplies it without **B5**. Applying Lemma MPP and the direct-averaging argument of Step 4 in Appendix A.4 componentwise to the four one-sided component processes (left and right of each threshold) identifies the limit as a Poisson random measure on the disjoint union of four half-lines, with intensity $f_\mu(\gamma^{\mu,0})$ on the two μ-components and $f_\sigma(\gamma^{\sigma,0})$ on the two σ-components, and the mark laws of Step 3. The four driving observation sets are pairwise disjoint with probability approaching one (Step 2 for μ-vs-σ; the left/right bands are disjoint by construction), so the four-component martingale problem factorizes and the limiting components are mutually independent (Kallenberg, 2002, Ch. 12). Together with the tightness bound (A.34) (the pairwise band-occupancy bound of Step 3 in Appendix A.4, which by Lemma 3(i) holds verbatim for each of q^μ and q^σ, applied at each threshold) this gives the process convergence in part (ii), with $(\mathcal{J}^\mu, \mathcal{J}^\sigma)$ independent.

Step 5: drifts. On the μ side, using (V1), B1 ($\Sigma_\ell^{0,-1} \succeq \bar\sigma^{-1} I_n$) and the mean clause of **B2**,

$$\int u\, \mu^{\mu,+}(du) = -\tfrac{1}{2}\, E\Big[\delta_t^{\mu\prime} \Sigma_{\ell(t)}^{0,-1} \delta_t^\mu \,\Big|\, q_{t-d_\mu}^\mu = \gamma^{\mu,0}\Big] \;\le\; -\tfrac{1}{2\bar\sigma}\, E\Big[\|\delta_t^\mu\|^2 \,\Big|\, q^\mu = \gamma^{\mu,0}\Big] \;<\; 0,$$

strictly, since $P(\delta_t^\mu \neq 0 \mid q^\mu = \gamma^{\mu,0}) > 0$; symmetrically for $\mu^{\mu,-}$. On the σ side the drift is the deterministic constant $-\frac{1}{2}\sum_j(\lambda_j - \ln\lambda_j - 1) < 0$, strictly by the variance clause of **B2**. Finiteness from B6 (the σ-side mean is finite outright). Both pairs of random walks tend to $-\infty$ almost surely.

Step 6: uniqueness of the argmax intervals. The σ-jump laws are absolutely continuous with no atom: $\Sigma_1^0 \neq \Sigma_2^0$ implies $I_n - Q \neq 0$ (if $Q = I_n$ then $\Sigma_2^0 = \Sigma_1^0$), so at least one weight $1-\lambda_j$ in (V2) is nonzero and $\xi^{\sigma,\pm}$ has a Lebesgue density. The μ-jump laws may place an atom at zero of mass $P(\delta_t^\mu = 0 \mid q^\mu = \gamma^{\mu,0}) < 1$, with absolutely continuous nonzero part (conditionally on $\{\delta_t^\mu \neq 0\}$ and $\ell(t)$, the jump is Gaussian with variance $\kappa_t > 0$); the thinning argument of the limit-objects paragraph of Section 2.4 applies. Hence each of $\mathcal{J}^\mu, \mathcal{J}^\sigma$ attains its global maximum on an almost-surely unique interval, $[M_-^\mu, M_+^\mu)$ and $[M_-^\sigma, M_+^\sigma)$.

Step 7: argmax. Theorem 2(i) localizes: $T(\hat\gamma^\mu - \gamma^{\mu,0})$ and $T(\hat\gamma^\sigma - \gamma^{\sigma,0})$ are $O_p(1)$, so with probability approaching one the smallest maximizer of the profile criterion over $\Gamma^\mu \times \Gamma^\sigma$ lies in the box $[-K, K]^2$ for K large. By (A.36) with $R_T = 0$ on an event of probability approaching one, the criterion is *additively separable* on the box, so its smallest maximizer is the pair of coordinatewise smallest maximizers of $\mathcal{L}_T^\mu$ and $\mathcal{L}_T^\sigma$. Applying the Skorohod representation and the argmax argument of Step 7 in Appendix A.4 to each coordinate, and using the independence from Step 4,

$$\Big(T\big(\hat\gamma^\mu - \gamma^{\mu,0}\big),\; T\big(\hat\gamma^\sigma - \gamma^{\sigma,0}\big)\Big) \Rightarrow \big(M_-^\mu, M_-^\sigma\big), \qquad M_-^\mu \perp M_-^\sigma,$$

proving part (ii).

Step 8: part (iii). By the decoupled Lemma 7(i) and part (i), $\hat{\varphi}(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma}) = \hat{\varphi}(\gamma^{\mu,0}, \gamma^{\sigma,0}) + o_p(T^{-1/2})$, and by Steps 3–4 of Appendix A.5 the feasible one-step GLS estimator at the true thresholds is first-order equivalent to the infeasible GLS estimator, whose scores at θ_0, with $\varepsilon_t^0 = \Sigma_{\ell(t)}^{0,1/2}\varepsilon_t$,

$$s_t^{B_i} = \left(x_t \otimes I_n\right) \Sigma_{\ell(t)}^{0,-1}\, \varepsilon_t^0\, \mathbb{I}\{q_{t-d_\mu}^{\mu} \in \mathcal{R}_i^{\mu,0}\}, \qquad s_t^{\Sigma_\ell} = \tfrac{1}{2} D_n' \mathrm{vec}\Big(\Sigma_\ell^{0,-1}\big(\varepsilon_t^0 \varepsilon_t^{0\prime} - \Sigma_\ell^0\big)\Sigma_\ell^{0,-1}\Big)\, \mathbb{I}\{q_{t-d_\sigma}^{\sigma} \in \mathcal{R}_\ell^{\sigma,0}\},$$

are stationary ergodic martingale differences with finite second moments (Theorem A1), so the martingale-difference CLT (Billingsley, 1961) with Cramér–Wold gives joint asymptotic normality. The information is block diagonal across mean regimes (disjoint μ-indicators), across variance regimes (disjoint σ-indicators), and between the B- and Σ-blocks (the B–Σ cross moments mix indicators from different partitions, but the factor $E[\varepsilon_t \otimes (\varepsilon_t\varepsilon_t' - I)] = 0$ annihilates them regardless). The B_i-block information is M_i^{GLS} as defined in the theorem, giving avar $(M_i^{GLS})^{-1}$; the Σ_ℓ-block gives $\sqrt{T_\ell}\,\mathrm{vech}(\hat{\Sigma}_\ell - \Sigma_\ell^0) \Rightarrow N(0,\ 2D_n^+(\Sigma_\ell^0 \otimes \Sigma_\ell^0)D_n^{+\prime})$ with $T_\ell/T \to \pi_\ell^\sigma$ (the pooling across mean regimes within variance regime ℓ is reflected in T_ℓ), the stated law. Asymptotic independence from the threshold pair follows as in Step 8 of Appendix A.4: the Gaussian block is driven by full-sample score sums, the threshold pair by the $O_p(1)$ band observations, whose score contribution is $o_p(1)$ (Appendix A.5, Step 2). □

A.6 Discussion of the Assumptions and the Implementation

A.6.1 The feasible profile-GLS algorithm (Remark 1)

For any candidate pair $(\gamma^{\mu}, \gamma^{\sigma}) \in \Gamma^{\mu} \times \Gamma^{\sigma}$:

- Assign each observation to its mean regime i, variance regime ℓ, and joint cell $S_{i\ell} := \{t : q_{t-d_\mu}^{\mu} \in \mathcal{R}_i^{\mu},\ q_{t-d_\sigma}^{\sigma} \in \mathcal{R}_\ell^{\sigma}\}$.
- Estimate preliminary mean-regime coefficients $\{\widehat{B}_i^{(0)}(\gamma^{\mu})\}_{i=1}^{k_\mu}$ by OLS within each mean regime, ignoring heteroskedasticity at this step, and let $\widehat{u}_{it} = y_t - (x_t' \otimes I_n)\widehat{\beta}_i^{(0)}$.
- Construct preliminary variance-regime covariances by pooling these residuals across all mean regimes within variance regime ℓ: $\widehat{\Sigma}_\ell^{(0)}(\gamma^{\mu}, \gamma^{\sigma}) = t_\ell^{-1} \sum_{i=1}^{k_\mu} \sum_{t \in S_{i\ell}} \widehat{u}_{it}\widehat{u}_{it}'$, with $t_\ell := \sum_{i=1}^{k_\mu} t_{i\ell}$.
- Using the weights $(\widehat{\Sigma}_\ell^{(0)})^{-1}$, compute the feasible GLS estimator $\widehat{B}_i^{GLS}(\gamma^{\mu}, \gamma^{\sigma})$ from (4).
- With $\widehat{\varepsilon}_{it} = y_t - (x_t' \otimes I_n)\widehat{\beta}_i^{GLS}$, update the variance-regime covariances by (5).
- Substitute $\{\widehat{\beta}_i^{GLS}\}_{i=1}^{k_\mu}$ and $\{\widehat{\Sigma}_\ell(\gamma^{\mu}, \gamma^{\sigma})\}_{\ell=1}^{k_\sigma}$ into the concentrated log-likelihood (6); the threshold estimator is its maximizer (7).

Further GLS iterations may be performed; one covariance update defines a coherent feasible estimator and keeps the asymptotic analysis transparent.

A.6.2 State variables satisfying B5–B8 (Remarks 5 and 8)

B5–B7 beyond states that are elements of y_t. **B5** is kept in the statements of the theorems to preserve the parallel with Samia and Chan (2011); with general measurable transformations of the state it remains the operative condition, while for elements of y_t at fixed lags Lemma 4 discharges it. Beyond such states, **B5–B7** hold for fixed linear combinations of the components of y_t with a nonzero contemporaneous coefficient, finite moving averages of such combinations, and absolute values thereof, since each is, conditionally on the past, a nondegenerate Gaussian functional with density bounded by a multiple of $C_* = (2\pi\sigma_*)^{-1/2}$, the constant of Lemma 3 (for absolute values this is the folded-Gaussian bound: if $W \mid \mathcal{F} \sim N(m, s^2)$ with $s^2 \geq \underline{s}^2$, the density of $|W|$ is at most $2(2\pi\underline{s}^2)^{-1/2}$).

B8 for distinct elements of y_t. When the two state variables are distinct elements of y_t at (possibly different) lags, **B8** holds automatically: conditioning first on the information preceding the earlier of the two dates and then sequentially, each state has a one-step conditional density bounded by C_* (Lemma 3), so the pair's joint density is bounded by C_*^2; when the two dates coincide, the pair is conditionally jointly Gaussian with a nonsingular 2×2 covariance submatrix of some Σ_ℓ^0 (**A3**), with the same conclusion.

B Additional Tables from Monte Carlo Exercise

This appendix collects the full statistics from the Monte Carlo experiments of Section 3.

Table A-1: Joint MC – full threshold statistics.

		RMSE		MAD		$\hat{\gamma}$: bias / std	
DGP	T	SSR	Joint	SSR	Joint	SSR	Joint
DGP2	200	0.025	0.025	0.017	0.017	−0.013 / 0.021	−0.013 / 0.022
	1000	0.004	0.004	0.003	0.003	−0.002 / 0.004	−0.002 / 0.004
DGP3	200	2.004	0.080	1.752	0.050	+1.728 / 1.015	−0.001 / 0.080
	1000	3.204	0.013	2.872	0.008	+2.865 / 1.435	+0.001 / 0.013
DGP4	200	0.081	0.039	0.035	0.025	+0.006 / 0.081	−0.012 / 0.037
	1000	0.010	0.006	0.006	0.004	+0.001 / 0.010	−0.001 / 0.006
T-rate ratio DGP2		SSR 5.50, Joint 5.62					
T-rate ratio DGP3		SSR 0.63, Joint 6.01					
T-rate ratio DGP4		SSR 8.42, Joint 6.00					

Notes. DGP1 omitted because the threshold is unidentified in the no-regime DGP, so RMSE and MAD are not meaningful quantities. T-rate ratio is RMSE($T = 200$)/RMSE($T = 1000$); target is 5 for super-consistency at rate T.

Table A-2: DGP1 (no regime): parameter RMSE under estimated and fixed partitions

		Estim. split, equal weights		Estim. split, share weights		Median split	No split
	T	SSR	joint	SSR	joint		
RMSE $\hat{B}_i$	200	1.758	1.908	0.796	0.799	0.345	0.164
	1000	2.259	2.590	0.488	0.502	0.152	0.072
RMSE $\hat{\Sigma}_i$	200	0.360	0.495	0.247	0.291	0.245	0.174
	1000	0.263	0.426	0.111	0.135	0.110	0.078
$\|\hat{\Phi}_{1,0} - \hat{\Phi}_{2,0}\|$	200	1.704	1.849			0.299	
	1000	1.856	2.294			0.133	
$\hat{\gamma}$ outside central 80%	200	0.257	0.414				
	1000	0.489	0.633				
Median share, smaller regime	200	0.188	0.125				
	1000	0.104	0.049				

Notes: $S = 5{,}000$; true parameters common across the two sub-samples. "Estimated split": the SSR or joint threshold estimate, with RMSE averaged over regimes with equal weights (as in Table 2) or with weights equal to each regime's share of observations. "Median split": regimes defined by the sample median of $y_{1,t-1}$. "No split": the constant-parameter VAR. The intercept gap has true value zero. The smaller regime's minimum share is $11/T$ at both T, the estimator's feasibility bound.

Table A-3: Joint MC – classification accuracy

		Classification accuracy	
DGP	T	SSR	joint
DGP2	200	0.9987	0.9986
	1000	0.9997	0.9997
DGP3	200	0.7257	0.9904
	1000	0.6583	0.9985
DGP4	200	0.9943	0.9975
	1000	0.9991	0.9995

Notes: $S = 5{,}000$. Fraction of observations assigned to their true regime under $\hat{\gamma}$. DGP1 has no true regime, so accuracy is undefined; its parameter RMSEs are in Tables 2 and A-2. Parameter RMSEs for DGP2–DGP4 are in Table 2.

Table A-4: Decoupled MC – full threshold statistics.

DGP	T	Estimator	RMSE	MAD	bias / std	T-rate ratio
DGP5	200	joint q^μ	0.092	0.036	-0.019 / 0.090	see below
		joint q^σ	0.138	0.082	$+0.008$ / 0.138	
		decoupled $\hat{\gamma}^\mu$	0.040	0.026	-0.012 / 0.038	
		SSR q^μ	0.051	0.031	-0.014 / 0.049	
	1000	joint q^μ	0.009	0.005	-0.002 / 0.008	
		joint q^σ	0.022	0.014	$+0.002$ / 0.022	
		decoupled $\hat{\gamma}^\mu$	0.008	0.006	-0.001 / 0.008	
		SSR q^μ	0.008	0.005	-0.002 / 0.008	
DGP5 decoupled $\hat{\gamma}^\sigma$: $T = 200$ RMSE $= 0.103$, $T = 1000$ RMSE $= 0.018$.						
DGP5 T-rate ratios: joint q^μ 10.75, joint q^σ 6.30, decoupled μ 5.29, decoupled σ 5.67.						
DGP6	200	joint q^μ	0.033	0.021	-0.009 / 0.032	
		joint q^σ	0.094	0.050	$+0.007$ / 0.094	
		decoupled $\hat{\gamma}^\mu$	0.027	0.018	-0.009 / 0.025	
		SSR q^μ	0.032	0.021	-0.009 / 0.030	
	1000	joint q^μ	0.006	0.004	-0.001 / 0.006	
		joint q^σ	0.013	0.008	$+0.002$ / 0.013	
		decoupled $\hat{\gamma}^\mu$	0.005	0.003	-0.001 / 0.005	
		SSR q^μ	0.006	0.004	-0.001 / 0.005	
DGP6 decoupled $\hat{\gamma}^\sigma$: $T = 200$ RMSE $= 0.052$, $T = 1000$ RMSE $= 0.009$.						
DGP6 T-rate ratios: joint q^μ 5.92, joint q^σ 7.18, decoupled μ 5.63, decoupled σ 5.97.						

Notes. "joint q^μ" is the joint estimator with state variable q^μ, comparable to the decoupled estimator on the mean dimension; analogously for "joint q^σ". The full decoupled estimator returns both $\hat{\gamma}^\mu$ and $\hat{\gamma}^\sigma$; their statistics are reported separately. Statistics for $\hat{\gamma}^\sigma$ are summarized in the comment rows for readability.

Table A-5: Decoupled MC – classification accuracy

			Classification accuracy	
DGP	T	Estimator	mean regime	variance regime
DGP5	200	joint q^{μ}	0.9946	—
		joint q^{σ}	—	0.9861
		decoupled	0.9970	0.9906
		SSR q^{μ}	0.9956	—
	1000	joint q^{μ}	0.9992	—
		joint q^{σ}	—	0.9979
		decoupled	0.9991	0.9983
		SSR q^{μ}	0.9993	—
DGP6	200	joint q^{μ}	0.9955	—
		joint q^{σ}	—	0.9829
		decoupled	0.9969	0.9907
		SSR q^{μ}	0.9957	—
	1000	joint q^{μ}	0.9992	—
		joint q^{σ}	—	0.9974
		decoupled	0.9992	0.9983
		SSR q^{μ}	0.9992	—

Notes: $S = 5{,}000$. Fraction of observations assigned to their true mean regime (under $\hat{\gamma}^{\mu}$) and true variance regime (under $\hat{\gamma}^{\sigma}$); "—" marks the dimension an estimator does not model. For the decoupled estimator the joint $(\mu \times \sigma)$ cell accuracy is 0.9946 at $T = 200$ and 0.9988 at $T = 1000$ on DGP5, and 0.9944 and 0.9989 on DGP6. Parameter RMSEs are in Table 4.

Table A-6: Threshold estimators: finite-sample vs. limiting distributions

		Quantiles of $T(\hat{\gamma} - \gamma^0)$					
		q_{10}	q_{25}	q_{50}	q_{75}	q_{90}	KS
DGP7, $\hat{\gamma}$ (joint)	limit	−6.06	−3.32	−1.29	−0.12	2.93	
	$T = 200$	−6.05	−3.38	−1.29	−0.10	3.04	0.010
	$T = 1000$	−5.62	−3.12	−1.22	−0.06	3.44	0.024
DGP8, $\hat{\gamma}^{\mu}$	limit	−6.73	−3.80	−1.64	−0.40	1.34	
	$T = 200$	−6.73	−3.83	−1.61	−0.38	1.47	0.009
	$T = 1000$	−6.78	−3.79	−1.60	−0.34	1.23	0.025
DGP8, $\hat{\gamma}^{\sigma}$	limit	−9.95	−3.58	−0.02	6.14	15.64	
	$T = 200$	−12.42	−4.31	−0.13	6.49	17.89	0.031
	$T = 1000$	−10.02	−3.55	0.03	6.15	16.57	0.012

Notes: limits simulated with $R = 200{,}000$ draws of the smallest maximizer M_{-} of the corresponding two-sided compound Poisson process; finite-sample distributions from $S = 5{,}000$ ($T = 200$) and $S = 2{,}000$ ($T = 1000$) replications. KS is the two-sample Kolmogorov–Smirnov distance to the simulated limit; because S differs across columns, the entries have different sampling floors (5% critical values 0.019 and 0.030 respectively) and are not comparable across T.

Table A-7: Regime parameters, independence, and plug-in coverage

		$T = 200$	$T = 1000$
Asymptotic standard deviations: empirical/theoretical ratio, $\sqrt{T}\,\mathrm{vec}(\hat{B}_1 - B_1^0)$			
DGP7	mean [range]	1.04 [1.02, 1.06]	1.00 [0.98, 1.01]
DGP8	mean [range]	1.05 [1.03, 1.06]	1.00 [0.99, 1.03]
Independence checks (correlations)			
DGP7	$\max_j \lvert\mathrm{corr}(T\Delta\hat{\gamma},\ \sqrt{T}\,\mathrm{vec}(\hat{B}_1)_j)\rvert$	0.080	0.026
DGP8	$\max_j \lvert\mathrm{corr}(T\Delta\hat{\gamma}^{\mu},\ \sqrt{T}\,\mathrm{vec}(\hat{B}_1)_j)\rvert$	0.072	0.053
DGP8	$\mathrm{corr}(T\Delta\hat{\gamma}^{\mu},\ T\Delta\hat{\gamma}^{\sigma})$	−0.007	+0.013
Remark 15 plug-in confidence intervals for $\gamma^{\sigma,0}$*: empirical coverage*			
nominal 90%	oracle / feasible	0.859 / 0.860	0.894 / 0.899
nominal 95%	oracle / feasible	0.916 / 0.915	0.942 / 0.940

Notes: $\Delta\hat{\gamma} := \hat{\gamma} - \gamma^0$. Theoretical standard deviations from $S_1^{-1} \otimes \Sigma_1^0$ (DGP7) and the GLS information inverse (DGP8), with S_1 computed from a long simulated path. Feasible coverage: each replication re-simulates the variance-side limit law from its own $(\hat{\Sigma}_1, \hat{\Sigma}_2)$ (1,500 draws), with the exogenous-state density known; evaluated on 1,000 replications (binomial standard error ≈ 0.010).

Table A-8: Nesting experiment: selection frequencies under alternative penalties and quantiles of the likelihood gap

		Restriction selected, $c_T =$			Quantiles of $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}})$					
DGP	T	$\ln T$	$2\ln T$	$3\ln T$	min	50%	90%	95%	99%	max
DGP1	200	0.610	0.872	0.964	0.00	4.17	11.77	14.62	20.00	32.08
	1000	0.761	0.948	0.992	0.00	3.85	10.98	14.01	19.98	28.79
DGP2	200	0.813	0.990	0.999	0.00	3.35	6.40	7.83	10.55	18.05
	1000	0.946	0.998	1.000	0.00	2.73	5.93	7.06	9.99	19.01
DGP3	200	0.800	0.997	1.000	0.00	3.54	6.36	7.35	9.20	15.29
	1000	0.920	1.000	1.000	0.00	3.74	6.61	7.55	9.89	14.92
DGP4	200	0.992	1.000	1.000	0.00	0.54	2.38	3.24	5.03	11.46
	1000	0.999	1.000	1.000	0.00	0.00	1.61	2.40	4.19	12.89
DGP5	200	0.000	0.000	0.001	13.77	37.39	46.66	49.45	55.25	64.95
	1000	0.000	0.000	0.000	139.2	197.1	217.8	223.4	235.6	257.8
DGP6	200	0.000	0.000	0.000	18.82	43.36	55.76	59.47	65.92	75.49
	1000	0.000	0.000	0.000	157.6	221.3	245.9	253.1	268.7	301.7

Notes: $S = 5{,}000$ per cell. Frequencies are the share of replications with $T(\hat{L}_T^{\mathrm{F}} - \hat{L}_T^{\mathrm{J}}) < c_T$; $\ln T = 5.30$, 6.91 and $2\ln T = 10.60$, 13.82 and $3\ln T = 15.89$, 20.72 at $T = 200$, 1000. Monte Carlo standard errors are at most 0.007. The statistic is constructed as in Table 3; on DGP1–DGP4 it is non-negative by construction, and the frequencies are identical whether or not the unrestricted criterion is augmented with the restricted estimate. With $c_T = 3\ln T$ the restriction is wrongly selected in 3 of 5,000 DGP5 samples at $T = 200$ (minimum gap 13.8 against a cutoff of 15.9), the only loss of power in the table.

C The Interest-Rate (ZLB) Exercise

This appendix reports the interest-rate counterpart of the slack exercise of Section 4, with the lagged three-month T-bill rate as the mean state and RZ2018's imposed threshold of 0.25%. The post-WWII sample is infeasible (too few low-rate observations before 2009), so two samples are reported.

Thresholds. The joint estimator places the threshold at 3.48% on both samples (Table 5): a broad low-rate environment, not a zero lower bound, and again a variance-driven split (the SSR criterion gives 0.21% and 1.22%). Once the fiscal-event regime is isolated, the decoupled mean threshold on the full sample is 0.24% – RZ2018's 0.25% recovered from the data – while on the sample excluding WWII it is 2.73%; we therefore label the regimes low-rate and high-rate rather than ZLB and normal.

Table A-9: Decoupled TVAR, interest-rate exercise: thresholds and four-cell observation counts

	Thresholds		Low rate		High rate		
Sample	$\hat{\gamma}^{\mu}$ (%)	$\hat{\gamma}^{\sigma}$	normal	fiscal-event†	normal	fiscal-event†	$\lvert z\rvert$ share (%)
Full sample	0.24*	0.0243	50 (12.5)	7 (1.8)	317 (79.3)	26 (6.5)	37.3
Excl. WWII	2.73	0.0241	140 (37.0)	14 (3.7)	213 (56.3)	11 (2.9)	45.7
Imposed (RZ2018)	0.25	—			—		—

Notes: As Table 6, with the lagged T-bill rate as mean state. *Essentially RZ2018's imposed value.

Multipliers. Tables A-10 and A-11 and Figures A-1–A-2 report the normal-shock multipliers at the estimated and at the imposed thresholds. At the estimated thresholds, the full-sample multipliers are close to one in both regimes and the difference is never significant; on the sample excluding WWII the low-rate regime has the larger multiplier (0.98–1.37 against 0.15–0.23), significantly so only at $h = 3$. At the imposed 0.25% threshold the full-sample estimates essentially coincide with the free ones – the two thresholds differ by one basis point, an internal consistency check – while on the sample excluding WWII the low-rate regime's multiplier is 2.5–2.8 against 0.6 in the high-rate regime, with the difference significant at every horizon ($\Delta = -2.0$ to -2.2). This mirrors RZ2018's own finding of larger multipliers at the zero lower bound once WWII is excluded. Two caveats: the low-rate normal-shock cell contains 50 quarters (the 1930s and the post-2008 period), so the result is complementary to, not independent of, the slack results; and the difference between the two historical samples traces to the leverage points of Appendix D.

D Leverage Points and the Full Sample

On the full sample the imposed-threshold estimates are less precise than on the sample excluding WWII (Tables 8 and A-11), and in the interest-rate exercise the free estimates differ markedly between the two samples although the fiscal-event regime is isolated in both. A decomposition that swaps, one at a time, the regime coefficients and the regime covariances estimated on the two samples attributes the entire difference to the mean-regime coefficients of the low-rate (and, in the slack exercise, high-unemployment) regime, not to the covariances; and within those coefficients to exactly three quarters, 1941Q3–1942Q1, the U.S. entry into

Table A-10: Interest-rate exercise: normal-shock multipliers at the estimated thresholds

		Horizon h (quarters)			
Sample	Regime	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Full sample	low rate	0.96	0.57	0.41	0.31
		[0.44, 1.62]	*[0.08, 1.18]*	*[-0.34, 1.07]*	*[-1.63, 1.22]*
	high rate	0.58	0.60	0.63	0.67
		[0.29, 1.01]	*[0.37, 0.91]*	*[0.43, 0.93]*	*[0.46, 1.10]*
	Δ (high − low)	−0.38	0.02	0.22	0.36
		[-0.88, 0.22]	*[-0.55, 0.59]*	*[-0.43, 1.10]*	*[-0.50, 2.72]*
Excl. WWII	low rate	0.98	1.00	1.10	1.37
		[-0.49, 3.85]	*[0.04, 2.52]*	*[0.17, 2.36]*	*[0.39, 2.55]*
	high rate	0.15	0.13	0.17	0.23
		[-1.48, 0.84]	*[-3.78, 2.90]*	*[-3.83, 3.64]*	*[-1.96, 3.96]*
	Δ (high − low)	−0.83	−0.87	−0.93	−1.14
		[-3.95, -0.29]	*[-5.47, 2.06]*	*[-4.88, 2.66]*	*[-3.92, 2.55]*

Notes: As Table 7; mean regimes at $\hat{\gamma}^{\mu}$ of Table A-9. $\Delta < 0$ indicates larger multipliers in the low-rate regime.

Table A-11: Interest-rate exercise: normal-shock multipliers at the imposed 0.25% threshold

		Horizon h (quarters)			
Sample	Regime	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Full sample	low rate	0.97	0.58	0.42	0.33
		[0.42, 1.56]	*[0.21, 1.01]*	*[-0.07, 0.84]*	*[-0.62, 0.76]*
	high rate	0.58	0.60	0.64	0.68
		[0.25, 1.02]	*[0.37, 0.91]*	*[0.44, 0.92]*	*[0.48, 1.03]*
	Δ (high − low)	−0.38	0.02	0.22	0.35
		[-0.81, 0.11]	*[-0.36, 0.40]*	*[-0.19, 0.78]*	*[-0.10, 1.39]*
Excl. WWII	low rate	2.54	2.71	2.75	2.80
		[1.30, 4.12]	*[1.59, 4.24]*	*[1.57, 4.34]*	*[1.40, 4.71]*
	high rate	0.59	0.55	0.57	0.66
		[-0.45, 1.41]	*[-0.08, 1.20]*	*[0.02, 1.19]*	*[0.21, 1.32]*
	Δ (high − low)	−1.95	−2.16	−2.18	−2.15
		[-3.40, -0.93]	*[-3.69, -1.10]*	*[-3.82, -0.89]*	*[-4.04, -0.51]*

Notes: As Table 8, mean threshold imposed at 0.25%. Normal-shock cell counts (low / high rate): 51 / 316 (full), 50 / 303 (excl. WWII).

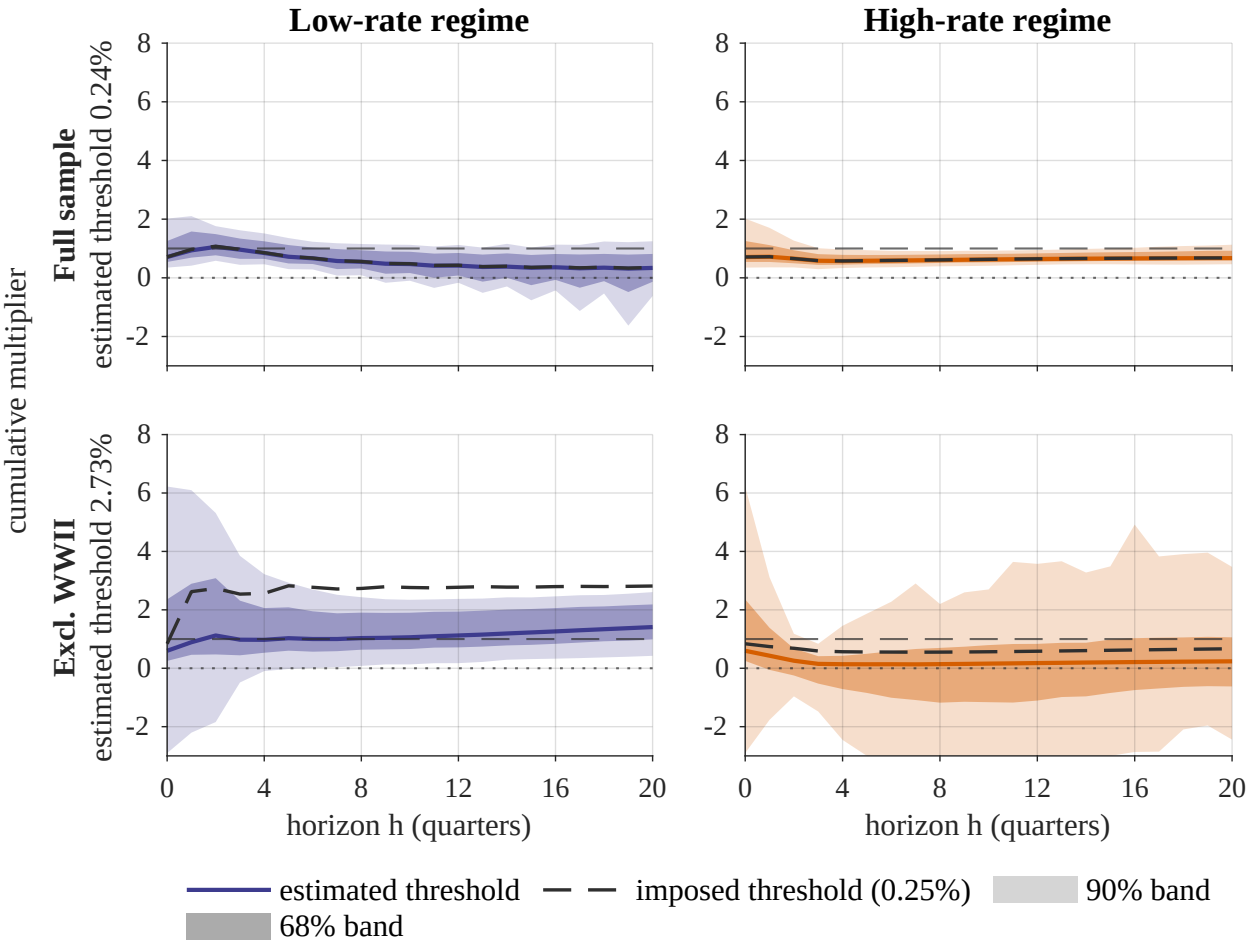


Figure A-1: Interest-rate exercise: cumulative multipliers in the normal-shock regime at the estimated thresholds (rows: full sample, excl. WWII; columns: low-rate, high-rate), 68% (dark) and 90% (light) bands; dashed line: point estimate at the imposed 0.25%.

WWII. These quarters carry the largest news realizations of the century while unemployment is still high and rates are at their floor, so they are extreme leverage points inside the regime. Two of the three fall in the fiscal-event regime, but the fiscal-event regime does not exclude them from the estimation of the mean-regime coefficients: under the decoupled model's GLS structure, observations in the fiscal-event regime enter $\hat{B}_i$ with a smaller precision weight, not a zero weight. Excluding the sample removes them; isolating the shock regime downweights them. This is why the sample excluding WWII – the sample RZ2018 themselves emphasize for the interest-rate exercise, given the rate peg and rationing of the war years – delivers the cleaner comparison on both state variables.

E Instrument Strength by Cell

In the internal-proxy SVAR the news series is ordered first and its innovation is the identified shock, so the strength of the identifying variation in each cell can be measured as in an instrumental-variables first stage: a regression of g_t on z_t and the VAR controls x_t within the cell, with z_t the news series. Table A-12 reports the

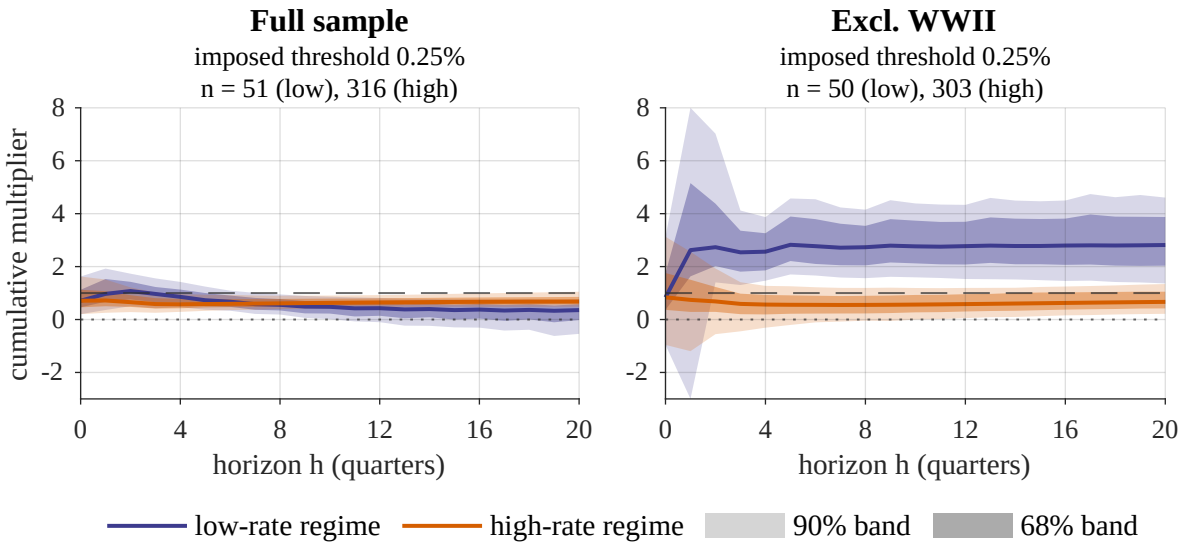


Figure A-2: Interest-rate exercise at the imposed 0.25% threshold, by sample: low-rate (violet) and high-rate (orange) regimes with 68% (dark) and 90% (light) bands conditional on the fixed threshold pair.

number of observations, the F-statistic on z_t with Newey–West (four lags) and with homoskedastic standard errors, the effective F of Montiel Olea and Pflueger (2013), which for a single instrument coincides with the HAC F, the HAC t-statistic, and the share of the cell's $\sum_t z_t^2$ accounted for by its three and five largest realizations. Four layers are compared on each sample: the whole sample without regimes; RZ2018's own partition, lagged unemployment at 6.5% with no variance regimes; the imposed 6.5% threshold with the fiscal-event regime isolated at the estimated variance threshold, which is the exercise of Table 8; and the estimated mean and variance thresholds of Table 7. Cells with fewer than 16 observations are not estimated.

Three facts emerge. First, isolating the fiscal-event regime strengthens the first stage in the low-unemployment regime of every sample: relative to RZ2018's partition the HAC F rises by factors of 5.3, 17.6 and 2.1 on the full, excluding-WWII and post-WWII samples (from 2.2, 0.8 and 1.5 to 11.6, 13.7 and 3.1; homoskedastic F from 16 to 84 on the full sample). The fiscal-event cells carry a contemporaneous news–spending coefficient of the opposite sign ($t = -5.3$ on the full sample), so the pooled regime's first stage is a cancellation. Isolating the fiscal-event regime also strengthens both post-WWII cells and the full-sample high-unemployment cell at 6.5%; it weakens the high-unemployment cell on the sample excluding WWII ($F < 0.1$ with 170 observations), where the news variation outside the fiscal-event regime is negligible, as RZ2018 note for their high-unemployment state on this sample. Second, no cell in any layer, RZ2018's own partition included, reaches the Montiel Olea and Pflueger (2013) critical value for a 5% worst-case bias (37.4; 23.1 for 10%): the largest effective F is 13.7, and the top-three shares, between 0.56 and 1.00, show that in every cell the identifying variation is a small number of quarters. This is the property of the military news series that RZ2018 document, and it is shared by every specification in this literature; a VAR first stage is also contemporaneous, whereas the news series is anticipatory and RZ2018's first stages cumulate spending over the horizon, so the statistics here are conservative. The identifying moment of the decoupled model is the normal-shock covariance Σ_1, which pools

the two mean regimes: on it the homoskedastic F is 39, 50 and 3.6 on the three samples (HAC: 3.0, 6.0 and 4.1). Third, the two cells whose bootstrap draws produce the denominator problem noted under Tables 7 and 8 – the post-WWII low-unemployment cell at the estimated threshold (47 quarters) and the post-WWII high-unemployment cell at 6.5% (79 quarters) – are both weak by every measure, but so are seven other normal-shock cells in which the denominator never crosses zero. Weak identification is pervasive in these data; the denominator problem arises where it meets a small cell.

F Posterior Bands as a Check on the Bootstrap

The wild-bootstrap intervals of Section 4 are wide in some cells, and a reader may ask whether the width is a property of the bootstrap or of the object. This appendix answers by computing the bands a second way. Conditional on the estimated threshold pair, the decoupled model is a set of linear regressions with cell-specific covariances, and a two-block Gibbs sampler – B_i given Σ from its Normal full conditional, Σ_ℓ given B from an inverse-Wishart – delivers a posterior for the regime parameters and, through the same impulse-response code, for the multipliers. We use two priors. The flat prior has no shrinkage and serves as the benchmark that should agree with the bootstrap if the bootstrap is doing its job. The second is a Minnesota prior with the usual tightness ($\lambda_1 = 0.2$, lag decay 1, cross-equation factor 0.5) on the macroeconomic equations, but a loose prior on the coefficients of the news series: the news column is the instrument, and shrinking it towards zero removes the identifying variation from the system. A standard Minnesota prior applied to all columns shrinks the summed coefficients of output on lagged news from 0.10 to 0.005 in the post-WWII low-unemployment cell at the estimated threshold, and the changes it then produces in the multipliers reflect that choice rather than the data. Each chain has 2,000 retained draws after 500 of burn-in, started at the FGLS estimates.

Three results. First, the posterior under the flat prior reproduces the bootstrap. Posterior medians are within a few hundredths of the point estimates in the well-populated cells, and the 90% posterior intervals are 0.81–0.98 times the bootstrap width on the two historical samples and 0.65–0.72 times post-WWII, where the wild bootstrap's resampling of heteroskedastic residuals produces heavier tails than a Gaussian likelihood. The two methods agree on which cells are not identified: in the post-WWII high-unemployment cell at the imposed threshold the 90% interval under the flat prior at $h = 3$ is $[-14.9, 13.6]$ against the bootstrap's $[-23.5, 21.6]$, and the cumulative spending response changes sign in 17–20% of posterior draws against 23% of bootstrap draws. The width is the object. Second, under the Minnesota prior the results of Section 4 survive where they are claimed (Table A-13). At the imposed threshold on the sample excluding WWII the difference between regimes is 1.0–2.1 with the 90% interval excluding zero at $h = 3$, 7 and 11; at the estimated thresholds on the same sample the ranking is unchanged and the 68% interval excludes zero at every horizon; on the post-WWII sample at the estimated thresholds the low-unemployment multiplier is 2.9 at $h = 3$ against 1.5 in the high-unemployment regime, the 68% interval for the difference excluding zero at $h = 3$, 11 and 19 and the 90% interval never, as under the bootstrap. The imposed post-WWII high-unemployment cell remains unidentified under every prior. Third, the shrinkage exposes one fragility the bootstrap does not: in the full-sample high-unemployment regime at the estimated threshold (84 quarters) the estimated within-regime dynamics are explosive, with a largest root of 1.58, and under a prior that shrinks the own lags while leaving the news column free the long-horizon denominator changes sign in 8–28% of draws for $h \geq 7$. The short-

Table A-12: First-stage strength of the news instrument by cell (slack exercise)

Sample	Cell	n	F_{HAC}	F_{homosk}	eff. F	t_{HAC}	top-3	top-5
L0: whole sample, no regimes								
Full sample	all quarters	499	3.75	19.73	3.75	1.94	0.56	0.73
Excl. WWII	all quarters	477	0.90	8.76	0.90	0.95	0.66	0.85
Post-WWII	all quarters	272	1.25	1.26	1.25	-1.12	0.95	0.97
L1: RZ2018 partition, lagged unemployment at 6.5%, no variance regimes								
Full sample	low unemployment	318	2.21	16.45	2.21	1.49	0.64	0.83
	high unemployment	181	2.87	0.83	2.87	1.70	0.93	0.99
Excl. WWII	low unemployment	300	0.78	7.11	0.78	0.88	0.82	0.95
	high unemployment	177	0.42	0.34	0.42	0.65	0.96	0.98
Post-WWII	low unemployment	189	1.51	1.64	1.51	-1.23	0.96	0.98
	high unemployment	83	0.64	0.67	0.64	0.80	0.75	0.92
L2: imposed 6.5% mean threshold, estimated variance threshold								
Full sample	low-u × normal-shock	294	11.62	84.18	11.62	3.41	0.72	0.88
	low-u × fiscal-event	24	27.59	9.75	27.59	-5.25	0.95	1.00
	high-u × normal-shock[a]	172	6.10	2.67	6.10	2.47	0.99	0.99
	high-u × fiscal-event	9	–	–	–	–	1.00	1.00
	normal-shock, pooled	466	2.96	39.29	2.96	1.72	0.72	0.84
	fiscal-event, pooled	33	22.98	5.47	22.98	-4.79	0.82	0.99
Excl. WWII	low-u × normal-shock	282	13.66	68.93	13.66	3.70	0.90	0.94
	low-u × fiscal-event	18	0.04	0.01	0.04	0.20	0.98	1.00
	high-u × normal-shock[a]	170	0.00	0.00	0.00	-0.05	0.95	0.97
	high-u × fiscal-event	7	–	–	–	–	1.00	1.00
	normal-shock, pooled	452	5.97	50.44	5.97	2.44	0.88	0.91
	fiscal-event, pooled	25	3.03	1.02	3.03	-1.74	0.94	0.99
Post-WWII	low-u × normal-shock[a]	176	3.11	2.48	3.11	1.76	0.58	0.71
	low-u × fiscal-event	13	–	–	–	–	1.00	1.00
	high-u × normal-shock[a]	79	1.32	0.91	1.32	1.15	0.76	0.93
	high-u × fiscal-event	4	–	–	–	–	1.00	1.00
	normal-shock, pooled	255	4.05	3.62	4.05	2.01	0.47	0.64
	fiscal-event, pooled	17	1.15	0.32	1.15	1.07	1.00	1.00
L3: estimated mean and variance thresholds								
Full sample	low-u × normal-shock	382	11.46	101.4	11.46	3.39	0.64	0.83
	low-u × fiscal-event	28	16.89	5.17	16.89	-4.11	0.95	1.00
	high-u × normal-shock[a]	84	1.03	0.78	1.03	1.02	1.00	1.00
	high-u × fiscal-event	5	–	–	–	–	1.00	1.00
	normal-shock, pooled	466	2.96	39.29	2.96	1.72	0.72	0.84
	fiscal-event, pooled	33	22.98	5.47	22.98	-4.79	0.82	0.99
Excl. WWII	low-u × normal-shock[a]	37	0.05	0.01	0.05	-0.23	1.00	1.00
	low-u × fiscal-event	4	–	–	–	–	1.00	1.00
	high-u × normal-shock[a]	415	4.11	66.30	4.11	2.03	0.88	0.91
	high-u × fiscal-event	21	0.09	0.02	0.09	0.30	0.99	1.00
	normal-shock, pooled	452	5.97	50.44	5.97	2.44	0.88	0.91
	fiscal-event, pooled	25	3.03	1.02	3.03	-1.74	0.94	0.99
Post-WWII	low-u × normal-shock[a]	47	4.98	2.27	4.98	2.23	0.56	0.76
	low-u × fiscal-event	5	–	–	–	–	1.00	1.00
	high-u × normal-shock[a]	208	4.59	3.74	4.59	2.14	0.55	0.74
	high-u × fiscal-event	12	–	–	–	–	1.00	1.00
	normal-shock, pooled	255	4.05	3.62	4.05	2.01	0.47	0.64
	fiscal-event, pooled	17	1.15	0.32	1.15	1.07	1.00	1.00

Notes: First-stage regression of g_t on z_t and the VAR regressors x_t (constant and four lags of z, y, g) within each cell; slack exercise, mean state lagged unemployment. L2 and L3 cells are (mean regime) × (variance regime); "pooled" rows pool the two mean regimes within a variance regime. $F_{\text{HAC}} = t^2_{\text{HAC}}$ with a Newey–West (Bartlett, 4 lags) variance, lags in calendar quarters; F_{homosk}: classical first-stage F. Eff. F: Montiel Olea and Pflueger (2013) effective F with the same HAC variance; with one instrument it equals F_{HAC}. Critical value (simplified procedure, 5% test, worst-case bias 5% of the benchmark): 37.42 (10%: 23.11). Top-3 / top-5: share of the cell's $\sum z_t^2$ due to its three / five largest $|z_t|$. – : cell too small for the regression ($n < 16$).
[a] Normal-shock cell with $F_{\text{HAC}} < 10$.

horizon multipliers in that cell are unaffected; the $h = 19$ values should be read with that caveat, which we note under Table 7. Where the posterior and the bootstrap differ, the difference traces to the prior; where they agree, the bands are what the data support.

G The Stability Condition at the Estimates

Assumption **A4** requires a common Lyapunov matrix for the mean-regime companion matrices. It restricts the data-generating process, not the estimates, but it can be checked at the point estimates: for the two estimated companion matrices $\hat{A}_1, \hat{A}_2$, bisection on λ over the feasibility problem $P \succeq I$, $\hat{A}_i' P \hat{A}_i \preceq \lambda^2 P$, $i = 1, 2$, solved as a semidefinite program. The smallest feasible λ is the common contraction factor; a regime whose companion matrix has a root of modulus one or more rules out any $\lambda < 1$. Two things the check does and does not establish. A root above one rules out **A4**, which is a sufficient condition for geometric ergodicity, not ergodicity itself: a threshold process can be globally stationary with a locally explosive regime (Remark A.1), and the point estimates cannot distinguish that case from genuine nonstationarity. What the check establishes is whether the premise under which Theorems 1–2 are proved holds at the estimated parameters; where it fails, the theory is silent, not contradicted. Table A-14 reports, for the slack exercise at the estimated thresholds of Table 6, the largest root of each estimated mean regime, its distribution across the wild-bootstrap draws behind Table 7 (thresholds and regime parameters re-estimated in each draw), and the verdict.

Three facts stand out. On the sample excluding WWII the condition holds, with a contraction factor equal to the largest within-regime root: the switching itself costs nothing. On the other two samples it fails at the point estimates in the smaller cell. In the post-WWII low-unemployment regime the failure is within sampling uncertainty: the root lies below one in a quarter of the draws, with a 90% range from 0.94 to 1.17, and even the 436-quarter regime of the sample excluding WWII has its root below one in only 39% of the draws. This is the near-unit-root persistence of a VAR in levels – the linear VAR's largest root is 0.91–0.97 – estimated regime by regime on few quarters. In the full-sample high-unemployment regime the failure is not sampling uncertainty: a root of 1.58, above one in every draw but one, says that a constant-coefficient law of motion is not an adequate description of the Great Depression, and that cell lies outside the premise of Theorems 1–2. Explosive within-regime roots are a known feature of regime-switching VARs on these data. Ramey and Zubairy (2018) report that their own threshold VAR had explosive roots under the slack definition of the state (their Section VI.B and footnote 42), which is what led them to NBER recessions as the state of their TVAR exercise; at their 6.5% threshold the high-unemployment regime is explosive on all three samples in our estimates as well. Where the failure is within sampling uncertainty, the thresholds and the regime parameters are unaffected, and so are short-horizon multipliers, which are finite sums of a few coefficients; the long-horizon multipliers of the small cells are the objects that depend on the roots, and Section 4 reads them accordingly. The Great Depression cell is the exception, and is read at short horizons only.

H Fiscal-Event Cells

The fiscal-event cells are too small for inference – 4 to 28 quarters in the slack exercise and 7 to 26 in the interest-rate exercise (Tables 6 and A-9) – and their multipliers are dominated by a small number of historical

Table A-13: Posterior bands for the normal-shock multipliers under a Minnesota prior with a loose news column

Sample	Regime	Horizon h (quarters) $h=3$	$h=7$	$h=11$	$h=19$
Panel A: estimated thresholds (Table 7)					
Full sample	low unemployment	0.41	0.44	0.46	0.45
		[0.15, 0.69]	*[0.26, 0.66]*	*[0.30, 0.66]*	*[0.30, 0.64]*
	high unemployment	0.76	1.08	1.19	1.14
		[0.21, 1.48]	*[-1.31, 3.90]*	*[-5.09, 5.99]*	*[-6.71, 7.87]*
	Δ (high − low)	0.34	0.65	0.73	0.70
		[-0.18, 0.98]	*[-1.77, 3.48]*	*[-5.49, 5.55]*	*[-7.10, 7.44]*
Excl. WWII	low unemployment	0.28	0.15	0.07	−0.07
		[0.08, 0.52]	*[-0.06, 0.39]*	*[-0.18, 0.43]*	*[-0.46, 0.76]*
	high unemployment	0.86	1.05	1.07	0.97
		[0.06, 1.85]	*[0.19, 2.12]*	*[0.21, 2.28]*	*[0.21, 2.08]*
	Δ (high − low)	0.57	0.89	0.99	1.01
		[-0.16, 1.47]	*[0.01, 1.99]*	*[0.02, 2.21]*	*[-0.17, 2.27]*
Post-WWII	low unemployment	2.85	2.38	2.50	2.98
		[0.52, 7.20]	*[0.54, 5.67]*	*[0.50, 7.71]*	*[-7.98, 21.32]*
	high unemployment	1.53	1.26	1.05	0.81
		[-0.38, 3.88]	*[-0.42, 3.29]*	*[-0.55, 2.88]*	*[-0.49, 2.30]*
	Δ (high − low)	−1.34	−1.14	−1.53	−2.27
		[-5.13, 0.79]	*[-4.21, 1.14]*	*[-6.84, 1.04]*	*[-20.91, 9.33]*
Panel B: imposed 6.5% mean threshold (Table 8)					
Full sample	low unemployment	0.47	0.48	0.49	0.47
		[0.18, 0.80]	*[0.23, 0.78]*	*[0.27, 0.78]*	*[0.29, 0.75]*
	high unemployment	0.45	0.67	0.78	0.82
		[-0.05, 0.92]	*[0.31, 1.03]*	*[0.37, 1.20]*	*[0.31, 1.37]*
	Δ (high − low)	−0.01	0.19	0.28	0.33
		[-0.46, 0.39]	*[-0.20, 0.57]*	*[-0.20, 0.74]*	*[-0.25, 0.92]*
Excl. WWII	low unemployment	0.40	0.27	0.22	0.17
		[-0.05, 1.01]	*[-0.04, 0.75]*	*[-0.04, 0.62]*	*[-0.04, 0.48]*
	high unemployment	1.43	2.05	2.27	2.31
		[0.38, 4.17]	*[0.56, 6.67]*	*[0.45, 8.44]*	*[0.20, 10.01]*
	Δ (high − low)	1.01	1.76	2.03	2.14
		[0.05, 3.61]	*[0.24, 6.44]*	*[0.22, 8.07]*	*[0.00, 9.86]*
Post-WWII	low unemployment	1.17	0.73	0.59	0.49
		[-0.68, 3.29]	*[-0.46, 1.93]*	*[-0.43, 1.57]*	*[-0.42, 1.38]*
	high unemployment	0.57	0.77	0.71	0.59
		[-10.36, 11.07]	*[-6.53, 8.49]*	*[-5.00, 6.24]*	*[-2.84, 3.51]*
	Δ (high − low)	−0.66	0.06	0.16	0.14
		[-11.90, 9.31]	*[-7.23, 7.54]*	*[-5.71, 5.73]*	*[-3.60, 2.91]*

Notes: Posterior medians with 90% posterior intervals below, from a two-block Gibbs sampler conditional on the threshold pair (2,000 draws after 500 of burn-in, started at the FGLS estimates), Minnesota prior with $\lambda = (0.2, 0.5, 1)$ on the macroeconomic coefficients and prior standard deviation $2/j$ on the lag-j coefficients of the news series in the output and spending equations. Multipliers as in Table 7; Δ rows are posterior differences across mean regimes, draw by draw. Point estimates and 90% wild-bootstrap intervals for the same cells are in Tables 7 (Panel A) and 8 (Panel B). Normal-shock cell counts (low / high unemployment): Panel A, 382 / 84 (full), 374 / 415 (excl. WWII), 47 / 208 (post-WWII); Panel B, 294 / 172, 282 / 170, 176 / 79.

Table A-14: Assumption A4 at the estimated parameters, slack exercise

Sample	Mean regime	Quarters	Largest root	Share < 1	5th–95th pct.	A4
Full sample	$u_{t-1} \leq 8.42$	410	0.926	0.96	0.86–1.00	fails
	$u_{t-1} > 8.42$	89	1.582	0.00	1.26–1.75	
Excl. WWII	$u_{t-1} \leq 3.44$	41	0.963	0.59	0.90–1.11	holds, $\lambda^* = 0.985$
	$u_{t-1} > 3.44$	436	0.984	0.39	0.97–1.15	
Post-WWII	$u_{t-1} \leq 4.23$	52	1.045	0.24	0.94–1.17	fails
	$u_{t-1} > 4.23$	220	0.977	0.80	0.91–1.04	

Notes: Largest root is the spectral radius of the estimated companion matrix ($p = 4$, $n = 3$, thirteen coefficients per equation). Share < 1 and percentiles refer to the largest root across the 499 wild-bootstrap draws of Table 7. A4 is checked by bisection on λ over the semidefinite feasibility problem in the text; λ^* is the smallest feasible value. For reference, the linear VAR's largest root is 0.906 (full sample), 0.969 (excl. WWII) and 0.946 (post-WWII); at RZ2018's 6.5% threshold the high-unemployment regime's largest root is 1.03, 1.10 and 1.03 respectively.

episodes; the main text uses them only to isolate the normal-shock regime. Table A-15 documents what distinguishes the fiscal-event regime from the normal-shock regime in the second moments the likelihood uses: the residual standard deviation of each equation in the two variance regimes, the correlation between the output and spending innovations, and the log-determinant difference that enters the profile likelihood.

Table A-15: Residual volatility by variance regime at the estimated thresholds

			S.d., normal-shock (VR1)			S.d., fiscal-event (VR2)			$\text{corr}(y, g)$		
Exercise, sample	n_1	n_2	first	y	g	first	y	g	VR1	VR2	$\Delta \ln \det$
Slack, full sample	466	33	3.05	1.68	0.80	19.04	1.95	2.92	0.08	0.51	6.15
Slack, excl. WWII	452	25	2.39	1.70	0.60	14.82	1.55	1.57	−0.03	0.44	5.04
Slack, post-WWII	255	17	1.08	0.77	0.22	14.61	1.22	0.27	0.22	0.29	6.13
Interest rate, full	367	33	3.52	1.43	0.90	16.61	2.02	2.91	0.11	0.67	5.51
Interest rate, excl. WWII	353	25	2.57	1.34	0.69	15.56	1.89	1.25	0.01	0.61	4.80
AG2012, 1949–2008	225	15	0.79	2.60	1.24	1.51	3.72	1.52	0.13	−0.26	1.39

Notes: $\Delta \ln \det = \ln \det \hat{\Sigma}_2 - \ln \det \hat{\Sigma}_1$. $\hat{\Sigma}_\ell$ is the pooled residual second-moment matrix of variance regime ℓ at the stored thresholds and coefficients, the object the estimator uses. Standard deviations are $\sqrt{\text{diag}\, \hat{\Sigma}_\ell}$ in percent of potential GDP (slack and interest rate) and 100×log units (AG2012). "First" is the news equation in the slack and interest-rate exercises and the tax equation in the AG2012 exercise, whose VAR does not contain the news series. The news residual is large in the fiscal-event regime by construction, since the state is the lagged news; the spending and output columns and the correlation are not. No confidence intervals are reported: the fiscal-event regime has 15–33 quarters. For comparison, at the joint split of Figure 2 (3.56%, full sample) the low-state/high-state ratios are 1.04, 1.13 and 2.58 for news, output and spending.

For completeness, Table A-16 reports the fiscal-event cells of the AG2012 exercise, where the cell sizes are documented in Table A-18: every 90% interval includes zero, the fiscal-event environment carries larger point multipliers, the ranking across mean regimes matches the normal-shock cells, and the $h = 0$ multiplier is identical across mean regimes within the variance regime, as implied by the common impact vector.

Table A-16: Decoupled TVAR on AG2012 data: fiscal-event (VR2) cell multipliers

	Horizon h (quarters)					
Cell	$h=0$	$h=3$	$h=7$	$h=11$	$h=15$	$h=19$
MR1×VR2 (low growth, $n=11$)	0.84	0.20	0.12	0.40	0.80	1.20
	[-1.77, 10.25]	*[-3.50, 10.30]*	*[-3.66, 8.17]*	*[-3.94, 6.84]*	*[-4.45, 6.28]*	*[-5.29, 5.63]*
MR2×VR2 (high growth, $n=4$)	0.84	1.79	2.50	3.05	3.58	4.02
	[-1.77, 10.25]	*[-0.93, 9.25]*	*[-1.53, 8.65]*	*[-2.21, 8.26]*	*[-2.79, 7.83]*	*[-3.12, 7.68]*

Notes: Cumulative dollar multipliers in the fiscal-event variance regime (VR2) of the decoupled specification with $q_t^\sigma = |\text{news}_{t-1}|$; 90% wild-bootstrap intervals in brackets ($B = 499$, all draws valid). Estimates are descriptive only, given the cell sizes documented in Table A-18. The identical $h = 0$ entries across the two cells reflect the common impact vector within a variance regime.

I Spending Growth as the Variance State

This appendix re-estimates the decoupled model of Section 4.3 with the absolute lagged change in government spending, $|\Delta g_{t-1}|$, in place of the absolute lagged news series as the variance state, on the same samples, grid rule and bootstrap. Table A-17 compares the two states. The spending-growth state does not recover the fiscal-event regime: it isolates the build-ups *and* demobilizations of WWI, WWII and Korea together with pre-1940 peacetime spending volatility, misses the announcement quarters that precede spending and every fiscal event after 1968, and agrees with the news-based regime on 15 of its 92 quarters ($\kappa = 0.16$; 0.12 excluding WWII; 0.00 post-WWII, where peacetime spending changes are too small for the state to separate a shock environment and the above-threshold regime contains 91% of the sample). This is the pattern the timing argument of Section 4.3 predicts for a state defined on the response rather than on the identifying shock. Multipliers are interpretable only on the full sample, where the mean threshold moves to 3.56% – the joint estimate of Table 5, since the state leaves the wartime variance where the joint estimator found it – and the difference between the normal-shock cells vanishes ($\Delta = 0.06$ at $h = 19$); the other two samples yield boundary partitions with cells too thin for inference.[2]

J The Auerbach–Gorodnichenko Exercise

This appendix documents the robustness exercise summarized in Section 4.7.

Data, identification and states. We use the AG2012 replication data: real government purchases G_t, net real taxes T_t and real output Y_t, in log levels, 1947Q1–2008Q4 ($T = 248$), with $p = 4$ lags. Identification follows AG2012 and Blanchard and Perotti (2002): a Cholesky decomposition of the regime-specific covariance matrix with G ordered first identifies the government-spending shock by the restriction that G does not respond to within-quarter innovations in T or Y. We report cumulative integral dollar multipliers, $\sum_{s\le h}\hat{\phi}_{Y,s}/\sum_{s\le h}\hat{\phi}_{G,s}$ converted with the sample ratio $\bar{Y}/\bar{G} = 5.71$. AG2012's state variable is a seven-quarter

[2]The grid is built by the rule used for $|z_{t-1}|$, which has an atom at zero; $|\Delta g_{t-1}|$ is continuous, and a stricter minimum regime size would rule out the post-WWII corner solution. We keep one rule so that the comparison uses one code path.

Table A-17: Fiscal-event regime from news vs spending growth: thresholds, overlap and normal-shock multipliers ($h = 19$)

	$(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma})$		VR2 quarters			News state, $h = 19$			Spending state, $h = 19$		
Sample	News	Spending	News	Spending	κ	Low-u	High-u	Δ	Low-u	High-u	Δ
Full sample	(8.42, 2.434)	(3.56, 0.651)	33	92	0.16	0.45	0.98	0.53 [-0.69, 1.62]	0.49	0.55	0.06 [-0.88, 0.72]
Excl. WWII†	(3.44, 2.412)	(12.06, 0.674)	25	71	0.12	0.24	0.55	0.31 [-0.90, 1.61]	0.16	-36.75	-36.91 [-16.04, 15.34]
Post-WWII†	(4.23, 2.111)	(4.72, 0.023)	17	247	0.00	4.47	0.72	-3.75 [-25.59, 13.77]	1.37	-0.13	-1.50 [-11.66, 4.53]

Notes: Decoupled TVAR, mean state lagged unemployment (%). News state: $|z_{t-1}|$ (% of GDP); spending state: $|\Delta g_{t-1}|$, first difference of the VAR's spending series g (% of potential GDP). $\hat{\gamma}^{\sigma}$ in %. VR2 = fiscal-event regime (above the variance threshold). κ: Cohen's kappa between the two VR2 indicators, quarter by quarter on the common sample. Normal-shock (VR1) cumulative multipliers at $h = 19$; Δ = high-u − low-u with its 90% wild-bootstrap interval in brackets ($B = 499$).
† Spending state, Excl. WWII: point estimates lie outside their own 90% band at $h = 11, 19$ (cumulative-spending denominator near zero in a 17-quarter cell); these multipliers are not interpretable.
† Spending state, Post-WWII: the smallest normal-shock cell has only 6 quarters, so the multipliers are imprecise.

centered moving average of output growth dated at $t-1$, which embeds $\Delta \ln Y_{t+1}$ and $\Delta \ln Y_{t+2}$ and is therefore not known at $t-1$; since our consistency results require a predetermined state (Assumption B2), we use the backward seven-quarter moving average, $q_t^{\mu} = \frac{1}{7}\sum_{j=0}^{6} \Delta \ln Y_{t-j}$. The variance state is the same Ramey (2011) military-news series as in Section 4, here expressed as a percent of GDP (the two normalizations differ by a factor of 100, verified on the common quarters). Inference uses the same wild bootstrap as in Section 4 ($B = 499$).

Table A-18: Decoupled TVAR on AG2012 data (1947Q1–2008Q4): thresholds and four-cell observation counts

	Thresholds		Low growth		High growth		
Sample	$\hat{\gamma}^{\mu}$ (%)	$\hat{\gamma}^{\sigma}$	normal	fiscal-event†	normal	fiscal-event†	q^{σ} share (%)
Panel A: Ramey news, $q_t^{\sigma} = \lvert news_{t-1}\rvert$ (% of GDP)							
$\lvert news_{t-1}\rvert$	1.245	2.036	184 (76.7)	11 (4.6)	41 (17.1)	4 (1.7)	79.8
Panel B: G-equation residual, $q_t^{\sigma} = \lvert\hat{u}_{t-1}^{G}\rvert$ (log-difference)							
$\lvert\hat{u}_{t-1}^{G}\rvert$	1.245	0.00326	46 (19.2)	149 (62.1)	5 (2.1)‡	40 (16.7)	97.3

Notes: $\hat{\gamma}^{\mu}$ is the threshold on the backward seven-quarter moving average of $\Delta \ln Y$ (quarterly percent; $1.245 \approx 5\%$ annualized); mean regimes: low-growth ($q_{t-1}^{\mu} \le \hat{\gamma}^{\mu}$) and high-growth. $\hat{\gamma}^{\sigma}$ is the threshold on the variance state; variance regimes: normal-shock (VR1) and fiscal-event (VR2) in Panel A, quiet and dispersed in Panel B (Panel B's split has no event structure and serves as a specificity check). The last column is the share of $\sum_t |q_t^{\sigma}|$ in the above-threshold variance regime. †Cells too small for inference; see Appendix H. ‡Five observations; descriptive only.

Thresholds and cells. The news candidate selects $\hat{\gamma}^{\sigma} = 2.04\%$ of GDP, isolating 15 quarters that coincide, quarter for quarter, with the fiscal-event regime of Section 4 on the common 1949Q1–2008Q4 window (Cohen's $\kappa = 1$; the two RZ2018 quarters outside this window, 2011Q4 and 2013Q2, fall after AG2012's

sample).[3] The mean threshold is 1.25% per quarter (≈ 5% annualized), isolating a high-growth regime of 45 quarters.

Table A-19: Decoupled TVAR multipliers on AG2012 data, normal-shock regime

	Horizon h (quarters)			
Cell	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Panel A: Ramey news variance state ($\hat{\gamma}^{\sigma} = 2.04$% of GDP)				
low-growth × normal-shock	0.18	−0.15	−0.06	0.31
	[−0.90, 1.17]	*[−1.61, 0.85]*	*[−2.06, 0.96]*	*[−2.98, 1.71]*
high-growth × normal-shock	1.11	1.54	2.08	3.12
	[0.32, 1.82]	*[0.08, 3.18]*	*[−0.22, 4.17]*	*[−0.59, 4.78]*
Δ (high − low)	0.93	1.69	2.14	2.80
	[0.05, 1.79]	*[0.15, 3.76]*	*[0.06, 5.04]*	*[−0.56, 5.89]*
Panel B: G-residual variance state ($\hat{\gamma}^{\sigma} = 0.00326$)				
low-growth × normal-shock	−0.11	−0.57	−0.66	−0.67
	[−1.56, 1.12]	*[−2.21, 0.81]*	*[−2.61, 0.83]*	*[−3.57, 1.40]*
high-growth × normal-shock‡	0.84	1.11	1.55	2.47
	[−0.19, 1.69]	*[−0.52, 2.92]*	*[−0.87, 3.90]*	*[−1.67, 4.63]*

Notes: Cumulative integral dollar multipliers in the below-threshold variance regime only. 90% wild-bootstrap intervals in brackets ($B = 499$). Δ: difference across mean regimes, interval from the same bootstrap draws as the cell bands (the two cells share the normal-shock covariance, so the interval is tighter than a comparison of bands suggests); it excludes zero through $h = 11$. ‡Cell with $n = 5$ observations; descriptive only. Fiscal-event cells in Appendix H.

Multipliers. Table A-19 and Figure A-3 report the normal-shock cells. The high-growth multiplier exceeds the low-growth multiplier at every horizon – 1.11, 1.54, 2.08, 3.12 against 0.18, −0.15, −0.06, 0.31 at $h = 3, 7, 11, 19$ – and the difference is significant through $h = 11$ ($\Delta = 0.93$, 1.69, 2.14, with 90% intervals $[0.05, 1.79]$, $[0.15, 3.76]$, $[0.06, 5.04]$), even where the individual bands overlap, since the two cells are positively correlated through the shared normal-shock covariance.

Mean regimes across the two exercises. The two mean states measure different things – a labor-market stock in RZ2018, a growth flow in AG2012 – and their partitions overlap only partially on the common window: 45 and 48 quarters respectively, 16 shared ($\kappa = 0.19$), with the growth state leading the labor-market state by about six quarters through 1970 and the two selecting different episodes afterwards. The ranking of multipliers is nonetheless the same under both (Section 4.7). Figure A-4 shows the three regime indicators on the common window.

[3]The agreement is not near-automatic: the news candidate induces 59 distinct nontrivial partitions on this sample, so agreement by chance is below two percent; the fifteen matched quarters span six episodes across six decades; and Panel B shows that a dense volatility candidate – the G-equation's own absolute residual, free to isolate the largest spending surprises – is not steered toward an event partition but selects a low-versus-high dispersion split. Recovering the fiscal-event regime as the variance break therefore requires a candidate that encodes the timing of fiscal events, and it is that specific partition that the macroeconomic covariance confirms.

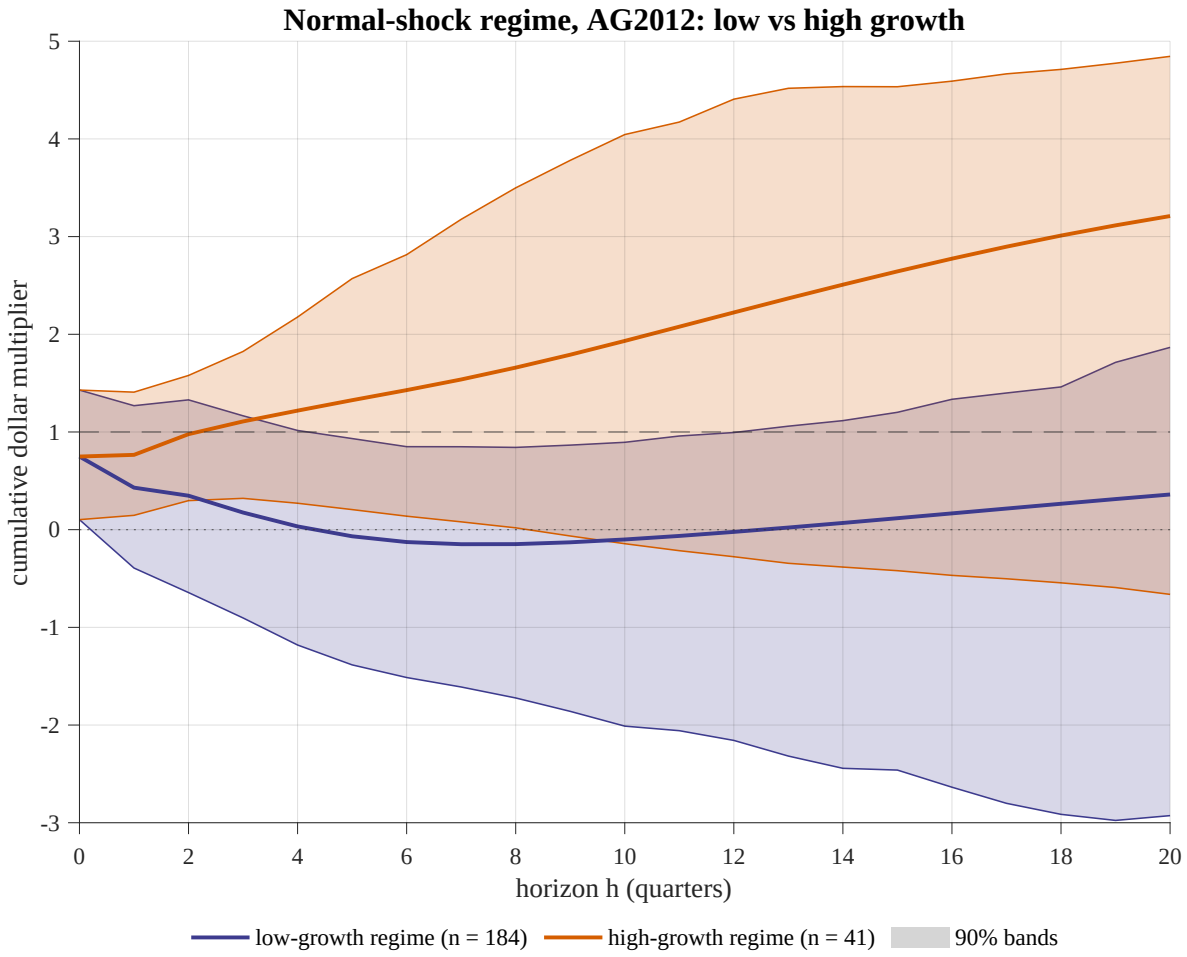


Figure A-3: AG2012 exercise: cumulative dollar multipliers in the normal-shock cells (high-growth and low-growth regimes) with 90% wild-bootstrap bands.

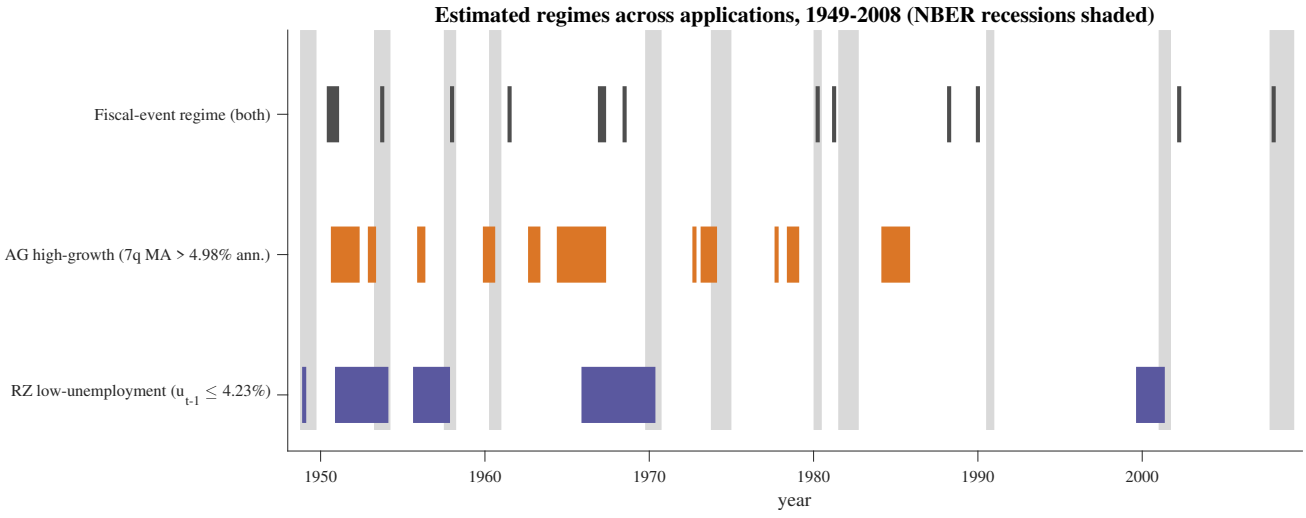


Figure A-4: Estimated regime assignments across the two exercises on the common window (1949Q1–2008Q4), NBER recessions shaded. *Bottom:* RZ2018 low-unemployment regime ($u_{t-1} \leq 4.23\%$). *Middle:* AG2012 high-growth regime. *Top:* the fiscal-event quarters, identical across the two exercises.